Innovative Approaches to Data Curation and Retrieval-Augmented Generation: From Annotation and Preparation to Retrieval.

by

Yushi Sun

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ABSTRACT

This thesis investigates the challenges and advancements in data curation and Retrieval-Augmented Generation (RAG), addressing critical issues that impact the effectiveness of Large Language Models (LLMs).

In recent years, RAG has garnered significant attention for its potential to reduce hallucination problems inherent in LLMs by integrating relevant external data sources. However, the success of RAG relies heavily on the quality of the curated data, which must undergo rigorous annotation, and preparation processes.

The first work addresses the complexities of data annotation by presenting a novel cross-domain task allocation scheme to optimize annotator training and selection in crowdsourced scenarios. The second work shifts focus to data preparation, introducing an innovative framework for column semantic type annotation that considers inter-table context, thereby enhancing the effectiveness of data preparation. Finally, the third work explores the implementation of knowledge-based RAG approaches in the context of LLMs. We investigate strategies to effectively manage knowledge overloading and the complexities of multi-hop queries, developing a retrieval content summarization method tailored for knowledge base question answering. Collectively, these contributions aim to enhance the reliability and effectiveness of RAG for more robust AI systems capable of delivering accurate and contextually relevant responses.

Through these contributions, this thesis not only elucidates the multifaceted nature of data curation and RAG but also offers practical solutions to the pressing challenges faced in the field. The findings underscore the importance of effective data management practices, ultimately paving the way for enhanced data accessibility and utility in an increasingly data-driven world.

We thoroughly evaluated the effectiveness of our proposed advancements against existing state-of-the-art approaches. Finally, we conclude the thesis by raising promising future research directions related to data curation and RAG research.

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by

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This is to certify that I have examined the above PhD thesis
and have found that it is complete and satisfactory in all respects,
and that any and all revisions required by
the thesis examination committee have been made.

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CHAPTER 1

INTRODUCTION

1.1 Background

Retrieval-Augmented Generation (RAG) has become a prominent area of exploration in recent years, attracting considerable interest from both academic researchers and industry practitioners. This surge in attention is primarily driven by the need to address the hallucination issues that frequently afflict Large Language Models (LLMs) [5, 6, 7, 8]. Hallucinations refer to instances where LLMs generate information that is either fabricated or inaccurate, which can significantly undermine the trustworthiness of the outputs. RAG offers a compelling solution by enhancing the capabilities of LLMs through the incorporation of relevant, context-specific data from external sources. This integration not only enriches the generated content but also provides a solid foundation of factual knowledge that helps align the model's outputs with real-world information. As a result, RAG represents a crucial step towards creating more reliable AI systems, capable of delivering responses that are not only coherent and contextually appropriate but also grounded in verifiable facts. This innovative framework holds promise for a wide range of applications, from customer service to content creation, thereby positioning itself as a transformative technology in the evolving landscape of natural language processing.

Data curation plays a vital role in the effectiveness of Retrieval-Augmented Generation (RAG), as it ensures that raw data is transformed into a usable format suitable for enhancing Large Language Models (LLMs). The process of data curation involves several critical steps, including data annotation, data preparation, and data organization. Data annotation is essential for labeling and categorizing information, allowing models to understand the context and significance of the data. Following this, data preparation involves cleaning and formatting the data, ensuring it is ready for integration into RAG systems. Finally, data

organization focuses on structuring the curated data in a way that facilitates efficient retrieval during the generation process. By meticulously curating data, we not only improve the quality of the information fed into RAG models but also enhance their ability to generate accurate, context-aware responses. Thus, effective data curation is foundational to the success of RAG, ultimately leading to more reliable and informative outputs.

However, there exists challenges in the process of data curation and RAG: Firstly, the process of data annotation is non-trivial, especially when we consider the necessity of domain knowledge and the inherent difficulties of the annotation tasks. Under the context of RAG, many data annotation tasks require a deep understanding of specific fields, such as biology, law, or engineering, making it essential for annotators to possess sufficient specialized expertise. This requirement can limit the availability of qualified annotators and lead to inconsistencies in the quality of the annotations. Moreover, the tasks involved in data annotation can be inherently complex and demanding. Annotators may face challenges such as ambiguous data, nuanced contextual meanings, or intricate relationships within the data. These factors can make it difficult for annotators to provide accurate annotations for RAG. Therefore, determining how to effectively train and select annotators with domain-specific expertise remains a challenging issue in the data annotation process [9].

Besides, data preparation, especially on tabular data also present significant challenges. One of the most significant issues lies in semantic differences. When standardizing data lakes from various sources, it is common to encounter columns that may seem similar but have distinct meanings or representations. For instance, a "Name" column in one data lake may represent the names of actors, while in another, it might indicate the names of products. Naively formatting the two "Name" columns into the same column type would lead to confusion in downstream applications such as training a machine learning model and RAG and thus there is a need for a deeper understanding of table context to clarify the subtle differences of column semantic meaning during the data preparation process. As a result, column semantic type annotation becomes a crucial task. By clearly defining the meaning and context of each column, we can ensure that the prepared data is accurate and meaningful for RAG, and thus guarantee the performance of RAG. This annotation process

helps to standardize the interpretation of data across different sources, facilitating a more seamless integration. However, how to guarantee the effectiveness of column semantic type annotation remains a challenge in the data preparation phase [10].

Last but not least, while Retrieval-Augmented Generation (RAG) offers significant advantages in enhancing the performance of Large Language Models (LLMs), it also faces several challenges, particularly in effectively utilizing curated data for efficient question answering. Even with well-annotated and prepared datasets, the process of leveraging this information to address user queries can be quite complex. As pointed out by CRAG [6], a notable challenge arises when dealing with intricate queries, such as range queries, set queries, and multi-hop queries. These types of queries require not only retrieving relevant information from external databases, but also ensuring that the extracted data is both accurate and concise. For instance, a range query might necessitate filtering data within specific bounds, while a multi-hop query demands synthesizing information from multiple sources to form a coherent response. The need for sophisticated retrieval mechanisms and effective data integration strategies becomes apparent, as the success of RAG hinges on its ability to navigate these challenges and deliver precise, contextually relevant answers.

This thesis presents my exploration of effective data curation and RAG in response to the three challenges mentioned above. My first work (Chapter 2) resolves the difficulty in designing a proper cross-domain annotator training and selection scheme in crowdsourced data annotation. Specifically, we proposed a novel cross-domain task allocation scheme to identify the most appropriate crowd workers upon facing new data annotation tasks. My second work (Chapter 3) introduces a new table column semantic type annotation framework that innovatively considers the inter-table context in the data lake to provide accurate column semantic type annotation performance. My third work (Chapter 4) presents my exploration on effective KB-based RAG approaches in the era of LLMs. Specifically, we explored how to adaptively handle the knowledge overloading and multi-hop challenges of KBQA with LLMs and developed an effective retrieval content summarization scheme for KB-based RAG.

1.2 Motivation

In this section, we would like to provide some exploration on the motivation to explain why we RAG on curated data instead of directly RAG on raw data. In general, the RAG process involves parsing, retrieval, and summarization. The user query is first parsed, and then we use the parsing content to retrieve relevant information and send for summarization to generate answers.

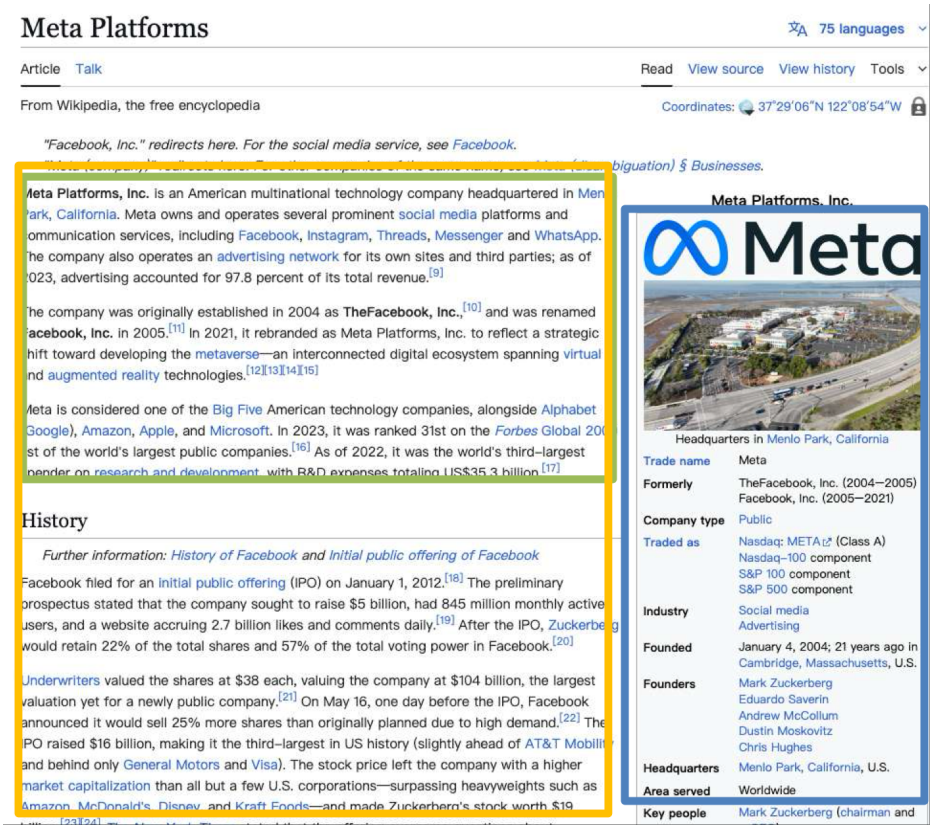


Figure 1.1: An exemplar Wiki page data.

We validate the effectiveness of different data sources on the Head2Tail dataset [11] by conducting a brief case study. Specifically, we would like to explore the effectiveness of the summarization module on different data sources to illustrate the importance of data curation in RAG. We considered two different data sources: Wikipedia pages and KB. As shown in the Figure 4.1, we present the wikipedia page of Meta as an example. The raw web page contains summary of the entity (colored green), paragraph of the entity (colored in yellow), and infobox of the entity (colored in blue). For the Wiki page data source method, we first use the LLM to detect the topic entity and then link to wikipedia pages and obtain the page

summary, paragraph, and infobox (marked green, yellow, and blue in our example), and then send the retrieved content to an LLM-based summarizer to get the answer. For the KB-based method, we parse the topic entity and get the one-hop triples from the KB and then send the obtained triples to summarizer to get the answer.

We present the experimental results in Table 1.1. Wikipedia pages improve recall compared to the KB data sources. However, the summarization accuracy drops significantly. Although the curated KB data source has a lower recall, it leads to overall highest accuracy. We attribute this to the fact that although raw web data provide better coverage in terms of answers, they are messier, and the signal-to-noise ratio is significantly lower than the curated KBs, which poses challenges to the answer generation process of the summarizer. As a result, the summarizer performs worse on raw web data despite the fact that the data source itself has a better recall. This case study validates the importance of data curation in effective RAG.

Table 1.1: Dataset statistics

Data Sources	**Recall**	**Accuracy**
KB	0.952	0.908
Wikipedia Infobox	0.980	0.872
Wikipedia Page	0.990	0.845

CHAPTER 2

CROSS-DOMAIN-AWARE WORKER SELECTION WITH TRAINING FOR CROWDSOURCED ANNOTATION

Chapter 2 introduces Cross-domain-aware Worker Selection with Training for Crowdsourced Annotation, a novel cross-domain-aware worker selection with training framework that properly allocates the training tasks and selects and trains the top-performing data annotators iteratively. Existing methods propose selecting workers based on their performance on tasks with ground truth, while two important points are missing. 1) The historical performances of workers in other tasks. In real-world scenarios, workers need to solve a new task whose correlation with previous tasks is not well-known before the training, which is called cross-domain. 2) The dynamic worker performance as workers will learn from the ground truth. In this work, we consider both factors in designing an allocation scheme named cross-domain-aware worker selection with training approach. Our approach proposes two estimation modules to both statistically analyze the cross-domain correlation and simulate the learning gain of workers dynamically. A framework with a theoretical analysis of the worker elimination process is given. To validate the effectiveness of our methods, we collect two novel real-world datasets and generate synthetic datasets. The experiment results show that our method outperforms the baselines on both real-world and synthetic datasets.

2.1 Introduction

The quality of the labeled data is of great importance for the performance of machine learning, especially for supervised learning models [12]. To get high-quality annotations for large-scale datasets, recruiting domain experts is too expensive and thus unacceptable. With a limited budget, annotation through selecting crowdsourcing workers is preferable and has

drawn attention in recent years [13, 14]. Worker selection is one of the most important issues in the quality control consideration of crowdsourcing [15], which focuses on identifying workers with high performance from the worker pool. How to design an allocation scheme to effectively and efficiently select high-performance crowd workers remains a challenging problem.

In order to select workers through worker quality estimation, existing methods [16, 15, 17, 18] consider different factors in the crowdsourcing process: 1) workers' responses to the golden questions [16]; 2) additional social network interactions for worker trustworthiness estimation [15, 17]; 3) assumption of worker skills, which are hidden states between worker performance and tasks [18].

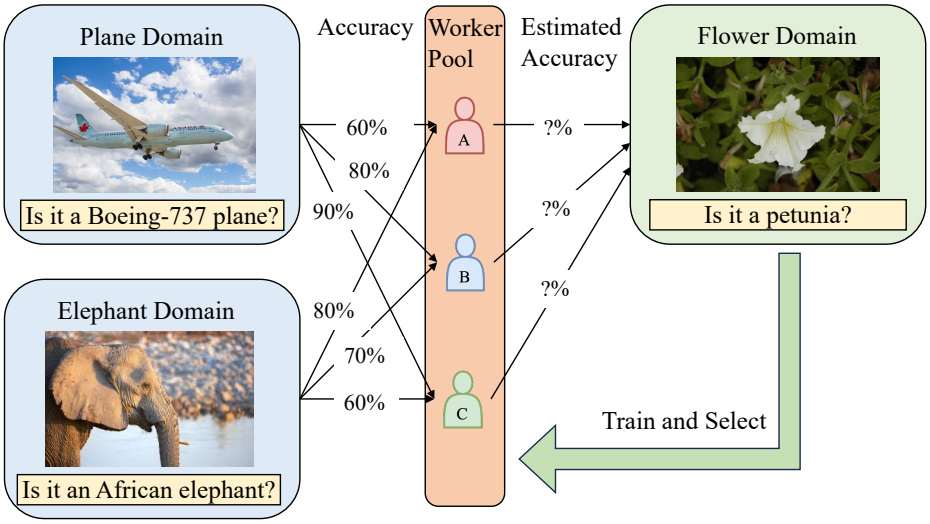


Figure 2.1: Cross-domain worker selection. The left shows the two prior domains: plane and elephant. The right shows the target domain: flower. We record workers' historical accuracy on the two prior domains and estimate the accuracy on the target domain, to effectively train and select desired workers.

In order to obtain large-scale manually labeled data for business, many companies such as JD.com, Inc. [19], Alibaba [20], and Baidu [21] have their own commercial crowdsourcing platforms with worker pools. The answering history of workers stored in commercial crowdsourcing platforms can be helpful in selecting high-quality workers to complete tasks in a new domain [22], which is not well-explored in the existing worker selection methods [16, 15, 17, 18]. We refer to the tasks of new topics requiring workers to annotate as *target domain tasks*, while the tasks of historical topics are *prior domain tasks*. In the beginning, the correlations between the target domain and these prior domains are not well-

known, which is called *cross-domain*. The performance of workers in the prior domain can help predict their performance in the target domain. As shown in Figure 2.1, given the classification performance on the elephants and planes of workers A, B, and C, we can obtain a rough idea of their domain knowledge of distinguishing living creatures and distinguishing machines, which is helpful for us to select the proper workers to work on tasks on other domains, such as flowers. Intuitively, workers with good performance in distinguishing elephants are likely to be sensitive to color and shape differences (since different kinds of elephants are similar in size but different in color and shape). In contrast, workers who perform well in distinguishing planes are likely to be good at identifying size differences (since different kinds of planes are similar in color and shape yet different in size). Given this prior domain knowledge, workers who are potentially good at distinguishing flowers (relying on color and shape differences) can be identified. We can train these identified workers by demonstrating the ground truth answers to them so that they can actively learn the characteristics of petunia and achieve better performance on the annotations. After that, we can select the best workers as the desired worker candidates for the target domain. In this way, the *golden questions* from the target domain are fully utilized: not only used for estimating the cross-domain worker quality to select the best candidates but also used to boost the annotation performance of workers on the target domain through worker training.

However, transferring and incorporating workers' performance profiles across different domains is challenging. Explicitly defining the mappings between the domains and the skill sets requires a comprehensive understanding of the domain tasks, which needs expert effort and thus fails to scale well in reality. Therefore, we propose automatically and inherently capturing the relationship between each domain and the required skills to ensure feasibility and scalability in real-world applications. Workers' performances are modeled based on reasonable assumptions for inner- and inter-domain. To be more specific, we apply normal distributions to model workers' performances on each domain following the modeling done by previous studies [23, 24] and adopt a multivariate normal distribution to model the correlation between workers' performances on different domains to achieve the goal of cross-domain worker selection.

As correlation is not well-known for the cross-domain problem, we apply a worker learning stage to train and select workers while simultaneously extracting their correlations. During the learning stage, limited *golden questions* are given with accurate labels from experts. Previous approaches [25] treat the assignment of golden questions as a sampling process to get a static estimate of worker quality. However, workers' knowledge of the target domain can be dynamic [26, 27]. For instance, in Figure 2.1, workers are asked whether the flower is a petunia. Initially, workers may have no idea what a petunia is. However, after we assign multiple golden (learning) questions regarding the petunia and reveal the answers to the workers, they can gradually learn about the characteristics (such as shape and color) of petunia and thus perform better when answering new questions on the same domain.

Unfortunately, previous work has not studied the dynamic worker knowledge change in worker selection. We fill the research gap by simulating workers as trainable to handle the dynamic worker selection instead of a static one. In this chapter, we propose to model the learning gain of workers based on the Item Response Theory (IRT) [28] from the Knowledge Tracing field to fully use the golden questions in selecting workers. Our allocation algorithm can select potential workers who can improve quickly during the learning stage, which are filtered out by static methods.

The contributions of this chapter are as follows:

- We incorporate the cross-domain knowledge information and propose a novel Median Elimination-based worker selection with training algorithm to find high-quality workers.
- We comprehensively consider the learning gain of workers during the learning task worker training process over the new domain to get a better estimate of the dynamic change in worker quality.
- We collect two novel cross-domain worker selection datasets for the crowdsourcing research community to study the problem of cross-domain worker selection with training.

- We conduct extensive experiments on real-world and synthesized datasets to evaluate the performance of our proposed method comprehensively.

The following sections are arranged as follows. We first introduce the related work in Section 2.2. We then discuss the setup and problem formulation in Section 2.3. The methodology is introduced in Section 2.4. We demonstrate the experiment results in Section 2.5, and finally, we conclude the paper in Section 2.7.

2.2 Related Work

The general process of worker selection for quality control requires first estimating worker quality and then designing proper worker elimination algorithms to select the best workers. In this section, we first discuss the related works on worker selection from two aspects: worker quality estimation and worker elimination, which are the core components for worker selection with training task. Then we introduce the related works in Knowledge Tracing, which are related to the learning gain estimation process used for estimating worker quality and performing worker elimination.

2.2.1 Worker Quality Estimation

Worker Quality Estimation assesses the abilities and reliability of individual workers participating in tasks on a crowdsourcing platform. Previous studies proposed various worker quality estimation methods based on the information available in different crowdsourcing worker selection scenarios. Liu et al. [25] estimate worker quality based on workers' answers to golden questions. Li et al. [29] proposed a user discovery framework to select reliable workers based on general characteristics such as educational level, gender, and age. However, these characteristics may be too general for us to identify the best workers accurately. For instance, the approach would fail if the recruited workers were all college students with similar backgrounds. Zhao et al. [15], and Wu et al. [17] proposed integrating social network information into the worker quality estimation process. A major limitation of these approaches is that social network information is not necessarily avail-

able in many online crowdsourcing platforms, which limits the application domain of these approaches. Yadav et al. [18] proposed constructing a universal skill set with the mapping relationship between the skills and the crowdsourcing tasks. The skills of the workers are estimated based on the historical annotation performance. However, the universal skill set and the mapping relationship should be manually defined. Explicitly constructing the universal skill set and the mapping relationship between the skills and the tasks require domain knowledge, which brings overheads to real-world worker selection applications. Different from existing worker quality estimation approaches, we considered both cross-domain knowledge and the dynamic worker knowledge change to better capture the worker quality during the crowdsourcing process.

2.2.2 Worker Elimination

This section introduces the worker elimination algorithms used by existing works. Worker elimination is a process used to select the most qualified workers for a task while filtering out underperforming or unreliable workers. Even-Dar et al. [30] proposed a naive Uniform Sampling algorithm and a Median Elimination algorithm for the top-k selection multi-armed bandits problem, which can be adapted to perform worker elimination. Liu et al. [25] identify the best group of workers by assigning golden questions and selecting the workers that perform the best on those questions. Li et al. [29] selected the best workers based on the estimated performance generated from the general workers' profiles. Cao et al. [31] refined the theoretical bounds of [30]. They introduced budget constraints and proposed a greedy-based heuristic algorithm to sort the workers based on the error rate and the requirements. Zhao et al. [15] proposed forward and backward selection algorithms based on social network connections to gradually identify the best workers. Wu et al. [17] considered the interest similarity between the workers and the tasks based on social network information to identify the best workers for the tasks. Yadav et al. [18] proposed a team formation algorithm to gather the workers with the desired expertise for the target tasks. Building on the Medium Elimination algorithm introduced by [31], we additionally considered the worker learning gain during the worker elimination process, so as to achieve better

worker elimination results.

2.2.3 Knowledge Tracing

Knowledge tracing is a technique used in education to understand how well a student is learning a particular subject. It involves tracking and predicting a student's knowledge and understanding over time. By analyzing the student's responses to questions or tasks, knowledge tracing models can estimate the student's current level of knowledge, identify areas of strength and weakness, and provide personalized feedback and guidance to enhance learning [32]. As discussed in [33, 34], the knowledge tracing methods can be divided into three categories: Bayesian Knowledge Tracing, Factor Analysis Models, and Deep Knowledge Tracing based on the different types of inputs and application scenarios.

Bayesian Knowledge Tracing (BKT): The BKT model is first introduced by [32], where the skills behind the questions are considered. Four different types of probabilities associated with changes in skill mastery are modeled, and the Bayesian estimation of the final state skill mastery probability is used. Several subsequent works [35, 36, 37, 38, 39] propose to extend the original BKT model with student-specific modeling and inter-skill relationship.

Factor Analysis Models: The simplest and the most widely used Item Response Theory (IRT) model is Rasch's model [28], which defines a one-parameter logistic (1PL) IRT model. The probability that a worker answers a question correctly is modeled as a logistic function based on the worker's learning parameters and the difficulty of the question. Wilson et al. [40] extended the original 1PL IRT model to Hierarchical IRT and Temporal IRT by additionally considering the relatedness of parameters across different questions and times, respectively. Performance Factor Analysis (PFA) [41] is proposed to extend the IRT model by replacing the learning parameter with multiple learning skill parameters to model the relationship of multiple skills.

Deep Knowledge Tracing (DKT): DKT is first proposed in [42], which models the knowledge states (skills) of people with Long Short Term Memory (LSTM) [43]. The LSTM network contains many neurons to represent the hidden states of workers' answer history. The current knowledge states of workers can be learned from the training data. Several

Table 2.1: Notations.

Notations	Descriptions
T_l, T_w	learning tasks and working tasks sets
W	the worker pool
w_i	the i-th worker in the worker pool W
h_i	the historical profile of the worker w_i
n_i	number of annotation tasks completed by worker w_i on different domains
B	the total budget
k	number of workers we want to select
α_i	the learning parameter of worker w_i
β_d	the domain difficulty parameter for domain d
θ_i	the proficiency parameter of worker w_i
K_j	the cumulative number of learning tasks assigned to each remaining worker till round j
n	the number of elimination rounds
a_t	the initialized annotation accuracy of the target domain
Q	the number of learning tasks per batch

works [44, 45, 46] extend the idea of the original DKT model to achieve improved performance.

In our paper, we aim to model the worker learning gain without explicitly defining and modeling the relationship between the skills and the questions, so we adopt Rasch's IRT model [28] to model the learning process of workers while answering learning questions. Note that the focus of this chapter is to introduce the knowledge tracing techniques into the crowdsourcing worker selection process to achieve better worker quality estimation and elimination results, instead of developing novel knowledge tracing approaches.

2.3 Problem Formulation

We present the notations used in this chapter in Table 2.1. The general process of cross-domain-aware worker selection with training can be divided into three steps, as shown in Figure 2.2. We first discuss the tasks and workers considered in our paper, introduce the three steps generally, and formally define the cross-domain-aware worker selection with training problem.

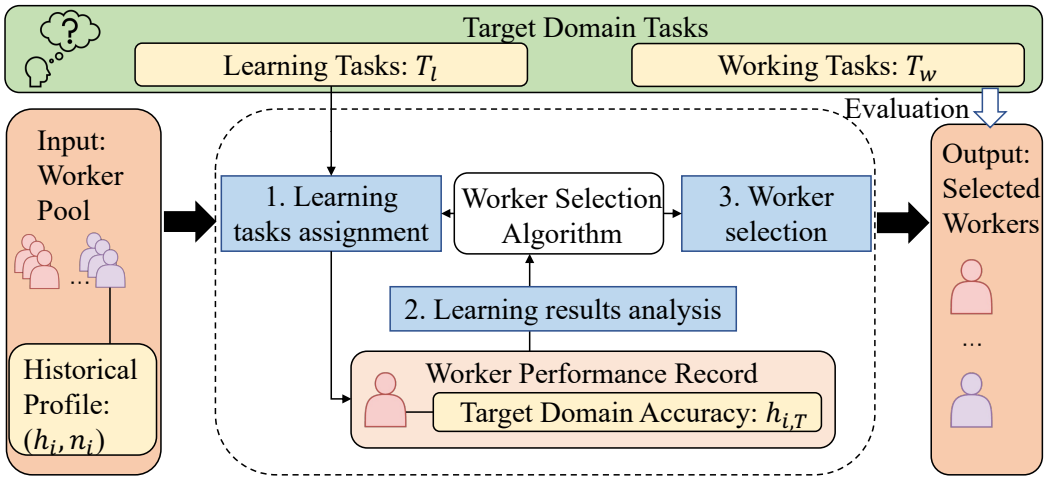


Figure 2.2: The definition of cross-domain-aware worker selection with training problem. The worker selection algorithm assigns learning tasks to workers, records and analyzes the learning task results, and performs worker selection. The performance of the selected workers is evaluated based on the target domain working tasks.

Definition 2.3.1 (Tasks). The crowdsourcing tasks on the target domain can be categorized into learning and working tasks. The learning tasks (golden questions) refer to the tasks that have gold labels. The working tasks are the tasks without gold labels. We denote the set of tasks on the target domain as T, the set of learning tasks as T_l, and the set of working tasks as T_w.

For simplicity, we consider Multiple Choice Question tasks in our paper. As suggested by [22], this selection of task type does not influence the generalizability of our approach since our approach is based on the answering accuracy, which can also be computed if other kinds of tasks are used.

Definition 2.3.2 (Workers). We denote the worker pool as W. Each worker w_i in W is associated with a historical profile $(h_i, n_i) = (\{h_{i,1}, h_{i,2}, ..., h_{i,D}\}, \{n_{i,1}, n_{i,2}, ..., n_{i,D}\})$ where D is the number of prior domains, $h_{i,j}$ is the annotation accuracy of worker w_i on the j-th prior domain, and $n_{i,j}$ is the number of annotation tasks completed by worker w_i on the j-th prior domain. The annotation accuracy of w_i on the target domain working tasks is denoted as $h_{i,T}$.

Definition 2.3.3 (Learning tasks assignment). Learning tasks assignment is the process of assigning learning tasks to the worker and recording the accuracy of each worker. The

answers to the learning tasks are revealed to each worker after he/she submits the answers so that he/she can learn the target domain knowledge from the revealed ground truths.

In our paper, we train workers in rounds. Each remaining worker w_i is assigned learning tasks and the annotation accuracy $a_{i,c}$ is recorded in round c.

Definition 2.3.4 (Learning results analysis). Learning results analysis is the process of estimating workers' performances based on their results on the learning tasks for the assignment of remaining learning tasks and the selection of workers.

Definition 2.3.5 (Worker selection). Considering the cross-domain performances and learning result feedback of workers, the worker selection step is to select the well-trained workers with the best performance in the target domain.

Definition 2.3.6 (Cross-domain-aware worker selection with training). Given target domain tasks $T = \{T_l, T_w\}$, the total budget B, and worker pool W with each worker w_i's historical profile h_i. Cross-domain-aware worker selection with training problem is to 1) properly assign no more than B tasks to $|W|$ workers for training based on workers' historical records and learning feedback and 2) select top k workers with the highest possible annotation accuracy on working tasks T_w.

2.4 Methodology

In this section, we first introduce our general framework for cross-domain-aware worker selection with training problem (Subsection 2.4.1). Then we demonstrate the process of worker training (Subsection 2.4.2) and discuss the details of the two core phases: Worker Quality Estimation (Subsection 2.4.3) and Worker Selection (Subsection 2.4.6) with theoretical analysis. A summary of the whole pipeline is presented in Subsection 2.4.7.

2.4.1 Framework

We display our framework in Figure 2.3. Workers are iteratively trained and selected through a 3-phase pipeline:

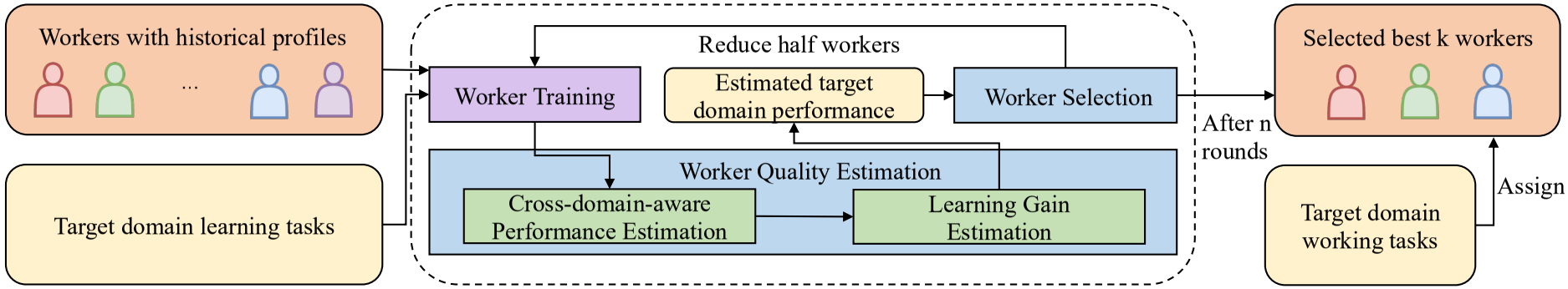


Figure 2.3: The general pipeline of our cross-domain-aware worker selection with training algorithm.

- Worker Training. In the target domain, workers answer questions and check answers to renew their knowledge.
- Worker Quality Estimation. The ability estimation of each worker will be updated according to his/her answers during worker training in addition to the historical records on target and other domains.
- Worker Selection. Based on the estimated worker quality, we select the best half of the workers to enter the next round.

Finally, after n rounds, we obtain the selected best k workers and assign the target domain working tasks for them to annotate. In our paper, we fix the budgets in each round and focus on the accuracy of dynamic worker estimation. Mathematically, $t = \lfloor \frac{B}{n} \rfloor$ is the fixed number of learning tasks per round, and $|W_c|$ is the number of remaining workers for the current round c. Then, $\lfloor \frac{t}{|W_c|} \rfloor$ is the number of tasks per worker for round c.

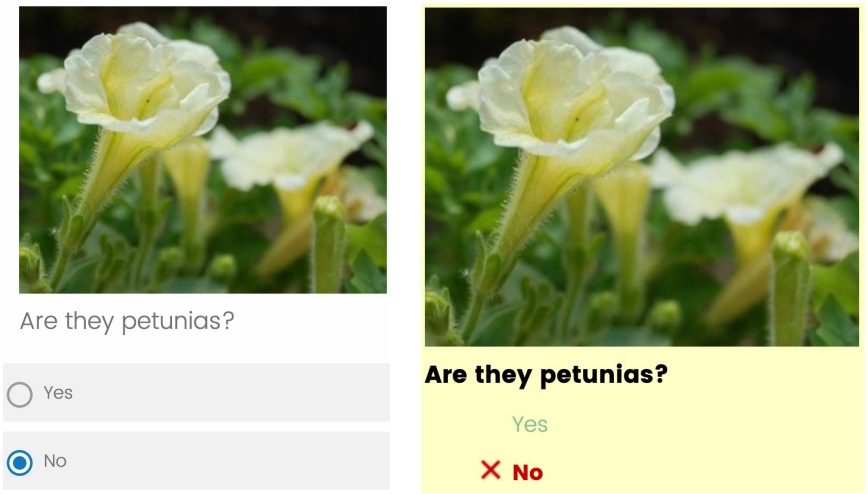


Figure 2.4: An illustration of the learning task (left) and its corresponding ground truth answer (right). The learning tasks will be displayed to the workers. After they complete their current round answers, the ground truth will be revealed for them to learn.

2.4.2 Worker Training

We can summarize the worker training as a simple "Answer and Learn" process for workers. To be more specific, after a worker completes one batch of learning tasks, their ground truth answers will be revealed to the worker. For example, as shown in Figure 2.4, the left shows

a learning task completed by a worker, and the right shows the ground truth answer to that task. A worker can learn from the ground truth answers and renew his/her target domain knowledge.

Formally, after each round of task assignment, we denote the answers of worker w_i in the current round c as $a_{i,c}$.

2.4.3 Worker Quality Estimation

To achieve high-quality worker selection, two factors are important: *cross-domain correlation* which can help us filter workers according to their performance on other domains; and *worker learning gain* where workers who improve more from training should be preserved and assigned with more training tasks. Thus, we divide the worker quality estimation phase into Cross-domain-aware Performance Estimation (CPE) and Learning Gain Estimation (LGE). CPE focuses on modeling the cross-domain correlation of workers, while LGE focuses on modeling the learning gain in the worker training process.

2.4.4 Cross-domain-aware Performance Estimation

As stated in Section 2.1, cross-domain information is important for worker selection. However, no correlation information between the prior domains and the target domain is available before worker training. Instead, worker feedback on learning tasks is accumulatively arriving, which requires a mining algorithm to dynamically capture the cross-domain correlation from scratch. In this subsection, we introduce our CPE estimation scheme, which 1) derives statistically analyzed expectation of accuracy and 2) supports online updates with a Maximum Likelihood Estimation as the base. We present the whole CPE estimation in Algorithm 1.

In order to model the correlation between workers' prior domain knowledge and the target domain knowledge, we adopt the multivariate normal distribution [47]. Precisely, to model the relationship between the D prior domains and the target domain effectively, we adopt a $(D+1)$-dimensional multivariate normal distribution $\mathcal{N}(\mu, \Sigma)$, where $\mu \in \mathbb{R}^{(D+1)}$ and $\Sigma \in \mathbb{R}^{(D+1)\times(D+1)}$:

Algorithm 1 Cross-domain-aware Performance Estimation (CPE)

Input:

The answers of workers in current round A_c

The learning tasks ground truth in current round G_c

The historical accuracy of workers in current round H_c

The number of workers remaining in current round $|W_c|$

Output:

The predicted accuracy of remained workers p_c

1: Initialize p_c to be an empty array

2: Initialize the multivariate normal distribution $\mathcal{N}(\mu, \Sigma)$

3: Compute the number of correct and wrong tasks of each worker $C_{i,c}, X_{i,c}$ according to Equations (2.3) and (2.4)

4: Compute updated distribution $\hat{\mathcal{N}}(\hat{\mu}, \hat{\Sigma})$, where

$$\hat{\mu}, \hat{\Sigma} = \arg\max_{\mu,\Sigma} \log L(\mathcal{N}, \{C_{i,c}, X_{i,c}\}_{i=1}^{|W_c|})$$

5: **for** each h_i in H_c **do**

6: Compute $p_{c,i}$ via Equation (2.8) and append to p_c

7: **end for**

8: **return** p_c

$$\mu = [\mu_1, \mu_2, ..., \mu_D, \mu_T]^{\mathrm{T}}, \tag{2.1}$$

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \rho_{1,2}\sigma_1\sigma_2 & ... & \rho_{1,D}\sigma_1\sigma_D & \rho_{1,T}\sigma_1\sigma_T \\ \rho_{2,1}\sigma_2\sigma_1 & \sigma_2^2 & ... & \rho_{2,D}\sigma_2\sigma_D & \rho_{2,T}\sigma_2\sigma_T \\ ... & ... & ... & ... & ... \\ \rho_{D,1}\sigma_D\sigma_1 & \rho_{D,2}\sigma_D\sigma_2 & ... & \sigma_D^2 & \rho_{D,T}\sigma_D\sigma_T \\ \rho_{T,1}\sigma_T\sigma_1 & \rho_{T,2}\sigma_T\sigma_2 & ... & \rho_{T,D}\sigma_T\sigma_D & \sigma_T^2 \end{bmatrix}, \tag{2.2}$$

the $\mu_i, \sigma_i, \rho_{i,j}$ where $i \neq j$ and $i, j \in \{1, 2, ..., D, T\}$ are the mean and standard deviation of worker accuracy on each domain and the correlation parameters between any pair of domains respectively.

The annotation accuracy for each worker w_i on each domain is modeled as a $(D+1)$-dimensional random vector $v_i = [h_{i,1}, h_{i,2}, ..., h_{i,D}, h_{i,T}]^{\mathrm{T}} \in \mathbb{R}^{(D+1)}$, where $v_i \sim \mathcal{N}(\mu, \Sigma)$.

As shown in Figure 2.3, we perform CPE (Algorithm 1) in each elimination round. Specifically, each worker is assigned $(\lfloor \frac{t}{|W_c|} \rfloor)$ learning tasks and we record the answers for each worker w_i: $a_{i,c} = [a_{i,c,1}, a_{i,c,2}, ..., a_{i,c,\lfloor \frac{t}{|W_c|} \rfloor}]$ and store to A_c. For each worker, we

compute the number of correct and wrong answers as follows:

$$C_{i,c} = \sum_{j=1}^{\lfloor t/|W_c| \rfloor} \mathbb{1}(a_{i,c,j} = g_{j,c}), \tag{2.3}$$

$$X_{i,c} = \lfloor (t/|W_c|) \rfloor - C_{i,c}. \tag{2.4}$$

Given each worker's correct and wrong answers in each round, we adopt Maximum Likelihood Estimation to estimate μ and Σ. The log-likelihood function L is formulated as follows:

$$\begin{aligned}
\log L &= \sum_{i=1}^{|W_c|} \log P(h_{i,T}|h_i) \\
&= \sum_{i=1}^{|W_c|} \log \int_0^1 h_{i,T}^{C_{i,c}} (1 - h_{i,T})^{X_{i,c}} \frac{e^{-\Psi}}{\sqrt{2\pi|\bar{\Sigma}|}} \mathrm{d}h_{i,T} \\
&= \sum_{i=1}^{|W_c|} \big[\log \int_0^1 h_{i,T}^{C_{i,c}} (1 - h_{i,T})^{X_{i,c}} e^{-\Psi} \mathrm{d}h_{i,T} \\
&\quad + \log \frac{1}{\sqrt{2\pi}} - \frac{1}{2} \log |\bar{\Sigma}| \big],
\end{aligned} \tag{2.5}$$

where $\bar{\mu}$ and $\bar{\Sigma}$ are the conditional distribution of the multivariate normal distribution $\mathcal{N}(\mu, \Sigma)$:

$$\bar{\mu} = \mu_T + \Sigma_{1\times D} \Sigma_{D\times D}^{-1} (h_i - \mu_{1\sim D}),$$

$$\bar{\Sigma} = \Sigma_{1\times 1} - \Sigma_{1\times D} \Sigma_{D\times D}^{-1} \Sigma_{D\times 1},$$

and $\Psi = \frac{(h_{i,T} - \bar{\mu})^{\mathrm{T}} (h_{i,T} - \bar{\mu})}{2\bar{\Sigma}}$.

In real-world applications, these parameters are updated under a large amount of streaming data, where directly calculating the optimal parameters is unacceptable. To enable an incremental parameter estimation, we update μ and Σ by maximizing Equation (2.5) with gradient descent in each round:

$$\mu' = \mu - r_1 \nabla_\mu \log L, \tag{2.6}$$

$$\Sigma' = \Sigma - r_2 \nabla_\Sigma \log L, \tag{2.7}$$

where r_1 and r_2 are the learning rates of gradient descent for μ and Σ; μ' and Σ' are the updated μ and Σ at the current gradient descent step. After obtaining the Maximum Likelihood Estimation of the mean $\hat{\mu}$ and standard deviation $\hat{\Sigma}$, we obtain the predicted annotation

accuracy for each worker w_i with the updated multivariate normal distribution $\hat{\mathcal{N}}(\hat{\mu}, \hat{\Sigma})$:

$$\begin{aligned} p_{c,i} &= E[h_{i,T}|h_i] \\ &= \int_0^1 h_{i,T} P(h_{i,T}|h_i) \mathrm{d}h_{i,T} \\ &= \int_0^1 h_{i,T} \frac{P(h_i, h_{i,T})}{P(h_i)} \mathrm{d}h_{i,T}, \end{aligned} \tag{2.8}$$

where $[h_i, h_{i,T}]^{\mathrm{T}} \sim \hat{\mathcal{N}}(\hat{\mu}, \hat{\Sigma})$ and $[h_i]^{\mathrm{T}} \sim \hat{\mathcal{N}}(\hat{\mu}_{1\sim D}, \hat{\Sigma}_{D\times D})$. $p_{c,i}$ is used as the estimated worker accuracy in the target domain instead of a coarse observation on $C_{i,c}$ and $X_{i,c}$.

2.4.5 Learning Gain Estimation

Through the interactive mode displayed in Figure 2.4, workers not only provide feedback but also *learn* from the results. This training process is important for crowdsourcing [26, 27], but is hardly studied together with worker selection. This motivates us to enhance worker estimation with training, which aims at capturing the changes in estimated target domain performance for workers. This further enables us to quantize each worker's learning gain after assigning a certain amount of learning tasks to get a more accurate dynamic estimation of workers' performance on T_w.

In order to capture the learning gains, we adopt the item response theory model used for capturing the student learning process from the previous work [40, 34]. In the original IRT model [40, 34], considering a worker w_i, the probability that w_i answers question q from domain d correctly is:

$$p_d(\theta_i) = (1 + e^{-(\theta_i - \beta_d)})^{-1}, \tag{2.9}$$

where θ_i is the proficiency parameter of the worker and β_d is the difficulty parameter of the question q from domain d. In our setting, $\theta_i = \alpha_i \ln(K_j + 1)$, which is proportional to the logarithm of the cumulative number of learning tasks ($K_j = \frac{(2^j-1)*t}{|W|}$ for the target domain) assigned to worker w_i, while α_i is computed through least square regression and will be discussed in Equation (2.11). Different difficulty parameters are assigned to tasks in different domains, denoted as $\beta_{1\sim D}$ for tasks in prior domains and β_T for tasks in the target

domain. The modified item response theory model for a single worker w_i at the learning stage j on the domain d is:

$$\begin{aligned} \hat{p}_{j,i,d} &= g(\alpha_i, \beta_d, K_j) \\ &= \frac{1}{1 + e^{-(\alpha_i \ln(K_j+1) - \beta_d)}}. \end{aligned} \tag{2.10}$$

The last step before one can further estimate the dynamic performance after training instead of a static performance is to get the update formula for the intrinsic learning parameter α_i of each worker in each round. We update the learning parameter α_i by minimizing the following least square regression objective:

$$\alpha_i = \arg\min_{\alpha_i} \left[\sum_{d=1}^{D} (\hat{p}_{1,i,d} - h_{i,d})^2 + \sum_{j=1}^{c} (\hat{p}_{j-1,i,t} - p_{j,i})^2 \right], \tag{2.11}$$

which comprises two parts: the first part minimizes gaps between learning gain estimations and accuracy on D prior domains; the second part minimizes each pair of worker accuracy estimated by Equation (2.10) in round $j-1$ and CPE in round j. The round index is different because the CPE estimation is based on the workers' performances in round j, where workers are only shown with $j-1$ rounds of answers (trained with $j-1$ rounds). The regression is conducted in each round to update each α_i.

According to the results of each round (e.g., the jth round), we assign tasks to workers and expect the best performance after the training of the next round (e.g., the $j+1^{th}$ round), which can be obtained through Equation (2.10) (e.g., compute $\hat{p}_{j+1,i,d}$), which is intractable for static methods. We argue that such an estimation is closer to the actual annotation performance on the working tasks after n rounds of training and thus can help us get a more accurate estimate of the actual value of $h_{i,T}$ for each worker w_i. We display the LGE in Algorithm 2.

2.4.6 Worker Selection

Based on the above estimations, we propose our algorithm for worker selection in this subsection, where a theoretical guarantee is given. Compared with the intuitive but effective

Algorithm 2 Learning Gain Estimation (LGE)

Input:

The workers remained in the current round W_c

The historical accuracy of workers in the current round H_c

The historical task numbers of workers in the current round N_c

The predicted accuracy arrays $p_1, p_2, ..., p_c$ at the current stage

Output:

The updated predicted accuracy with learning gains $\hat{p}_c$

Initialize $\hat{p}_c$ as an empty array
Initialize the difficulty parameters in the target domain (β_T) and prior domains ($\beta_1, \beta_2, \cdots, \beta_D$)
for each $w_i \in W_c$ **do**
    Initialize the learning parameter α_i
    for domain $d = 1, 2, ..., D$ **do**
        Compute the historical accuracy $h_{i,d}$
        Compute the historical task numbers $n_{i,d}$
        Compute the IRT score $\hat{p}_{1,i,d} = g(\alpha_i, \beta_d, n_{i,d})$
    end for
    for stage $j = 1, 2, ..., c$ **do**
        Compute the predicted accuracy for w_i: $p_{j,i}$
        Compute $\hat{p}_{j-1,i,T} = g(\alpha_i, \beta_T, K_{j-1})$
    end for
    Update α_i according to Equation (2.11)
    Compute $\hat{p}_{c,i,T} = g(\alpha_i, \beta_T, K_c)$
    Append $\hat{p}_{c,i,T}$ to $\hat{p}_c$
end for
return $\hat{p}_c$

Algorithm 3 Median Elimination (ME)

Input:

The predicted accuracy $\hat{p}_c$

The workers remained in the current round W_c

Output:

The selected workers W_{c+1}

$w_1, w_2, ..., w_{|W_c|}$ = the workers sorted in non-increasing order of their predicted accuracy $\hat{p}_c$
$W_{c+1} = \{w_1, w_2, ..., w_{\lceil \frac{|W_c|}{2} \rceil}\}$
return W_{c+1}

design of the Median Elimination algorithm discussed in [30], we have a fixed amount of budget to allocate tasks, where the original algorithm and theory cannot be directly applied. Here, we propose an adapted version, displayed in Algorithm 3. Specifically, ME is called in rounds, with the worst half of workers eliminated in each round. The algorithm terminates with k workers remaining as its output. With a limited budget, we reversely derive

Algorithm 4 General Algorithm

Input:

A set of workers $w_i \in W$, a set of learning tasks $t_j \in T_l$
Workers' historical accuracy $h_i = \{h_{i,1}, h_{i,2}, ..., h_{i,D}\}$
Workers' historical task number $n_i = \{n_{i,1}, n_{i,2}, ..., n_{i,D}\}$
Total budget B
Probability δ

Output:

The set of selected top k workers W_T

Set n, t as Equations (2.12), (2.13), $W_1 = W$, and $\delta_1 = \delta$
Initialize the current learning task index $r_1 = 1$
for $c = 1, 2, ..., n$ **do**
  Set A_c to be an empty set
  Set the ground truth labels of tasks t_{r_c} to $t_{r_c+(\lfloor \frac{t}{|W_c|} \rfloor)}$ as $G_c = [g_{1,c}, g_{2,c}, ..., g_{\lfloor \frac{t}{|W_c|} \rfloor, c}]$
  Set the historical accuracy of W_c as $H_c = \{h_1, h_2, ..., h_{|W_c|}\}$
  Set the historical task number of W_c as $N_c = \{n_1, n_2, ..., n_{|W_c|}\}$
  for each $w_i \in W_c$ **do**
    Assign learning tasks t_{r_c} to $t_{r_c+(\lfloor \frac{t}{|W_c|} \rfloor)}$ to w_i in batches, reveal the correct answers after w_i submits
    Get the answers $a_{i,c}$ of w_i store to A_c
  end for
  $r_{c+1} = r_c + (\lfloor \frac{t}{|W_c|} \rfloor)$
  The predicted accuracy $p_c = \text{CPE}(A_c, G_c, H_c, |W_c|)$
  The updated predicted accuracy with learning gains $\hat{p}_c = \text{LGE}(W_c, H_c, N_c, p_1, p_2, ..., p_c)$
  $W_{c+1} = \text{ME}(\hat{p}_c, W_c)$, $\delta_{c+1} = \frac{\delta_c}{2}$
end for
Set W_T to be the top k workers with highest $\hat{p}_n$ in W_{n+1}. If $|W_{n+1}| < k$, set W_T to be the top k workers with highest $\hat{p}_{n-1}$ in W_n.
return W_T

the number of rounds needed for elimination and allocate the budget. Specifically, given worker pool W, number of k, total budget B, we can get the total round n and the budget allocated in each round t as:

$$n = \lceil \log(|W|/k) \rceil, \tag{2.12}$$

$$t = \lfloor B/n \rfloor. \tag{2.13}$$

Unlike the original (ϵ, δ) bound formulation in [30], we constrain the total budget used for the task and prove a theoretical bound over error ϵ_c in each round. Specifically, given a fixed total budget of B, the algorithm aims at finding the top k workers. It satisfies that the best worker outputted by the algorithm in round $c + 1$ is an ϵ_c-optimal worker with respect to the best worker outputted in round c, with a probability of least $1 - \delta_c$. The error at each

round c is bounded by $O(\sqrt{(\frac{nk}{B})\ln(\frac{1}{\delta_c})})$.

Adapting from the proof of Lemma 11 in [30], we have the following theoretical results:

Theorem 1 *By applying our adapted ME algorithm, we have:*

$$P[\max_{w_j \in W_c} h_{j,T} \leq \max_{w_i \in W_{c+1}} h_{i,T} + \epsilon_c] \geq 1 - \delta_c, \tag{2.14}$$

where each worker is assigned $1/(\frac{2}{\epsilon_c^2})\ln(\frac{3}{\delta_c})$ tasks in round c.

Proof We present the proof in our technical report [48].

According to the above theorem, we can derive the following bound for our allocation scheme:

Theorem 2 *In each round c of Algorithm 4, the error ϵ_c is bounded by $O(\sqrt{(\frac{nk}{B})\ln(\frac{1}{\delta_c})})$.*

Proof We present the proof in our technical report [48].

2.4.7 Summary

The algorithm regarding the whole pipeline is summarized in Algorithm 4. Workers with historical profiles (prior domain performance) are first assigned target domain learning tasks for training purposes (Line 9 of Algorithm 4). The annotation accuracy is recorded, then we enter the worker quality estimation phase (Lines 13-14 of Algorithm 4): we perform Cross-domain-aware Performance Estimation to generate an estimation of the worker accuracy, and we further use Learning Gain Estimation to estimate the performance gains. Finally, we perform worker selection (Line 15 of Algorithm 4) by applying Median Elimination in each round to select the best half of workers. After n rounds of looping, we obtain the selected best k workers on the target domain tasks. As for the time complexity, as shown in Algorithm 4, we have n iterations. Let G be the number of gradient descent epochs performed to maximize Equation 2.5. In each iteration, we perform CPE, LGE, and ME, which take $O(G|W_c|)$, $O(|W_c|\log(|W_c|/k))$, and $O(|W_c|\log(|W_c|))$, respectively. Therefore, the overall time complexity for the worker quality estimation and selection process is $O(n|W|(G+\log(|W|)))$. We do not consider the time for workers to complete the learning tasks when analyzing the time complexity and we will discuss this in Section 2.5.8.

Table 2.2: Dataset statistics

Datasets	|W|	Q	k	total # of batches	B
RW-1	27	10	7	3	540
RW-2	35	10	9	3	700
S-1	40	20	5	7	2400
S-2	50	20	5	7	3000
S-3	80	20	5	15	6400
S-4	160	20	5	31	16000

Note that our solution is not restricted to the case where workers have been working on all the D domains. For each domain d, if worker w_i does not have historical record $h_{i,d}$, we can remove the corresponding d^{th} row and line in μ and $\sum$ of Equation (2.5) for worker w_i and remove the addition term $(\hat{p}_{1,i,d} - h_{i,d})^2$ in Equation (2.11), so that our approach can still work if any workers have not been working on all the prior domains. In this way, one can easily adapt our approach to suit the general cases.

2.5 Experiments

2.5.1 Datasets

Currently, no publicly available dataset records both the cross-domain worker historical profiles and the worker training process. Therefore, we have to construct new datasets that cover the two aspects of information to evaluate the performance of our method and the baselines. To this end, we build real-world and synthetic datasets, summarized in Table 2.2. We denote the number of learning tasks per batch as Q. The total budget $B = \lceil \log(\frac{|W|}{k}) \rceil * Q * |W|$, # of batches $= 2^{\lceil \log \frac{|W|}{k} \rceil} - 1$. Note that Q and k are the independent variables, while # of batches and B are dependent variables. We generate different synthetic datasets with different $|W|$ to study the influence of the size of the worker pool.

Real-world datasets: We invited 27 and 35 workers to complete the Qualtrics survey [49] through volunteer recruitment and gMission [50] and denoted as RW-1 and RW-2 datasets. The tasks are Yes/No questions regarding image classification on three prior domains and one target domain. We chose Yes/No questions since many complex question types such as

Table 2.3: Details of real-world datasets

Dataset	Domain	Features	Knowledge	Sources
RW-1 prior-1	Elephant	Color, Shape	Animal	[53]
RW-1 prior-2	Clownfish	Color, Shape	Animal	[54, 55, 56]
RW-1 prior-3	Plane	Size	Machine	[57]
RW-1 target	Petunia	Color, Shape	Plant	[58]
RW-2 prior-1	Peruvian lily	Color	Plant	[58]
RW-2 prior-2	Red fox	Shape	Animal	[59]
RW-2 prior-3	English marigold	Shape	Plant	[58]
RW-2 target	Lenten rose	Shape	Plant	[58]

MCQs can be transformed from them [51, 52]. We present the detailed information of the two real-world datasets in Table 2.3. Specifically, RW-1 examines workers' prior domain knowledge of animals (elephant and clownfish) and machines (plane) and evaluates their performance on plants (petunia). The key features that workers need to distinguish petunias from other flowers are color and shape. We also include RW-2 as a complement to RW-1: RW-2 focuses on finer-grained domains where the Peruvian lily, English marigold, and Lenten rose are all flowers. The key features that workers need to focus on differ: Peruvian lilies can be distinguished based on their color, while English marigolds and Lenten roses require detailed observation of petal and stamen shapes. By conducting experiments on both RW-1 and RW-2, we can comprehensively evaluate the performance and robustness of our approach and obtain interesting insights into cross-domain worker training. On each prior domain, each worker is asked to complete two batches of tasks. Each batch consists of 5 learning tasks and 5 working tasks. The answers are recorded to form the historical profiles of the workers. In the target domain, each worker needs to answer 30 learning tasks and 30 working tasks for us to record the worker training process. The learning and working tasks are assigned to each worker in batches. In each batch, workers are required to complete 10 learning tasks first, check the ground truth answers of the learning tasks, and then complete 10 working tasks. The learning tasks are used to train the workers gradually, while the working tasks are applied to test workers' annotation performance in the target domain. A sample learning task is shown in Figure 2.4. Only the answers to the learning tasks are used as the algorithm input. The answers to the working tasks are used to evaluate the performance.

Synthetic datasets: We further constructed synthetic datasets based on the distribution of the RW-1. We considered the synthetic datasets with worker pool sizes of $40, 50, 80$, and 160 to simulate the different supply conditions. We set the number of learning tasks per batch on primal domains and the target domain to 10 and 20, respectively, on S-1, S-2, S-3, and S-4 datasets. We started by modeling the relationship among the four domains with a truncated multivariate normal distribution $N(\mu, \Sigma)$ within $(0, 1)$, where the mean and standard deviation of the three prior domains are computed based on the learning task result of the workers on the three corresponding domains, while the mean and standard deviation of the target domain is calculated based on the first batch learning task results in the RW-1 dataset. The correlation parameters shown in Equation (2.2) are uniformly random initialized within $(0, 1)$. Each synthetic worker was sampled from N as $[h_1, h_2, h_3, h_T]^{\mathrm{T}}$, where $h_T \in (0, 1)$ denotes the probability that the worker answers the target domain tasks correctly. We can thus obtain the annotation accuracy on the target domain learning tasks with the following *answering rule: randomly select a number x in $(0, 1)$ if $x < h_T$, then the worker answers correctly. Otherwise, the worker answers wrongly.* We obtained the annotation accuracy of synthetic workers on the first batch of learning tasks and applied the modified IRT model in Equation (2.10) to get the learning parameter α_i for each worker. Then we updated each worker's h_T after each batch based on the modified IRT model with each worker's α_i and the annotation accuracy generated with the *answering rule*. The top-k high-quality workers were selected based on the value of h_T in the last batch.

Table 2.4: Mean and standard deviation of RW-1 and synthetic datasets

Dataset	**Prior 1**	**Prior 2**	**Prior 3**	**Target**
RW-1	(0.70, 0.22)	(0.88, 0.10)	(0.58, 0.25)	(0.55, 0.17)
S-1	(0.72, 0.23)	(0.86, 0.13)	(0.53, 0.29)	(0.49, 0.18)
S-2	(0.64, 0.27)	(0.83, 0.15)	(0.51, 0.25)	(0.51, 0.20)
S-3	(0.66, 0.26)	(0.87, 0.13)	(0.54, 0.27)	(0.50, 0.18)
S-4	(0.68, 0.25)	(0.87, 0.13)	(0.54, 0.27)	(0.50, 0.18)

Consistency: Notice that the synthetic datasets are generated based on RW-1; we will now study the consistency between their distributions. As shown in Table 2.4, the computed mean and standard deviation of the multivariate normal distributions for RW-1 and synthetic datasets are close in the target and prior domains. Besides, we present the distribution of

workers' annotation accuracy in the target domain for RW-1, S-1, S-2, S-3, and S-4 in our technical report [48]. The real-world dataset RW-1 and the four synthetic datasets generated have similar distributions on the target domain. Specifically, we bucket the annotation accuracy, compute the Pearson correlations between RW-1 and each synthetic dataset, and find that all Pearson correlations ρ's are larger than 0.75, which validates the consistency.

2.5.2 Baselines

In our experiment, we compared our proposed method with the general worker selection algorithms Median Elimination (ME), Uniform Sampling (US) discussed by [30, 31] and Li et al.'s method [29]. We chose these baselines because they do not require additional social interaction information (required by [15, 17]) and are comparable in terms of tasks and goals (Liu et al. [25] focus on optimizing the number of golden questions used, while the worker selection algorithm used is US [30]; Yadav et al. [18] aim at forming high-performance worker teams, which is not comparable under our problem setting). We introduce the baselines as follows:

- Uniform Sampling (US) [30, 31]: Assign each worker the same amount of learning tasks and select the top-k workers that have the highest accuracy.
- Median Elimination (ME) [30, 31]: Assign each worker $\frac{t}{|W_c|}$ learning tasks in round c, after each round, apply Algorithm 3 to select the best half of workers.
- Li et al. [29]: Adopt linear regression on the multiple features of workers and then select workers based on the regressed values. In our experiment, we use the historical profiles as the features for the regression process.

US and ME focus on selecting high-quality workers based on their annotation performance in the worker training process, while the Li et al. approach focuses on employing the historical profile features to identify high-quality workers.

2.5.3 Experiment Setting

To ensure fairness, we allocated the same amount of budget for our method and the baselines. The methods' performances are evaluated with respect to the average annotation accuracy of the selected workers on the target domain working tasks in the last round. For example, in RW-1 and RW-2 datasets, the worker medium elimination process terminates in two rounds. We used the annotation accuracy on the working tasks in the second round as the performance criteria. The difficulty parameters of prior domains β_d were initialized as $\beta_d = \ln(\frac{1}{a_d} - 1)$, where a_d is the averaged annotation accuracy for all the real workers on the domain d. For the target domain, we set $\beta_T = 0$ so that Equation (2.10) would be 0.5 when $K_j = 0$. Since the tasks are Yes/No questions, we believe that $a_T = 0.5$ is a good choice when no prior knowledge regarding the target domain tasks is given. We further conducted a parameter sensitivity experiment in Section 2.5.6 to validate our choice. We adopted the same initialization setting regarding the difficulty parameters for the synthetic datasets. The initial multivariate normal distribution $N(\mu, \Sigma)$ was generated as follows: $\mu_1, \mu_2, ..., \mu_D$ and $\sigma_1, \sigma_2, ..., \sigma_D$ were initialized based on the mean and variance of workers' annotation accuracy on the corresponding prior domains; μ_T was initialized to 0.5, and σ_T was initialized as $\frac{1}{D}\sum_{d=1}^{D}\sigma_d$. The correlation parameters were uniformly random initialized in the $(0, 1)$ range. The gradients with respect to $\bar{\mu}$ and $\bar{\Sigma}$ in the log-likelihood function L (Equation (2.5)) was computed based on backpropagation [60]. We set the learning rates used for gradient descent as $r_1 = 1e-7$, $r_2 = 1e-4$, and the epochs for gradient descent as $G = 50$. The average accuracy of each method was recorded.

2.5.4 Experiment Results

Table 2.5: Experiment results

	RW-1	**RW-2**	**S-1**	**S-2**	**S-3**	**S-4**
US [31, 30]	0.764	0.956	0.765	0.775	0.815	0.865
ME [31, 30]	0.771	0.944	0.720	0.785	0.795	0.880
Li et al. [29]	0.771	0.936	0.780	0.805	0.845	0.870
ME-CPE	0.781	0.950	0.785	0.790	0.838	0.875
Ours	**0.798**	**0.961**	**0.830**	**0.828**	**0.850**	**0.886**
Ground Truth	0.914	1.000	0.885	0.875	0.915	0.975

We present the experimental results regarding the average annotation accuracy of the best-k workers in Table 2.5. The ground truth results are presented in the bottom line of Table 2.5. In general, we observe that our method performs the best baseline approach on real-world and synthetic datasets. Specifically, on RW-1 and RW-2 datasets, our method outperforms US [31, 30] by 4.5% and 0.5%, while boosting the performance of ME [31, 30] by 3.5% and 1.8%, and Li et al. [29] by 3.5% and 2.7%, respectively. On the four synthetic datasets, our method outperforms US, ME, and Li et al. by 5.5%, 7.1%, and 2.9% on average. We attribute the performance uplift of our method over US and ME to the fact that we additionally consider the cross-domain historical profile information of workers and apply proper simulation of the learning gain of workers during the worker training process. The reason why our approach outperforms Li et al. can be attributed to the appropriate worker elimination process applied and the proper simulation of the learning gain of workers. We further notice that the average relative performance uplifts of our approach over the three baselines are 10.1%, 5.1%, 3.9%, and 1.6%, which decrease as the number of workers increases. We attribute this phenomenon to the fact that the number of high-performance workers increases as the size of the worker pool gets large. As a result, the accuracy difference induced by different worker selection strategies of different approaches is likely to decrease, and thus the performance uplift becomes smaller as the size of the worker pool increases. Overall, our method performs persistently well on both real-world and synthetic datasets, which implies the effectiveness and robustness of our approach for cross-domain worker selection.

2.5.5 Ablation Study

We conducted an ablation study regarding the following variants to understand the mechanism of different components:

- ME [31, 30]: ME is the backbone of our method, where the CPE and LGE (Worker Quality Estimation) are removed.
- ME-CPE: ME-CPE is a variant of our method where the LGE component is removed.

We present the experiment results in Table 2.5. We compare ME [31, 30] with ME-CPE to demonstrate the effect of the Cross-domain-aware Performance Estimation. The comparison between ME-CPE and our method shows the influence of the Learning Gain Estimation. First, we note that ME-CPE outperforms ME by 1.3% and 0.6% and our method further boosts the performance of ME-CPE by 2.2% and 1.2% on RW-1 and RW-2 datasets. This shows the effectiveness brought by CPE and LGE in estimating worker quality. CPE helps the algorithm capture the cross-domain information to estimate the annotation ability of workers, while LGE captures the learning gain of workers during the worker training process. On S-1, S-2, S-3, and S-4 datasets, our method improves over ME-CPE by 5.7%, 4.8%, 1.4%, and 1.3% respectively. We attribute the performance improvement of our method over ME-CPE to the LGE component, which obtains a more accurate estimation of workers' performance in the target domain during training. ME-CPE outperforms ME by relatively 9.0%, 0.6%, and 5.4% on S-1, S-2, and S-3 datasets while performing slightly worse than ME on the S-4 dataset. On average, ME-CPE relatively improves the performance of ME by 3.6% on the four synthetic datasets. In general, ME-CPE can improve or achieve comparable performance as ME. The CPE component can effectively capture the cross-domain information, which is helpful for identifying high-quality workers.

2.5.6 Method Parameter Sensitivity

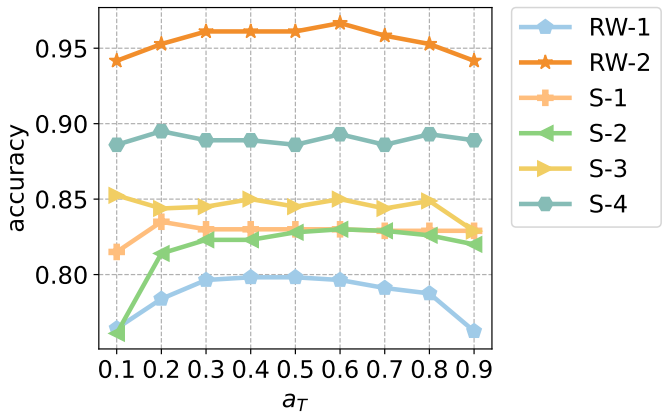


Figure 2.5: The sensitivity analysis regarding the initialized annotation accuracy of the target domain: $a_T = \frac{1}{1+e^{\beta_T}}$ on different datasets.

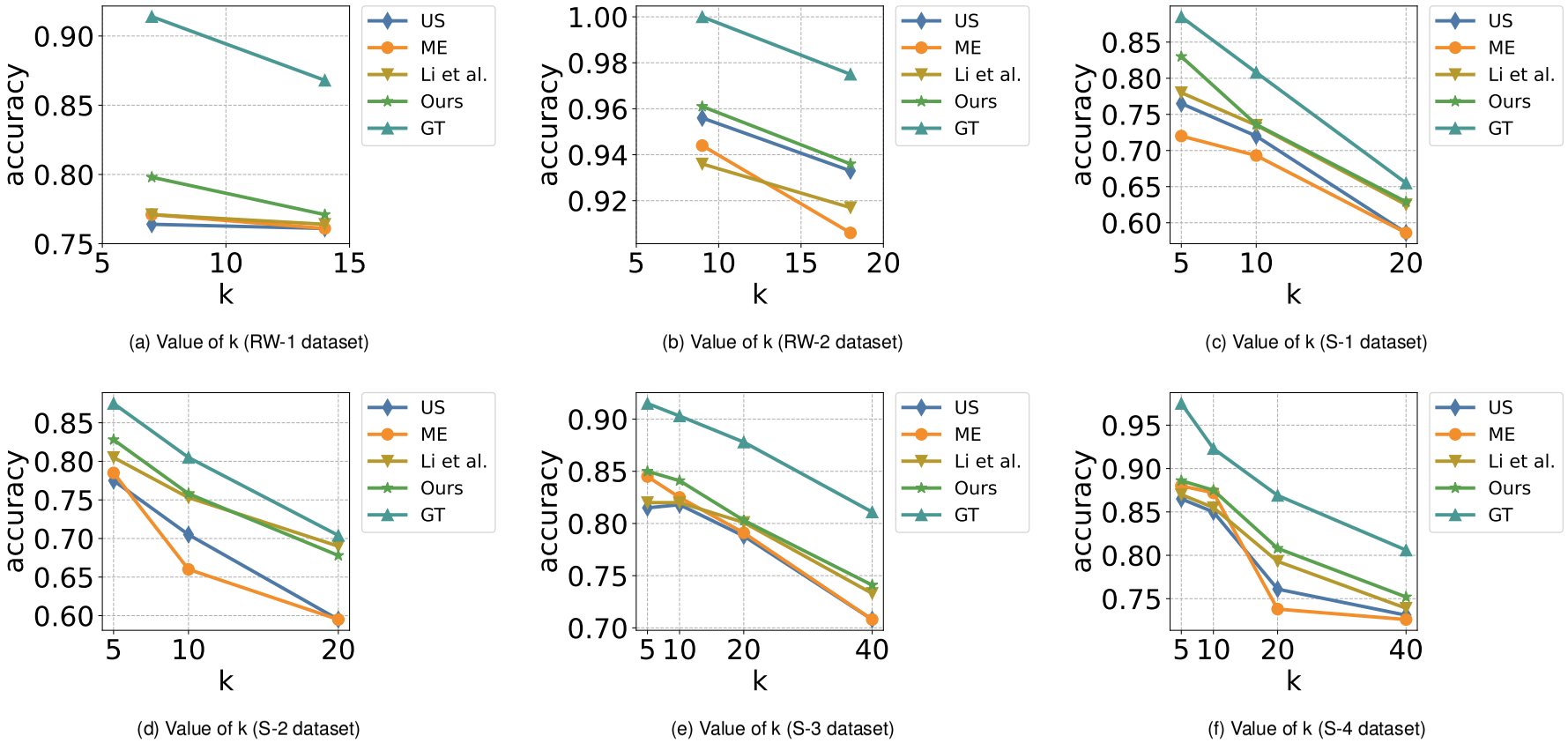
We analyzed the impact of the critical parameter of our method a_T, which is the ini-

tialized annotation accuracy of the target domain and is related to the initialization of the difficulty parameter β_T ($a_T = \frac{1}{1+e^{\beta_T}}$). Figure 2.5 presents the results. We notice that the performance of our method is relatively stable when the value of a_T is set within the range $[0.2, 0.8]$. In practice, we suggest setting a_t according to the difficulty level of the tasks. If the tasks are relatively easy, a large value of a_T can be adopted. Otherwise, a small value of a_T should be considered. If no domain knowledge regarding the difficulty level is available, we can set the value of a_T based on the nature of the tasks. Since our tasks are Yes/No quuestions, a natural choice for a_T would be 0.5. The sensitivity analysis coincides with our selection of a_T in Section 2.5.3: our method achieves a stable and good performance when $a_T = 0.5$.

2.5.7 Dataset Parameter Sensitivity

To comprehensively compare the performance of our method and baselines, we further conducted experiments regarding the important dataset parameters: the number of selected workers k and the number of learning tasks per batch Q. Since the number of learning tasks per batch cannot be changed once the RW datasets are collected, we conducted experiments on the four synthetic datasets to analyze its effect.

The value of k determines the number of rounds n required to obtain the top-k workers. As shown in Table 2.2, the values of k used in our main experiments are 7 and 9 for the RW-1 and RW-2 datasets and 5 for the four synthetic datasets, which leads to 2 rounds for RW-1 and RW-2 datasets, 3 rounds for the S-1 and S-2 datasets, 4 rounds for the S-3 dataset, and 5 rounds for the S-4 dataset. In the parameter sensitivity experiment presented in this section, we further increased the value of k to 14 and 18 for RW-1 and RW-2 datasets, 10 and 20 for the S-1 and S-2 datasets, and 10, 20, and 40 for the S-3 and S-4 datasets. The reason for experimenting with the increased number of k is to present a comprehensive view of the performance of our approach from the beginning stage of worker selection (when the value of k is large) to the ending stage of worker selection (when the value of k is small). For example, when changing the value of k from 14 to 7 on the RW-1 dataset, we can analyze the performance change of our method when conducting one round and two rounds



(a) Value of k (RW-1 dataset)

(b) Value of k (RW-2 dataset)

(c) Value of k (S-1 dataset)

(d) Value of k (S-2 dataset)

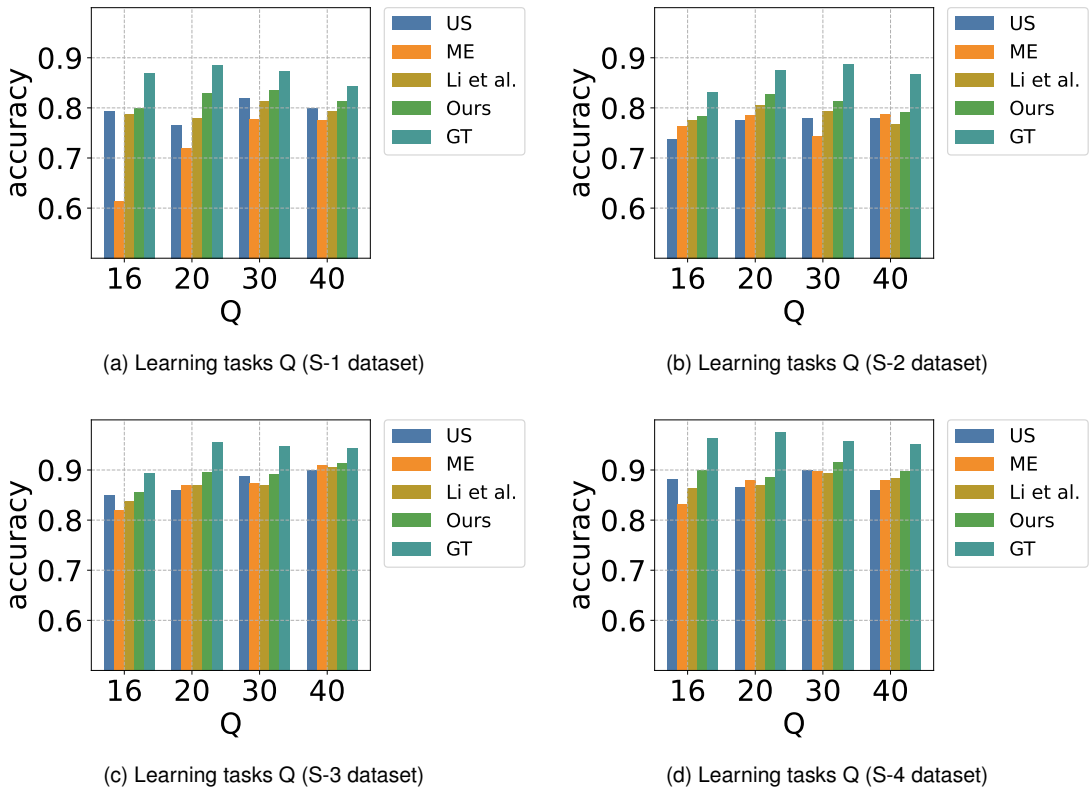
(e) Value of k (S-3 dataset)

(f) Value of k (S-4 dataset)

Figure 2.6: Parameter sensitivity experiments of the number of selected workers.

of eliminations. As shown in Figures 2.6a and 2.6b, on RW-1 and RW-2 datasets, our method consistently outperforms all the baseline approaches, when we increase the value of k. On the synthetic datasets S-1, S-3, and S-4, as shown in Figures 2.6c, 2.6e, and 2.6f, our approach still outperforms all the baselines when the value of k increases. On the S-2 dataset (Figure 2.6d), our approach outperforms all the baselines when the value of k is set to 5 and 10, while is slightly worse than the performance of Li et al. [29] when the value of k further increase to 20. We further observe that on the RW-1 dataset, when $k = 14$, our approach and Li et al. have similar performance. Similar phenomena can also be observed on the S-1, S-3, and S-4 datasets when we set the value of k to 20, 40, and 40. We attribute this to the fact that when the value of k is large (i.e., the model is at the beginning stage of elimination), the long-term learning improvement of the workers on the target domain is not yet significant, the linear regression approach introduced by Li et al. [29] can capture the static cross-domain information. However, as the value of k decreases (i.e., the elimination process proceeds), the dynamic cross-domain performance estimation and the learning gain estimation help our approach to outperform Li et al. [29].

As for the number of learning tasks Q, the default value is 20. We changed the value of Q to 16, 30, and 40, while keeping the value of k unchanged with a changing total budget B in this section to evaluate its influence and presented the experimental results in Figure 2.7. We first notice that our approach consistently outperforms all baselines on four synthetic datasets with different Q. We further observe that on four synthetic datasets, the performance of our approach and baselines tends to get close when Q increases. We attribute this phenomenon to the fact that when the budget is arbitrarily large, the improvement brought by adopting cross-domain information is reduced since the algorithm can get an accurate estimation of workers' target domain knowledge based on a large amount of learning tasks assigned to workers. However, when the total budget is small, our approach efficiently utilizes the cross-domain information to boost the worker selection performance of ME. In real-world applications, selecting workers effectively with a relatively small number of learning tasks is crucial, since in reality, the ground truth answers of golden questions in each domain require manual collection and thus are hard to obtain. In this sense, the performance uplift of our approach over other baselines when Q is small is favored and useful



(a) Learning tasks Q (S-1 dataset)

(b) Learning tasks Q (S-2 dataset)

(c) Learning tasks Q (S-3 dataset)

(d) Learning tasks Q (S-4 dataset)

Figure 2.7: The parameter sensitivity experiments of the number of learning tasks per batch Q.

for real-world applications.

2.5.8 Discussion

We recorded the running time results on one Intel Xeon Gold 6240 CPU @ 2.60GHz. Our method takes 3.9s, 5.0s, 6.3s, 7.8s, 13.4s, and 28.9s to select the best workers on the RW-1, RW-2, S-1, S-2, S-3, and S-4 datasets. Compared with the median completion times of our two surveys (1185s and 986s), the running time cost of our method is acceptable. As suggested by [61], the average completion time of tasks on AMT usually takes hours; we believe that the running time can facilitate the needs for real-world crowdsourcing applications.

The time used for workers to complete learning tasks inside the survey is approximately 500s for the two surveys. We observe that the average accuracy of all workers increases from 0.55 to 0.79 and from 0.65 to 0.85 on RW-1 and RW-2 respectively after a single round of worker training with 10 questions. Despite the additional training time required,

we believe that by introducing worker training into the worker selection process of crowdsourcing, we can significantly improve the overall worker performance in the target domain. As for the cost of the worker training process: denote the number of learning tasks and working tasks assigned as $|T_l|$ and $|T_w|$, the annotation accuracy before and after the worker training as a_t and a'_t. For simplicity, we consider the effect on one single worker with one round of training, assume the worker's accuracy is the same as the average accuracy of all workers, and consider the same monetary cost for completing each learning and working task. If $|T_w|/|T_l| > \frac{a_t}{a'_t - a_t}$, under the same total worker learning and working budget, the number of correctly annotated samples in T_w for the worker with worker training process would be greater than that without worker training. In our case, once $|T_w|/|T_l|$ is greater than 2.3 and 3.3 for RW-1 and RW-2 respectively, then the additional monetary cost can be counteracted. Furthermore, our worker training phase interacts with the workers by revealing the correct answers to learning tasks to workers promptly. Throughout multiple worker training rounds, workers can learn about their overall improved performance in the target domain, have a sense of accomplishment, and obtain new skills related to the target domain. As discussed by previous works [62, 63, 64, 65], with timely feedback received and new skills learned, workers tend to have improved engagement and performance. Therefore, we believe the worker training phase improves the engagement of workers, and stimulates workers to explore more useful features in the target domain.

We further report the estimated correlation between domains on the RW-1 and RW-2 datasets. Specifically, the correlation parameters estimated by our method are 0.50, 0.69, and 0.65 for Plane-Flower (P-F), Fish-Flower (F-F), and Elephant-Flower (E-F) on RW-1 and 0.23, 0.10, and 0.68 for Peruvian lily-Lenten rose (P-L), Red fox-Lenten rose (R-L), and English marigold-Lenten rose (E-L) on RW-2. The correlation parameters for F-F and E-F are larger than that for P-F, which coincides with our intuition that workers sensitive to color and shape differences (good at fish and elephant domains) are likely to perform well in distinguishing flowers. The correlation parameters for P-L and E-L are larger than that for R-L, which means the workers who are good at distinguishing other flowers are good at distinguishing Lenten roses. Besides, the correlation for E-L is larger than P-L. To distinguish English marigold from its counter-parts, workers need to notice the small

differences in petals and stamen (shape), which is close to the requirement of distinguishing Lenten rose; while to identify the Peruvian lily from its counter-parts, workers only need to pay attention to color difference. As a result, the correlation for E-L is larger than P-L. More details of the prior and target domains are presented in Table 2.3.

2.6 Future Work Discussion

We discuss whether we can integrate LLM into the process as a hybrid approach to further improve performance. A straightforward design is to use LLMs as a supplement when the number of human annotators is insufficient. For example, if we need to label 10,000 data entries and each human annotator can handle a maximum of 500 entries, but we only have 10 human annotators, we can introduce LLMs to take over part of the workload. Intuitively, LLMs help alleviate some of the workload for human annotators, allowing them to focus more and achieve better labeling results. Additionally, LLMs may possess superior knowledge in some specialized domains compared to layman human annotators. Finally, LLMs can provide labeling efficiency that exceeds that of human annotators, which can reduce the total time required for the labeling task. Therefore, the introduction of LLMs can enhance the overall annotation performance.

2.7 Conclusion

In this chapter, we formulated the cross-domain-aware worker selection with training problem and proposed a novel algorithm based on Medium Elimination to resolve it. Specifically, two estimation components CPE and LGE are designed to incorporate cross-domain knowledge information and capture the learning gains during worker training. Real-world and synthetic cross-domain-aware worker selection with training datasets were collected to evaluate different approaches. We conducted extensive experiments on real-world and synthetic datasets to show that our method outperforms all the state-of-the-art baselines on real-world and synthetic datasets. We confirm that applying CPE and LGE can capture cross-domain knowledge information and estimate the learning gain during the worker

training process. As a future direction, we aim to construct a unified multi-domain taxonomy that optimizes the worker training and selection process.

CHAPTER 3

RECA: RELATED TABLES ENHANCED COLUMN SEMANTIC TYPE ANNOTATION FRAMEWORK

Chapter 3 introduces the Related Tables Enhanced Column semantic type Annotation framework (RECA), which innovates on the table column semantic type annotation. Existing methods on column semantic type annotation are challenged by the difficulties of handling wide tables and incorporating complex inter-table context information. Failure to handle wide tables limits the usage of column type annotation approaches, while failure to incorporate inter-table context harms the annotation quality. Existing methods either completely ignore these problems or propose ad-hoc solutions. Therefore, we proposed RECA, which incorporates inter-table context information by finding and aligning schema-similar and topic-relevant tables based on a novel named entity schema. The design of RECA can naturally handle wide tables and incorporate useful inter-table context information to enhance the annotation quality. We conduct extensive experiments on two web table datasets to comprehensively evaluate the performance of RECA. Our results show that RECA achieves support-weighted F1 scores of 0.853 and 0.937 with macro average F1 scores of 0.674 and 0.783 on the two datasets respectively, which outperform the state-of-the-art methods.

3.1 Introduction

In various data mining applications, column type annotation plays a crucial role. In schema matching, column type annotation can help identify semantic relationships [66, 67]; Accurate annotation of data types can efficiently facilitate automated data cleaning [68, 69]; The integration of tabular data from different sources also benefits from accurate annotation of column types [70, 71]. In general, table column types can be divided into two categories:

atomic and *semantic* [71]. *Atomic* types, such as **Integer** and **Boolean**, provide fundamental and general information about the data type of the column content. *Semantic* types, such as **BirthPlace** and **Person**, provide fine-grained information of the column content. The fine-grained *semantic* type is helpful for various downstream applications that require a deep understanding of the column content semantics. However, although most existing systems can accurately annotate the *atomic* types, they are challenged by the difficulty of accurately annotating the *semantic* types solely based on the column content. As shown in Figure 3.1, the first columns in all three tables have similar content, which are the names of artworks. However, the semantic label of the first columns in Table T_1 and T_2 is **TelevisionShow**, while that in Table T_3 is **VideoGame**. The ambiguity of the column content makes the column semantic type annotation a great challenge to the database community.

In response, various approaches have been proposed to utilize additional context information in the tables to resolve the ambiguity and improve the annotation quality: Hulsebos et al. proposed Sherlock [66] based on hand-crafted statistical features. Sato [71] incorporates table topic features and inter-column co-occurrence relationships into generating the column representations. TaBERT [72] utilizes BERT [73] as a base model to capture the table content features. TABBIE [74] refines the idea of TaBERT by encoding rows and columns respectively to provide a more comprehensive view of tables. DODUO [75] designs a Transformer-based structure to encode the intra-table contextual information inside each table. Among these methods, TABBIE and DODUO achieve state-of-the-art performance. Although these methods achieve state-of-the-art performance on the column semantic type annotation task, they are challenged by the following difficulties: **(1) Difficulty in handling wide tables:** In the era of big data, the size of tables is growing. As shown in [76], the average number of columns of tables in Open Data is 16, with a large variance: some tables have hundreds of columns. Therefore, column semantic type annotation approaches that effectively handle these wide tables are needed. However, the existing state-of-the-art column semantic type annotation methods cannot handle the wide tables well due to the maximum input length limits of the language models (LMs) used by these methods. For instance, TABBIE [74] and DODUO [75] use BERT as the core LM, which has a hard maximum input length limit of 512 tokens [73]. In response, TAB-

?	?	?	?
Amorcito corazón	L. Suárez	D. Olivera	2012-06-10
A Nero Wolfe Mystery	S. M. Kaminsky	M. Chaykin	2002-08-18

WPPD

?	?	?	?
Chōriki Sentai Ohranger	T. Inoue	T. Satō	1996-02-23
Chōjin Sentai Jetman	T. Inoue	T. Wakamatsu	1992-02-14
Brewster Place	M. Angelou	O. Winfrey	1990-05-30
Anne of Green Gables: The Continuing Story	K. Sullivan	J. Crombie	2000-07-30
Angry Boys	C. Lilley	C. Lilley	2011-07-27
Alex Haley's Queen	A. Haley	Ann-Margret	1993-02-18
...	...	...	...

WPPD

?	?	?	?
Donkey Kong Country	Nintendo	2006-12-08	2006
F-Zero	Nintendo	2006-12-08	2006
SimCity	Nintendo	2006-12-29	2006
Super Castlevania IV	Konami	2006-12-29	2006
Street Fighter II: The World Warrior	Capcom	2007-01-19	2007
...	...	...	...

WODD

Figure 3.1: From the left to right we denote the tables as Table T_1, T_2, and T_3. T_1 and T_2 contain information about television series; T_3 is about video games. The named entities extracted from the tables by spaCy [1] are colored differently: Orange - WORK_OF_ART (W); Green - PERSON (P); Blue - DATE (D); Red - ORG (O). The bottom part shows the three tables' corresponding named entity schema strings.

BIE [74] ignores the issue of wide tables; DODUO [75] suggests designing user-defined splitting rules, splitting the wide tables into clusters, and then encoding each cluster to perform column semantic type annotation for wide tables. Unfortunately, DODUO [75] fails to provide a detailed design of the table splitting rules and the method of combining the annotations provided by each cluster. Besides, designing such splitting rules is non-trivial: rich domain knowledge is required to design rules that are both effective and generalizable. As a result, handling wide tables properly remains a challenge to the existing LM-based column semantic type annotation methods. **(2) Difficulty in incorporating complex inter-table context information:** TCN [77] draws schema-related tables from various websites and extracts inter-table context information. As suggested by [77], inter-table context information is potentially useful for column semantic type annotation. By incorporating the inter-table context information, ambiguity in column content is expected to be reduced, and the annotation quality can be improved. As shown in Figure 3.1, the first columns of T_1 and T_2 refer to TV series. Annotating the first column of T_1 solely based on the content of T_1 is difficult: the number of rows in T_1 is few, so it's difficult for existing methods to resolve the ambiguity between the similar semantic types, such as **TelevisionShow**, **VideoGame**, etc. By aligning T_1 and T_2, we augment the content of T_1, the model receives more useful information for annotation and the annotation quality is thus improved. However, both TABBIE [74] and DODUO [75] ignore the useful inter-table context information. Incorporating inter-table context information is non-trivial. Since the table schema information is generally not available [71, 75, 74], it is difficult to align tables that potentially have similar schema solely based on the table content. Although TCN [77] incorporates inter-table context information, it is limited to handling relational tables with a known table schema[1] and page topic. Unfortunately, valid table schema and page topic annotations are generally unavailable in real-world web tables datasets [78, 79]. Therefore, effectively incorporating inter-table context information solely based on the table content without additional table schema and page topic annotations remains a challenge to the column semantic type anno-

[1]TCN [77] manually cleans the headers to form the table schema, based on which it develops the inter-table relationship. However, according to the common practice of previous work [71, 75, 74], table headers should not be provided as a piece of known information to prevent introducing noise or incurring potential information leaks.

tation task.

In order to handle wide tables, we first studied the reasons why previous approaches have failed. One key observation is that they incorporate intra-table context to enhance the annotation quality. However, since the intra-table context (columns except for the one that we aim at annotating) grows linearly with respect to the table width, the design of incorporating intra-table context without splitting can easily lead to the exhaustion of input tokens for LM-based approaches. As a result, previous approaches cannot process entire wide tables properly. To circumvent this problem, we suggest utilizing inter-table context instead to enhance the annotation quality. In order to handle the difficulty of incorporating complex inter-table context information, we define and adopt a novel named entity schema for web tables. Based on this novel named entity schema, we can successfully align tables that have similar schemata and extract useful inter-table context information from the tables aligned with the original table. In this way, the difficulty of incorporating inter-table context without accessing table schemata and page topics can be resolved.

In general, RECA is a self-contained data-driven representation learning framework, i.e. it requires no external linkage to a knowledge base nor additional manual annotations of table schema and page topic. RECA considers the target column only in the original table so that it can easily accommodate wide tables. We define a novel named entity schema with table filtering and alignment strategies for RECA to efficiently extract and process the inter-table context to obtain rich context information for the annotation. RECA further leverages BERT [73] as the core LM to encode the target column and the inter-table context information. Finally, RECA combines the embeddings generated and annotates the target column. Specifically, we have made the following contributions:

- We propose a novel self-contained data-driven representation learning framework called RECA for column semantic type annotation. RECA extracts and leverages inter-table context to enhance the annotation quality of the target column, thus resolving the wide table issue.
- We define a novel named entity schema for RECA to efficiently align related and sub-related table, which resolves the difficulty of incorporating inter-table context

without manual table schemata annotations and page topics.

- We conduct extensive experiments on two real-world web table datasets to show that RECA outperforms all the state-of-the-art methods. The result demonstrates the effectiveness of utilizing the inter-table context to annotate column semantic types accurately.

- We show that RECA is data efficient and learning efficient, since it requires shorter input token sequences and fewer training data to achieve high performance. The result demonstrates the robustness and power of RECA on the column semantic type annotation task.

3.2 Related work

In general, there are two lines of research related to our work: feature-based data interpretation and pre-trained model-based representation learning.

Feature-based data interpretation Research in feature-based data interpretation aims to capture the semantics in tabular data by extracting hand-crafted, statistical, and semantic features. The extracted features are then employed to resolve the column semantic type annotation problem.

Some early works extract the semantics of data by designing hand-crafted features. SemanticTyper [80] first separates the tabular data into textual and numeric types and then suggests using Term Frequency-Inverse Document Frequency (TF-IDF) and a two-sample Kolmogorov-Smirnov test [81] to extract the semantics from the tabular data. Pham et al. [82] include Mann-Whitney test [81] and Jaccard Similarity for textual data to further improve the performance of SemanticTyper.

Several previous works consider statistical and semantic features in column semantic type annotation. Sherlock [66] extracts character-level, word-level, paragraph-level, and global-level features and then employs deep learning classifiers to achieve high annotation quality. Sato [71] furthers the capability of Sherlock [66] by incorporating the table topic

features and intra-table context into the annotation task.

However, the hand-crafted, statistical, and semantic features used by these methods are too general to capture the rich semantics and form expressive representations from tables, which is a significant limitation. As suggested by [77], these methods cannot effectively capture the fine-grained semantics in tables because of the fact that the shallow networks of these methods have limited expressiveness.

Pre-trained model-based representation learning To annotate column types, pre-trained neural models are used by recent methods to learn representations of tables and generate predictions. In general, the pre-trained model-based representation learning approaches generally outperform the feature-based data interpretation approaches and thus serve as our main baselines.

TURL [83] employs a pre-training + fine-tuning framework to perform column semantic type annotation. A visibility matrix is used to mask out the components of tables that are structurally irrelevant to the transformer structure. The generated column embeddings are fed into a classifier for column semantic type annotation. TaPas [84] encodes the queries and the table content together. The positional embedding of the original BERT [73] structure is modified by TaPas to ensure a better understanding of tables. TaBERT [72] slices the web tables with the user queries and feeds them into the BERT structure to learn the representations of every column in the tables. Based on Sherlock [66], Zhou et al. [85] utilize a Star-Transformer [86] structure to encode the features extracted using Sherlock. TCN [77] suggests using both intra-table and inter-table information and applying multi-task training to perform column semantic type annotation. However, the information used by TCN such as table schema and page topic requires extra manual annotation and cleaning, so it cannot be widely applied on most web table datasets. (For instance, the Semtab2019 dataset [78] does not provide table schema annotations and page topic information for its tables, and the Webtables dataset [71] does not provide page topic information.) TABBIE [74] furthers the design of TaBERT [72] by considering two independent transformers to encode the rows and columns of the tables jointly. TABBIE [74] provides a more comprehensive

view of tables in comparison with TaBERT [72] and therefore achieves better annotation quality in comparison with TaBERT. DODUO [75] uses a transformer structure to encode all the columns in the tables in one pass. The representations of all the columns in the tables are considered together when training the classification module. It also suggests applying multi-task learning to generate representations that contain rich semantic information.

Among these methods, TABBIE [74] and DODUO [75] achieve state-of-the-art performance on the column semantic type annotation task. However, they neglect the inter-table context information, which is beneficial in generating high-quality annotations. Besides, they cannot handle wide tables well. TABBIE and DODUO models directly feed all the columns into a transformer structure to learn the representation of each column, which leaves space for further improvement in terms of boosting the annotation performance.

3.3 Notations and Problem definition

We first discuss the notations used in this chapter and then formally define the problem of column semantic type annotation.

3.3.1 Notations and Concepts

The general notations are introduced in Table 3.1. We further introduce some important concepts used in this chapter.

Definition 1: (Target Column): We define the target column to be the column that has a manual annotation of its semantic type in the table. Note that not all the columns necessarily have a manually annotated label.

Definition 2: (Named Entity Schema): Named Entity Schema is the table schema generated based on the named entity type extracted in each column. As shown in Figure 3.1, the named entity schemata are formed by selecting the most frequent named entity type in each column. Each named entity schema s is represented as a string.

Definition 3: (Edit distance): The edit distance $\Phi(s_i, s_j)$ between named entity schemata s_i and s_j is defined as the minimum number of changes required to convert s_i

Table 3.1: General notations with corresponding descriptions.

Notation	Description
$D = \{T_1, T_2, ..., T_n\}$	web table dataset that contains n tables
T, (T_i)	the (i-th) web table without headers
M, (M_i)	the number of columns in the table T, (T_i)
N, (N_i)	the number of rows in the table T, (T_i)
C^m, (C_i^m)	the m-th column in the table T, (T_i)
C, (C_i)	the set of all the columns in T, (T_i)
$c^{m,n}$, $(c_i^{m,n})$	the cell at the m-th column, the n-th row of the table T, (T_i)
E^m, (E_i^m)	the collection of named entities in the m-th column of the table T, (T_i)
y^m, (y_i^m)	the semantic type ground truth label of the m-th column in the table T, (T_i)
s, (s_i)	the named entity schema of table T, (T_i)
R, (R_i)	the set of related tables of table T, (T_i)
X, (X_i)	the set of sub-related tables of table T, (T_i)
I_R^t, $(I_{R_i}^t)$	the set of identified columns corresponding to the target column C_i^t in related tables R, (R_i)
I_X^t, $(I_{X_i}^t)$	the set of identified columns corresponding to the target column C_i^t in sub-related tables X, (X_i)
S	the pre-defined column semantic type set
Ψ	Dictionary-based mapping from each named entity type to a distinct English character

to s_j with the insertion, deletion and substitution editing operations [87, 88].

Definition 4: (Target Table): We define the table that contains the target column to be annotated as the target table. For instance, if we currently want to annotate the semantic type of column C_i^m, then the target table here refers to T_i.

Definition 5: (Candidate Table): A table T_j is a candidate table of T_i if the Jaccard similarity [89] between their content is larger than a threshold.

Definition 6: (Related Table): A candidate table T_j is a related table of the target table T_i if $s_j = s_i$.

Definition 7: (Sub-related Table): A candidate table T_j is defined as a sub-related table of the target table T_i if the edit distance $\Phi(s_i, s_j)$ is less than or equal to a threshold.

Definition 8: (Identified Column): An identified column is the column in the related or the sub-related table that has the same column index and the same named entity type as the target column.

Definition 9: (Inter-table context): Given a target table, the inter-table context refers to the collection of the identified columns in related tables and sub-related tables of the target table.

3.3.2 Problem definition

Aligning with the problem definition of previous methods [71, 66, 75, 74], the goal of column semantic type annotation is to predict the semantic type of the target column from a pre-defined semantic type set. The pre-defined semantic type sets are normally defined using existing ontologies to ensure the coverage of column semantic types [66, 90]. The prediction of the target column type should be solely based on the table content without accessing the table headers (table schema) in alignment with the common practice of previous work [71, 66, 75, 74]. The problem is formulated as a multi-class classification problem.

Problem 1: (Column semantic type annotation): Given a web table T (without table headers) from the dataset D, denote the target column as C^t in T. The column semantic type annotation model W annotates C^t with a semantic type $\bar{y}^t = W(C^t, T, D) \in S$, such

that $\bar{y}^t$ best fits the semantics of C^t (being the closest to the ground truth semantic type y^t).

3.4 Methodology

In order to handle the wide tables, a naive design encodes the target column only. However, such a design discards the table context information, which is potentially useful for annotating the target column [71]. In order to handle the wide table issue without compromising the quality of annotations, we incorporate inter-table context information.

In Figure 3.2, we aim at annotating the first columns, which have **Country** type. Tables T_5 and T_6 are the tables that are potentially related to table T_4. Since table T_4 is wide, it cannot be properly processed by existing methods, such as DODUO [75], without splitting. The point is that tables T_4, T_5, and T_6 are similar in table schema and table content, so they can thus be viewed as a holistic collection of data records. The inter-table context from tables T_5 and T_6 can extend the information in the original table T_4's target column and thus serve as a perfect replacement for the intra-table context, which constrains LMs' ability to handle the wide tables.

In order to incorporate inter-table context information without additional table schema and page topic annotations, designing effective related table alignment and filtering rules is necessary. As shown in Figure 3.1, table schema and page topic are not available for general web tables. Therefore, we need to design a special approach to filter out irrelevant tables and align the related tables with the original tables. To this end, we define a novel table schema called a named entity schema. As shown in Figure 3.1, named entity types can be detected from the table and we can form the so-called named entity schema for the web tables by selecting the most frequent named entity type for each column. Intuitively, web tables with similar named entity schemata are likely to describe the same collection of data records. For example, the named entity schema in tables T_1 and T_2 is [WORK_OF_ART, PERSON, PERSON, DATE], while the named entity schema in table T_3 is [WORK_OF_ART, ORG, DATE, DATE]. Despite the fact that the first columns in both T_2 and T_3 are the names of artwork (and the fact that the first columns in T_2 and T_3 look similar in terms of content and format), T_2 and T_3 are not likely to belong to the same data record collection because of the



?	?	?	?	?	...	?
Albania	27,398	$11,800	$2,949.57	2,994,667	...	Parliamentary Democracy
Algeria	2,381,740	$159,000	$3,948.01	34,994,937	...	Republic
Angola	1,246,700	$85,810	$5,003.43	13,338,541	...	Republic; Multiparty Presidential Regime
...	...	...	...	...	...	...

?	?	?	?	?	...	?
Bahamas	10,070	$7,538	$21,547.17	313,312	...	Constitutional Monarchy with a parliamentary system of government
Bangladesh	133,910	$100,100	$481.36	158,570,535	...	Parliame
Belgium	30,278	$461,300	$43,648.01	10,431,477	...	Federal F
...	...	...	...	...	...	...

?	?	?	?	?	...	?
Canada	9,984,670	$1,334,140	$40,457	34,733,000	...	Federal Parliamentary Democracy
Chile	748,800	$199,200	$10,058.50	16,888,760	...	republic
China	9,326,410	$5,745,000	$2,459.43	1,336,718,015	...	Communist State
...	...	...	...	...	...	...

Figure 3.2: We denote the table lying on the top as Table T_4; the two tables at the bottom are Table T_5 and T_6, respectively. Note that these tables describe the detailed information of countries, so it can potentially grow very wide and thus cause challenges to existing methods. We notice that Table T_4, T_5, and T_6 share the same table schema, so we link them together and develop the main idea of RECA based on this.

difference in their named entity schemata. As a result, the possibility that the first columns of T_2 and T_3 refer to the same column type is lower than that of the first columns of T_2 and T_1. Indeed, the first columns in both Tables T_1 and T_2 include the names of TV series, while that in T_3 includes the names of games. In this way, we can gather together the tables that are similar in schemata. Table column alignment can then be applied thereafter, and we can find columns that likely extend the target column and form the inter-table context for column semantic type annotation.

The general process of RECA is shown in Figure 3.3: RECA first extracts the named entities and assigns types from a pre-defined named entity type set, which is discussed in Section 3.4.1. Based on the named entity types extracted, it further generates the named entity schema for the input table. In Section 3.4.2, we discuss how RECA filters out the topic-irrelevant tables. Then it finds the related tables and sub-related tables of the target table based on their corresponding named entity schemata, which is introduced in Section 3.4.3. RECA then encodes the target column and its corresponding inter-table context information, which is discussed in Section 3.4.4. Finally, we introduce how RECA performs column type annotation based on the encoding generated, which is discussed in Section 3.4.5.

3.4.1 Named entity tagging

First note that under our problem formulation, the table schema (header) information is not provided as input to the model. Therefore, we propose forming an approximated schema based on named entities in each column to identify related and sub-related tables. In order to achieve this, the first step is to identify the named entities in the tables.

Given a table T from dataset D that has M columns and N rows, RECA utilizes tagging tools such as spaCy [1] to identify the named entities in each column $C^m \in C$ and tag them into basic named entity types as shown in Figure 3.4. The tagged named entities in each column are denoted as $\bar{E}^m$.

Apart from the basic named entity types, we notice that several named entity types have different representation formats, such as DATE and PERSON, which can be further divided

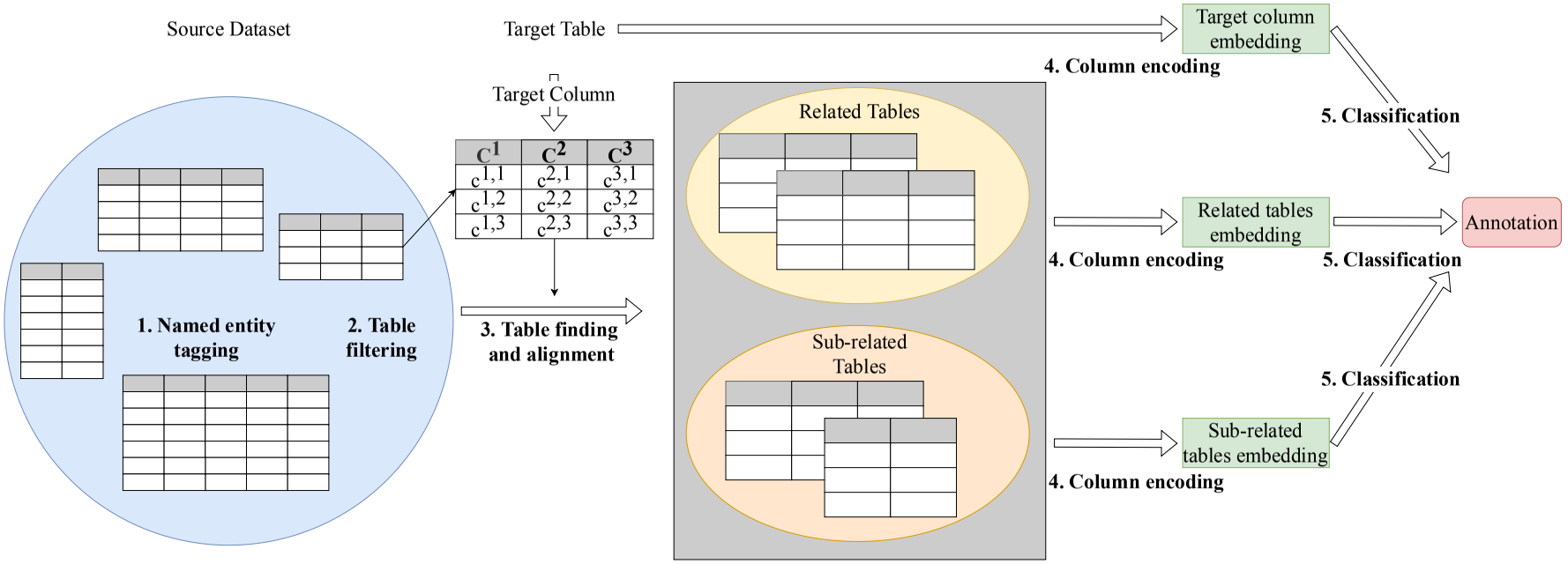


Figure 3.3: The general Column Semantic Type Annotation process of RECA.

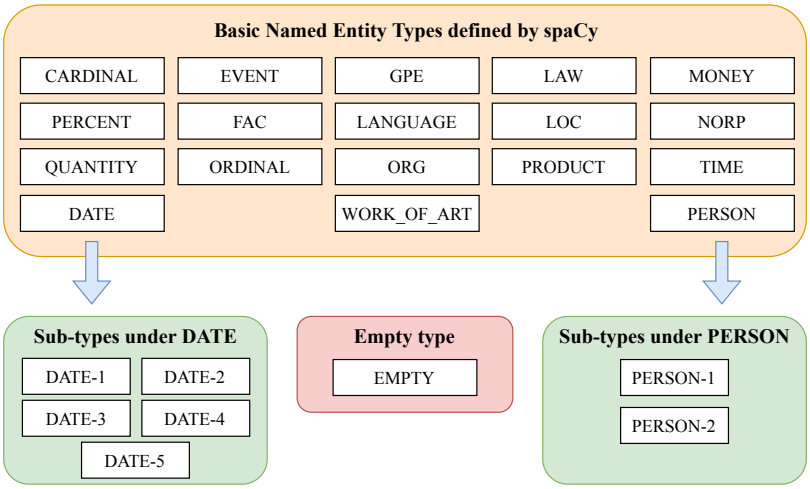


Figure 3.4: The basic named entity types defined by spaCy [1] (orange); the sub-types defined under DATE and PERSON (green); the EMPTY type that handles the case when no named entity is detected (red).

into sub-types based on simple hand-crafted rules. For instance, as shown in Figure 3.1, the fourth columns of both tables T_1 and T_3 are annotated as DATE, while the format of the content in the columns is different (YYYY-MM-DD & YYYY). The difference in the data format suggests that the two tables are likely to be from different sources. Therefore, the possibility that the two tables describe the same data collection is relatively low.

Table 3.2: Sub-types and the corresponding descriptions.

Sub-type	**Description**
DATE-1	dates in YYYY format. *e.g. 2022*
DATE-2	dates that contain month names or abbreviations of month names. *e.g. January 16th, 2022*
DATE-3	dates in DD-MM-YYYY or MM-DD-YYYY or YYYY-MM-DD format. *e.g. 18-11-1998*
DATE-4	dates that contain numerical months and days only. *e.g. 02-29*
DATE-5	dates in other formats.
PERSON-1	person names that contain abbreviations. *e.g. J. K. Rowling*
PERSON-2	person names in other formats.

To this end, we designed hand-crafted rules to further classify types DATE and PERSON into sub-types based on the data format[2]. Detailed descriptions of the sub-types are introduced in Table 3.2. We only divide the DATE and PERSON types into sub-types, since these two types are common, and designing hand-crafted rules to classify entities in

[2]In reference to the data formats in the Document Automation Language of Oracle

these two types is relatively simple[3]. Note that the sub-types we defined are general and may be further divided, but further dividing the sub-types may require more complicated hand-crafted rules, which would not be desirable for maintaining the generalizability of the RECA framework on other web table data. In other words, there is a trade-off between the generalizability of the hand-crafted rules and the granularity of the sub-types generated. Furthermore, we also include a new type called EMPTY to handle cases where no named entity is extracted from the column. As a result, we obtain the finer-grained named entity types, based on which we can generate the updated tagged named entities in each column m as E^m.

After obtaining the updated tagged named entities for each column, we further generate the named entity schema for the table T. For each E^m obtained, we denote the most frequent named entity type[4] in E^m as $\hat{e}^m$. Based on this, we obtain the named entity schema string s of table T: $s = (\Psi(\hat{e}^1), \Psi(\hat{e}^2), ..., \Psi(\hat{e}^M))$, where each entity type shown in Figure 3.4 is mapped to a distinct English character through a dictionary-based mapping Ψ[5]. In this way, we can generate the table schemata for all the tables in the dataset D (Lines 3 and 5, Algorithm 5).

3.4.2 Table filtering

In order to obtain candidate tables that are topically similar to the target table, we need to filter them based on the table content. In reference of [91], we adopt the Jaccard similarity [89] to achieve this goal. For each target table T_i and each table T_j, where $(j \neq i)$, we denote the set of words in T_i as A_i and the set of words in T_j as A_j. Then we compute the Jaccard similarity between T_i and T_j:

$$\text{Jaccard}(A_i, A_j) = \frac{|A_i \cap A_j|}{|A_i \cup A_j|} \tag{3.1}$$

[3]These rules can be applied as a complement on other tagging tools, since the rules are general and the types DATE and PERSON are also common in other tagging tools.

[4]In case a tie happens, for simplicity, we select the named entity type that occurs first in the column from the most frequent types.

[5]We constructed a dictionary that maps the named entity types and sub-types into 24 distinct English characters.

Algorithm 5 Training process of RECA

Input:

Number of epochs P
Training dataset D_{tr}
Loss function $\mathcal{J}$

Output:

Trained RECA model W_{tr}

for $i = 1, 2, ..., |D_{tr}|$ **do**
  for $j = 1, 2, ..., M_i$ **do**
    Named Entity tagging, obtain $\hat{e}_i^j$ from C_i^j
  end for
  Generate $s_i = (\Psi(\hat{e}_i^1), \Psi(\hat{e}_i^2), ..., \Psi(\hat{e}_i^{M_i}))$
end for
for $i = 1, 2, ..., |D_{tr}|$ **do**
  for $j = 1, 2, ..., i-1, i+1, ..., |D_{tr}|$ **do**
    Compute Jaccard(A_i, A_j), if Jaccard$(A_i, A_j) > \delta$, include T_j to T_i's candidate table set Q_i
  end for
end for
for $i = 1, 2, ..., |D_{tr}|$ **do**
  for each table T_k in Q_i **do**
    Compute edit distance $\Phi(s_i, s_k)$, if $\Phi(s_i, s_k) = 0$ add T_k to R_i; if $0 < \Phi(s_i, s_k) \leq \sqrt{M_i}$ add T_k to X_i
  end for
  for each target column t in T_i **do**
    Align and store $I_{R_i}^t$ and $I_{X_i}^t$ to D_{tr} as side information
  end for
end for
Initialize the LM core of W_{tr} with pre-trained BERT weights
for epoch $= 1$ to P **do**
  Randomly split D_{tr} into batches $\{B_1, B_2, ..., B_f\}$
  for $l = 1, 2, ..., f$ **do**
    Encoding and classification $L = \mathcal{J}(B_l, W_{tr})$
    $W_{tr} =$ Backpropagate(W_{tr}, L)
  end for
end for
return W_{tr}

We include T_j as a candidate table if the similarity is larger than a threshold δ. After performing table filtering, we obtain the candidate tables that are topically related to the target table (Line 9, Algorithm 5).

3.4.3 Table finding and alignment

Note that Section 3.4.2 considers the relatedness of tables mainly based on the table topics. We now consider the relatedness of tables based on the structural information (schema and location). Given a table T_i in the dataset D, we now want to find its corresponding related and sub-related tables based on the named entity schemata extracted. The alignment of the related tables is straightforward: we simply identify the candidate tables T_j that have the same schema s_j as the schema s_i of the table T_i, where $j \neq i$. We denote the set of these related tables T_j as R_i.

In order to identify sub-related tables that are 'approximately' aligned with the target table, we consider these requirements:

- **Schema similarity**: The named entity schema of the sub-related table should not be much different from that of the target table.
- **Column location alignment**: The named entity type of the target column matches with that of the column at the identical location in the sub-related table.

The intuition of schema similarity is that sub-related tables that share similar schemata with the target table are likely to have related or similar table content as the target table. For column location alignment, if the inner order of column types in a sub-related table differs much from the target table, then the inner schema structures of the two tables are not similar. Therefore, the possibility that the sub-related table and the target table describe the same data collection is low.

In considering the schema similarity, we compute the edit distance [88] between the table schema string of a candidate table T_k and that of the target table T_i. If the edit distance $\Phi(s_i, s_k)$ is larger than a threshold[6], then the candidate table cannot be regarded as a sub-

[6]We empirically set the threshold to be the square root of the width of the target table. i.e., $\sqrt{M_i}$.

related table of the target table. If $\Phi(s_i, s_k)$ is larger than 0 and less than or equal to the threshold, we add T_k to X_i. If $\Phi(s_i, s_k) = 0$, then T_k is a related table and is added to R_i. We denote the set of sub-related table candidates that are 'schema similar' to the target table as X_i (Line 14, Algorithm 5).

We obtain the identified columns as follows: For the target column t in T_i, we consider each related table T_k in R_i and include C_k^t in $I_{R_i}^t$. For each sub-related table in X_i, we consider the following exact alignment rule to achieve column location alignment: We denote the named entity column type of the target column as $\Psi(\hat{e}_i^t)$. For a sub-related table candidate T_k, if $\Psi(\hat{e}_k^t) = \Psi(\hat{e}_i^t)$, then the t-th column of T_k is the identified column, we include C_k^t in $I_{X_i}^t$ (Line 17, Algorithm 5).

3.4.4 Column encoding

After obtaining the inter-table context (identified columns in related and sub-related tables, i.e., $I_{R_i}^t$ and $I_{X_i}^t$), the next step is to generate embeddings for the annotation. As shown in Figure 3.3, the target column, the identified columns in the related tables, and the identified columns in the sub-related tables are encoded independently. Specifically, for the target column C_i^t, we concatenate the cell content in the t-th column to generate the input string L_i^t for BERT model:

$$L_i^t = \text{Concatenate}(c_i^{t,1}, c_i^{t,2}, ..., c_i^{t,N_i}) \tag{3.2}$$

Similarly, we concatenate the cells from the identified columns of the related table set $I_{R_i}^t$ (or the sub-related table set $I_{X_i}^t$) to obtain the concatenated column set Z_{R_i} (Z_{X_i}):

$$Z_{R_i} = \{\text{Concatenate}(c_k^{t',1}, c_k^{t',2}, ..., c_k^{t',N_k}) | \text{for table } T_k \in R_i\} \tag{3.3}$$

$$Z_{X_i} = \{\text{Concatenate}(c_k^{t',1}, c_k^{t',2}, ..., c_k^{t',N_k}) | \text{for table } T_k \in X_i\} \tag{3.4}$$

where t' is the identified column index in each related table (or sub-related table).

Then we generate the input strings $L_{R_i}^t$ and $L_{X_i}^t$ for related tables and sub-related tables:

$$L_{R_i}^t = \text{Concatenate}(Z_{R_i}) \tag{3.5}$$

$$L_{X_i}^t = \text{Concatenate}(Z_{X_i}) \tag{3.6}$$

Upon obtaining the input strings for the target column, identified columns of related tables, and sub-related tables, we tokenize the strings and append an additional [CLS] token at the front of each string. Each string is passed to BERT and transformed into a 768-dimension embedding. We denote the embeddings of the target column, the identified columns of related tables and the identified columns of sub-related tables as v_i^t, r_i^t, x_i^t (Line 24, Algorithm 5).

Note that BERT [73] has a maximum input sequence length limit B (which by default is 512). For target columns that exceed this limit, we truncate the sequence to fit within the limit. As for the identified columns in related and sub-related tables, we promote fair allocation of the input tokens (e.g., If there are four identified columns in the related table set, each identified column will take $\frac{B}{4}$ tokens. If the identified column exceeds this limit, we truncate it). The main reason why we adopt such a fair allocation is that we hold an optimal view of the related and sub-related tables aligned: All the related and sub-related tables should be of high quality in terms of relatedness and thus should be considered. Besides, if we consider only some but not all of the related and sub-related tables, then we are at risk of biasing the model towards some tables that have a skewed distribution of data.

3.4.5 Classification

The generated embeddings for the target column and the corresponding identified columns of related tables and sub-related tables are fed into the classification module for the annotation of the target column type as shown in Figure 3.3.

Note that the identified columns in the related and sub-related tables are expected to be of the same semantic type as the target column yet have different content to provide a broader view of context information for the target column. Therefore, we design a three-network classification module to compute the annotations based on the content of the target column, related tables, and sub-related tables. Specifically, we adopt the standard two-layer classification module [92] for each network: a dropout layer that avoids overfitting and a linear layer that generates the annotations based on the input embeddings. We denote the outputs of the three networks as $\hat{v}_i^t$, $\hat{r}_i^t$, $\hat{x}_i^t$ respectively. We then generate the final annotation

a_i^t of the target column by combining the three outputs:

$$a_i^t = \alpha * \hat{v}_i^t + \beta * \hat{r}_i^t + \gamma * \hat{x}_i^t \tag{3.7}$$

where α, β, and γ are trainable weights.

Cross-Entropy loss is used as the loss function. The formula for the loss $\mathcal{J}$ is as follows:

$$\mathcal{J} = -\sum_{k=1}^{|S|}(p_k * log(q_k)) \tag{3.8}$$

where $p_k \in \{0, 1\}$ is the ground truth label for the k-th type. q_k is the k-th entry of a_i^t. The predicted column type $\bar{y}_i^t = \arg\max_k q_k$ (Line 24, Algorithm 5).

3.4.6 Analysis

Algorithm 5 shows the complete training process of RECA. We perform named entity tagging in each table first and formulate the named entity schemata (lines 1 to 6). Then we compute the Jaccard similarity between each pair of tables to obtain candidate tables (lines 7 to 11). After that, we compute the edit distance between candidate table pairs (lines 13 to 15) so as to perform column alignment (lines 16 to 18). Then we adopt BERT to encode the columns and perform classification (line 24). For a dataset D with n tables, we denote the maximum number of columns of tables as $\hat{M}$; the maximum number of rows of tables as $\hat{N}$; BERT embedding dimension as d; BERT maximum input sequence limit as B; the size of the pre-defined label set as $|S|$; the number of epochs as P; the number of batches as f.

Time complexity For each table, named entity tagging is performed to generate its named entity schema: we need to go through each column and each row, which takes $O(\hat{M}\hat{N})$ time complexity. For all the n tables, we need $O(\hat{M}\hat{N}n)$ time. Computing the Jaccard similarity takes $O(\hat{M}\hat{N}n^2)$ time since there are $O(n^2)$ pairs to compute and computing the Jaccard similarity for each pair requires going through the table content and thus takes $O(\hat{M}\hat{N})$. Computing edit distance takes $O(\hat{M}^2n^2)$ time. Since we compute the pairwise edit distance, each pair needs $O(\hat{M}^2)$ time. Aligning identified columns takes $O(\hat{M}n^2)$ time, since each table contains at most $\hat{M}$ target columns, each target column has at most

n identified columns, in total there are n tables, aligning a pair of columns takes $O(1)$. Preparation for the input sequence takes $O(\hat{N}n)$ for each column so in total $O(\hat{M}\hat{N}n^2)$ time. Encoding each column with its related and sub-related identified columns takes $O(Bd(B+d))$ time [93, 73]. So preparing input and encoding the columns takes overall $O(\hat{M}\hat{N}n^2+Bd(B+d)nP)$ time. Classification takes $O(d^2|S|nP)$ time. So the overall time complexity is $O(\hat{M}\hat{N}n+\hat{M}\hat{N}n^2+\hat{M}^2n^2+\hat{M}n^2+\hat{M}\hat{N}n^2+Bd(B+d)Pn+d^2|S|Pn) = O(\hat{M}^2n^2+\hat{M}\hat{N}n^2+(B^2+Bd+d|S|)dPn)$.

Space complexity For named entity tagging, we need $O(\hat{M}\hat{N}+\hat{M}n)$ memory space to load each table and store the named entity schemata. The Jaccard similarity computation step requires $O(\hat{M}\hat{N}+n^2)$ space since $O(\hat{M}\hat{N})$ is used for computing the Jaccard similarity and $O(n^2)$ is used to store the pairwise Jaccard values between tables. Computing the edit distance requires $O(2\hat{M}+n^2)$ space: $O(2\hat{M})$ space to perform the computation and the edit distance results require $O(n^2)$ memory space. Aligning identified columns needs $O(\hat{M}n^2+\hat{M}\hat{N}n)$ space, where we need $O(\hat{M}n^2)$ to perform alignment and store the alignment relationship (since each table contains at most $\hat{M}$ target columns and each target column has at most n identified columns, there are n tables in total) and $O(\hat{M}\hat{N}n)$ to store the table content. Encoding columns requires $O(\hat{N}n+(B+d)d)$ space: $O(\hat{N}n)$ space for preparing the input sequence (since each column has at most $\hat{N}$ cells and each column has at most n identified columns), $O(Bd)$ space for the input and output embeddings and $O(d^2)$ for self-attention matrices. The classification step requires $O(d|S|)$ space. So the overall space complexity is $\max(O(\hat{M}\hat{N}+n^2), O(\hat{M}n(\hat{N}+n)), O(\hat{N}n+(B+d)d), O(d|S|))$.

3.5 Experiment

We compared RECA with the following baselines: Sherlock [66], TaBERT [72], TABBIE [74], and DODUO [75]. Comparison with Sato [71] is omitted, since it is outperformed by DODUO. We do not compare with TCN [77], since TCN requires additional manual annotations and cleaning for the table schemata and page topics.

3.5.1 Datasets

We selected the Semtab2019 dataset [78] and the WebTables dataset from the VizNet corpus [79] used in [71, 75] to evaluate the performance of RECA. These datasets contain vertical relational web tables with valid semantic labels of different levels of granularity. Specifically, we selected rounds 1, 3, and 4 in the Semtab2019 dataset. In total, there are 3,045 tables with 275 distinct semantic types. For the WebTables dataset, we selected the tables with multiple columns. There are 32,262 tables with 78 distinct types. As shown in Table 3.3, the tables in the Semtab2019 dataset tend to be larger in size than those in the WebTables dataset. Besides, the number of semantic types in the Semtab2019 dataset is larger than that of the WebTables dataset, while the WebTables dataset has more tables than the Semtab2019 dataset, and every column in the WebTables dataset is annotated. Considering these differences in the properties of the two datasets, conducting experiments on both datasets provided a comprehensive and deep understanding of the properties of RECA. All the table headers are excluded following the common practice of previous work [71, 66, 74, 75], the main reasons are: a) Several datasets [78, 94, 90] contain index-like headers, which do not contain useful semantic information to generate meaningful table schema. b) The commonly-used benchmark WebTables dataset utilizes the column headers as the ground truth semantic type labels. In this case, utilizing table headers as input is likely to incur information leaks since the model will learn to directly annotate the columns based on the headers instead of the table content. Incorporating column headers as input would be unfair when performing a comparison with other approaches in the evaluation process.

Table 3.3: Statistics of the Semtab2019 dataset and the WebTables dataset.

	# types	# tables	# cols	avg rows	avg cols
Semtab2019	275	3,045	7,603	69.0	4.5
WebTables	78	32,262	74,141	20.0	2.3

In order to select training, validation, and testing sets from the Semtab2019 dataset [78], we randomly sampled 10% of the annotated columns to form the testing set. Then we randomly divided the rest of the data into five folds to conduct 5-fold cross-validation (80%

for the training set and 20% for the validation set). We preserved the percentage of each type[7] when generating the training, validation, and testing sets. As for the WebTables dataset, we followed the 5-fold cross-validation setting provided by [71].

3.5.2 Baselines

In order to evaluate the performance of RECA, we selected the following methods for comparison:

- Sherlock [66]: Sherlock extracts character-level and global-level statistical features from the tables. Besides, it considers semantic features from word-level and paragraph-level to form vector representations for table columns.
- TaBERT [72]: TaBERT jointly considers queries and table content to identify three salient rows in a table to generate table content snapshots. Based on the table content snapshots generated, TaBERT utilizes BERT to process and formulate representations for each table column for classification purposes.
- TABBIE [74]: TABBIE adopts a dual-transformer structure to encode the table columns and rows. The generated embedding for the target column is then used to annotate the column semantic types.
- DODUO [75]: DODUO utilizes a transformer structure to encode all the columns in the target table together in order to consider the intra-table context.

Among these methods, TABBIE and DODUO achieve state-of-the-art performances. To be fair for the evaluation, we used the open-sourced official implementations of the above-mentioned baselines. We preserved the experimental settings mentioned by the original papers as much as possible. Specifically, for Sherlock [66], we followed the official implementation [96]. For TaBERT [72], we adopted the prescribed implementation [97]. Note that the original purpose of TaBERT was to facilitate the question-answering task on table content. We followed the suggestions in [98] and provided a blank space as the query

[7] We used StratifiedKFold in scikit-learn [95].

input of TaBERT. For TABBIE [74], we used the official implementation [99] in the paper. For DODUO [75], we adopted the implementation [100] released by the authors and maintained the experimental settings of the model.

3.5.3 Experiment Metrics

We adopted F1 scores as the evaluation metrics for RECA ($F1 = 2 \times \frac{precision*recall}{precision+recall}$). As suggested by [71], due to the imbalanced distribution of semantic types, we selected two different kinds of F1 scores, support-weighted F1 score and macro average F1 score to evaluate the performance of different methods comprehensively. Specifically, the Support-weighted F1 score is the weighted average of per-type F1 scores, where the weights are proportional to the support in each type. The macro average F1 score is the mean of all the per-type F1 scores, focusing on the types with fewer supports.

3.5.4 Experiment Settings

We trained RECA for 20 epochs on each fold of the Semtab2019 dataset and trained it for 15 epochs on each fold of the WebTables dataset. We used Adam optimizer [101], and we adopted spaCy [1] as the named entity tagging tool, which has 18 basic named entity types. After extending the named entity type set as shown in Section 3.4.1, we obtained 24 named entity types. The set of learning rates to be selected was $\{0.00005, 0.00001, 0.000005\}$. The batch size was selected from $\{8, 16, 32\}$. The weight decay of 0.01 was used to optimize the training process. The Jaccard threshold δ was set to 0.1. Following the settings of the classification module in the Huggingface [92] implementation of BERT [73], we set the dropout rate of the classification module to be 0.3. We followed a consistent model selection method with [75], where the best model was selected based on the performance on the validation set. The means and standard deviations of the F1 scores are reported in Table 3.4. We conducted all the experiments on Intel(R) Xeon(R) Gold 5220R @ 2.20GHz CPUs and GeForce RTX 3090 GPUs.

Table 3.4: Experimental results on the Semtab2019 dataset and the WebTables dataset.

	Semtab2019 dataset		WebTables dataset	
Model names	Support-weighted F1	Macro average F1	Support-weighted F1	Macro average F1
Sherlock [66]	0.646±0.006	0.440±0.009	0.844±0.001	0.670±0.010
TaBERT [72]	0.768±0.011	0.413±0.019	0.896±0.005	0.650±0.011
TABBIE [74]	0.799±0.013	0.607±0.011	0.929±0.003	0.734±0.019
DODUO [75]	0.820±0.009	0.630±0.015	0.928±0.001	0.742±0.012
RECA *target only*	0.808±0.017	0.586±0.039	0.911±0.001	0.688±0.014
RECA *w/o re*	0.836±0.012	0.641±0.037	0.927±0.001	0.748±0.024
RECA *w/o sub*	0.848±0.009	0.650±0.019	0.936±0.002	0.774±0.011
RECA	**0.853**±0.005	**0.674**±0.007	**0.937**±0.002	**0.783**±0.014

3.5.5 Experiment Results

We conducted a 5-fold cross-validation to evaluate the performance of RECA in comparison with the baselines. The mean F1 scores of the 5-fold are reported.$\pm$denotes the standard deviation recorded for the F1-scores in 5-fold.

As shown in Table 3.4, Sherlock [66] is significantly outperformed by RECA. The relatively poor performance of Sherlock can be attributed to the fact that Sherlock encodes the target column based on the cells in the target column only. Besides, Sherlock encodes the columns with statistical and simple semantic features, which have limited expressiveness to capture the rich semantics in tables.

We further noticed that TaBERT [72] performs poorly for macro average F1 scores on both datasets, which coincides with the experimental results shown by [85]. Besides, its performance for support-weighted F1 scores is also outperformed by RECA. As suggested by [85], the main reason for this phenomenon is that TaBERT focuses on table semantic parsing. The column embeddings generated by TaBERT are more suitable for understanding the alignment between the input text and table schema, while they are less powerful for column type annotation.

We also observed that RECA outperforms the state-of-the-art models TABBIE [74] and DODUO [75] for both support-weighted F1 scores and macro average F1 scores on the two datasets. Specifically, RECA outperforms TABBIE by 6.8% and 11.0% for the support-weighted F1 score and macro average F1 score, respectively, on the Semtab2019 dataset. For the WebTables dataset, RECA outperforms TABBIE by 0.9% and 6.7% for the two F1 scores. Furthermore, RECA achieves performance uplifts of 4.0% and 7.0% for the F1 scores on the Semtab2019 dataset and improvements of 1.0% and 5.5% on the WebTables dataset. Note that both TABBIE and DODUO utilize LMs to process intra-table context while ignoring the inter-table context information when generating the embeddings of the target columns. However, RECA mainly focuses on extracting useful inter-table context information to enhance the embeddings of the target columns. The performance uplifts of RECA over the state-of-the-art models show that the inter-table context information serves as a suitable replacement for the intra-table context information when generating the em-

beddings for the target columns. Besides, by incorporating inter-table context information in replacement of the intra-table context information, RECA can naturally handle wide tables. Furthermore, we noticed that the performance boost for macro average F1 scores is significant, which suggests that RECA is especially good at improving the performance for less-populated semantic types by incorporating useful inter-table context information.

3.5.6 Ablation Study

To better understand how different components of RECA benefit the model in annotating the column semantic type, we conducted an extensive ablation study to evaluate the performance of RECA and its following variants:

- RECA *target only*: The RECA model that only considers the target column without processing related and sub-related tables. This variant degenerates to the basic BERT model that only processes the cells in the target column.
- RECA *w/o re*: The RECA model that only considers the target column and the identified columns in the sub-related tables, without processing related tables.
- RECA *w/o sub*: The RECA model that only considers the target column and the identified columns in the related tables, without processing sub-related tables.

We present the results of RECA *target only* to establish a baseline performance for processing the content of the target column only, so as to analyze the importance of related and sub-related tables in generating target column representations. Then, we present the results of RECA *w/o re* and RECA *w/o sub* in order to see if related and sub-related tables provide useful context information that boosts the annotation quality of RECA.

As shown in Table 3.4, the performance of RECA drops significantly when both related tables and sub-related tables are removed (RECA *target only*). On the Semtab2019 dataset, 5.3% and 13.1% of the performance drops can be observed for the two metrics, while on the WebTables dataset, the F1 scores drop by 2.8% and 12.1%. These drops demonstrate the importance of incorporating inter-table context information when generating target column

representations. To further understand the effects of incorporating related and sub-related tables, we evaluated the performances of RECA *w/o re* and RECA *w/o sub*. Note that the performances of RECA drop slightly when the sub-related tables are removed: 0.9% and 3.6% on the Semtab2019 dataset; 0.1% and 1.1% on the WebTables dataset. But the performance drops are more prominent when the related tables are removed: 2.0% and 4.9% on the Semtab2019 dataset; 1.1% and 4.5% on the WebTables dataset. Since related tables are 'more related' compared with sub-related tables in terms of the similarity in the named entity schema, removing sub-related tables causes less harm to the model performance than removing related tables, as is shown in the experimental results. We further notice that the performance drop on the macro average F1 metric is much larger than that on the support-weighted F1 metric, which suggests that the related and sub-related table information can improve the annotation quality of RECA on the less-populated semantic types.

Table 3.5: Learning Efficiency evaluation on the Semtab2019 dataset and the WebTables dataset.

Datasets	[%]	Support-weighted F1	Macro average F1
Semtab2019	25	0.697±0.041	0.442±0.074
Semtab2019	50	0.792±0.020	0.566±0.045
Semtab2019	75	0.820±0.021	0.631±0.047
Semtab2019	100	**0.853**±0.005	**0.674**±0.007
WebTables	25	0.909±0.002	0.680±0.008
WebTables	50	0.924±0.004	0.738±0.019
WebTables	75	0.930±0.002	0.772±0.013
WebTables	100	**0.937**±0.002	**0.783**±0.014

3.5.7 Efficiency

Learning Efficiency We further evaluated the effectiveness of RECA with less training data. We trained RECA with training sizes ($25\%, 50\%, 75\%, 100\%$ of the full training set). As shown in Table 3.5, when using 75% of the training data, the F1 scores of RECA dropped by only 0.7% and 1.4% compared with training with the full training set on the WebTables dataset, while the F1 scores of RECA dropped by 3.9% and 6.4% compared with training with the full training set on the Semtab2019 dataset. RECA with 75% training data still slightly outperformed the state-of-the-art methods TABBIE [74] and DODUO [75] on two

datasets. While training RECA with 50% of the data, RECA achieves slightly worse performance than the state-of-the-art models TABBIE and DODUO. When RECA is trained with only 25% of the training data, it still outperforms Sherlock [66]. The learning efficiency results show that RECA has the advantage of training a high-performance model with a relatively small amount of training data.

Data Efficiency As suggested by [75], input data efficiency is an important issue worth considering when designing column semantic type annotation models. Due to the maximum sequence length limit of BERT, annotating columns accurately with only a small number of tokens is a crucial factor in evaluating the performance and robustness of column semantic type annotation models.

Table 3.6: Data Efficiency evaluation on the Semtab2019 dataset and the WebTables dataset.

Datasets	Max	Support-weighted F1	Macro average F1
Semtab2019	8	0.540±0.009	0.319±0.010
Semtab2019	16	0.654±0.013	0.436±0.006
Semtab2019	32	0.728±0.010	0.507±0.020
Semtab2019	128	0.816±0.017	0.620±0.033
Semtab2019	256	0.851±0.011	0.662±0.024
Semtab2019	512	**0.853**±0.005	**0.674**±0.007
WebTables	8	0.907±0.004	0.737±0.011
WebTables	16	0.923±0.002	0.762±0.011
WebTables	32	0.931±0.002	0.780±0.010
WebTables	128	**0.937**±0.002	**0.783**±0.014
WebTables	256	0.936±0.003	**0.783**±0.020
WebTables	512	0.936±0.001	0.780±0.011

We conducted experiments on two datasets with different maximum sequence length limits to evaluate the data efficiency of RECA. As shown in Table 3.6, there is not much difference when the maximum sequence length limit is set to 128, 256, and 512 on the WebTables dataset. While the performance dropped by only 0.2% and 1.8% when set from 512 to 256 on the Semtab2019 dataset. We noticed that RECA still outperforms all the state-of-the-art methods on the WebTables dataset even if we reduce the maximum sequence length to 32, while still achieving comparable performance to the state-of-the-art approaches even if we further reduce the maximum sequence length to 16. This phenomenon demonstrates the high data efficiency of RECA. Besides, it also shows the robustness of RECA when

dealing with data with small input size.

3.5.8 Discussion

We analyzed the effectiveness of the column location alignment requirement mentioned in Section 3.4.3 to see if aligning identified columns at the exact same location as the target column is indeed an effective strategy. Following the notations in Section 3.4.3, we compared RECA with the following alignment variants:

- RECA *set*: If not aligned by the exact alignment but if any $\Psi(\hat{e}_k^j) = \Psi(\hat{e}_i^t)$, $j = 1, 2, ..., M_k$ then the j-th column is the identified column. In case multiple j exist, select the one that is closer to t.
- RECA *prox*: If not aligned by the exact alignment but $\Psi(\hat{e}_k^{t-1})$ $= \Psi(\hat{e}_i^t)$ or $\Psi(\hat{e}_k^{t+1}) = \Psi(\hat{e}_i^t)$, then the t-1-th column or t+1-th column is the identified column. In case a tie happens, regard both columns as the identified columns.

Table 3.7: Alignment strategies evaluation on the Semtab2019 dataset and the WebTables dataset.

Datasets	F1	RECA *set*	RECA *prox*	RECA
Semtab2019	S	0.842±0.008	0.843±0.004	**0.853**±0.005
	M	0.660±0.014	0.663±0.009	**0.674**±0.007
WebTables	S	0.934±0.003	0.935±0.004	**0.937**±0.002
	M	0.778±0.016	0.780±0.017	**0.783**±0.014

S stands for support-weighted F1 scores; M stands for macro average F1 scores.

As shown in Table 3.7, the support-weighted F1 score of RECA drops by 1.2% and 1.3%, while the macro average F1 score drops by 1.6% and 2.1% when applying RECA *prox* and RECA *set* alignment strategies, respectively on the Semtab2019 dataset. Slight performance drops can also be observed on the WebTables dataset. We noticed that the performance drop when applying the RECA *set* alignment strategy is larger than applying the RECA *prox* alignment strategy, which is understandable since the requirements of the RECA *prox* strategy are stricter than that of the RECA *set* strategy: the shift of the identified column index from the target column index cannot be larger than 1. We attribute the

drop in performance to the reason that by applying alignment strategies that are 'looser', some noisy columns that should not be aligned may be included as the inter-table context and thus mislead the model. To this end, we conclude that the column location alignment requirement is necessary since it considers the inner schema structures of two tables. The similarity of the inner schema structures should be considered when designing the column alignment strategy for RECA.

3.5.9 Parameter Sensitivity

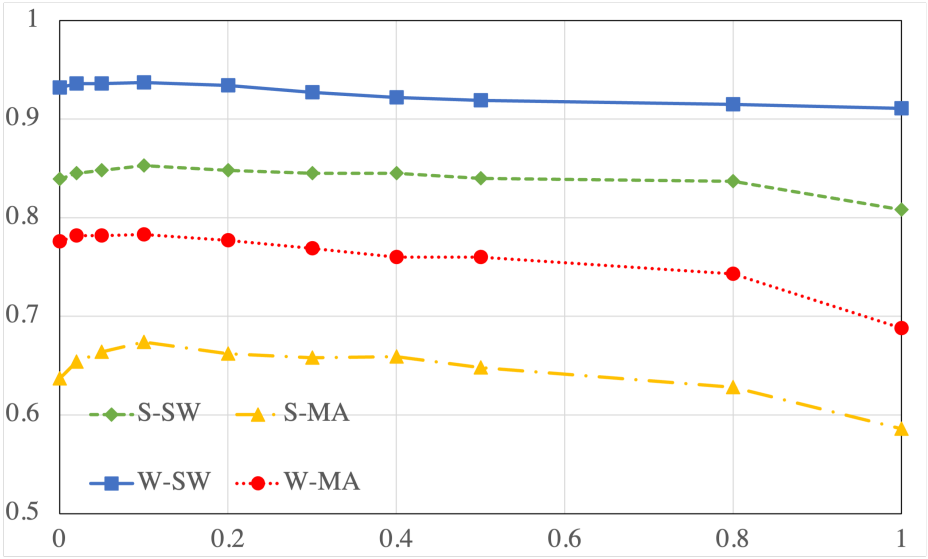


Figure 3.5: Sensitivity analysis of RECA on different values of the Jaccard threshold δ. The x-axis denotes the values of δ. The y-axis denotes the values of F1 scores. S-SW and S-MA stand for the support-weighted and macro average F1 scores on the Semtab2019 dataset; W-SW and W-MA stand for the support-weighted and macro average F1 scores on the WebTables dataset.

We analyzed the impact of the key parameter of RECA: the Jaccard threshold δ in Section 3.4.2. δ is chosen from $\{0, 0.02, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.8, 1\}$. Figure 3.5 presents the results. We notice that the performance of RECA increases as we increase δ from 0 to 0.1 since larger δ filters out the topically irrelevant inter-table context and thus reduces the noise. The performance decreases as we further increase δ to 1 because if the Jaccard threshold δ is too large, useful inter-table context is filtered out. We observe that the performance is generally stable for δ in the range of [0, 0.3]. In practice, we suggest finding the optimal values of δ based on the dataset.

3.5.10 Qualitative Evaluation

To better understand how inter-table context helps improve the annotation quality of column semantic types, we further conducted a qualitative evaluation of RECA and its state-of-the-art counterparts.

On the top left corner of Figure 3.6, we present Table #8 in the testing set of the Semtab2019 dataset. This table is about the rankings of films, the year of production, and the corresponding directors of the films. The second column is the target column in this table, and the ground truth label is **Film**. The intra-table content could be misleading since other similar semantic types describing WORK_OF_ART, such as **TelevisionEpisode** and **Musical**, can have very similar table content. TABBIE [74] and DODUO [75] are likely to be influenced by the intra-table context information since both of them annotate it as **TelevisionEpisode**. However, RECA finds the related tables and sub-related tables of the target tables and processes the identified columns as inter-table context information to provide more information for the semantic type of the target column. As shown in Figure 3.6, the bottom left shows one of the related tables, and the right shows two of the sub-related tables. We notice that the identified columns in the related tables and sub-related tables are all film names. By incorporating more film names as context information to RECA, the chance that it correctly annotates the target column as **Film** instead of other WORK_OF_ART types increases. Therefore, the inter-table context information can serve as an enhancement for the content in the target column. At the same time, it eliminates noisy information brought by the context in the target table.

3.6 Conclusion

In this chapter, we propose RECA, an inter-table context enhanced column semantic type annotation framework based on pre-trained language models. Extensive experiments on two web table datasets demonstrated the effectiveness of RECA in providing high-quality annotations for table columns. Specifically, RECA achieves new state-of-the-art performance on the two datasets. Through our evaluation, we also confirm that the inter-table

Fans' Rank	Title	Year	Director(s)	Overall Rank
201	The Grand Illusion	1937	Jean Renoir	90
202	Eternal Sunshine of the Spotless Mind	2004	Michel Gondry	176
203	Cinema Paradiso	1988	Giuseppe Tornatore	192
204	American Graffiti	1973	George Lucas	198
205	Traffic	2000	Steven Soderbergh	713
...	...	...	...	...

N	Title	Year	Director	IMDB Rating	IMDB Votes
1	The Social Network	2010	David Fincher	9.3	45993
2	Inception	2010	Christopher Nolan	8.9	333261
3	Alice in Wonderland	2010	Tim Burton	6.6	84086
4	Shutter Island	2010	Martin Scorsese	8.0	160301
5	How to Train Your Dragon	2010	Dean DeBlois	8.2	68159
...	...	...	...	...	

Rank	Title	Year	Director(s)	Overall Rank
1	Singin' in the Rain	1952	Stanley Donen and Gene Kelly	6
2	Some Like It Hot	1959	Billy Wilder	10
3	Dr. Strangelove or: How I Learned to Stop Worrying and Love the Bomb	1964	Stanley Kubrick	12
4	Annie Hall	1977	Woody Allen	25
5	The Graduate	1967	Mike Nichols	39
...	...	...	...	...

Date	Movie	Year	Director
2012-01-23	Blue Valentine	2010	Derek Cianfrance
2012-01-21	Dogtooth	2009	Giorgos Lanthimos
2012-01-20	Serendipity	2001	Peter Chelson
2012-01-19	Tinker Tailor Soldier Spy	2011	Tomas Alfredson
2012-01-15	Transit	2012	Antonio Negret
...	...	...	...

Figure 3.6: Top left is the target table, bottom left is one of the exemplars related tables, and the two tables on the right are the exemplar sub-related tables. The target column and the corresponding identified columns are marked in yellow. Note that the cells marked in grey are masked out from the model input during experiments. For ease of demonstration, we only present one related table and two sub-related tables in the figure.

context information can be a powerful replacement for the intra-table context information, thus giving the model the ability to annotate wide tables. Furthermore, we have shown that RECA is learning efficient and input data efficient. As a future direction, we would like to explore the effectiveness of inter-table context in other table understanding tasks, such as column relation prediction and row population.

CHAPTER 4

KERAG: KNOWLEDGE-ENHANCED RETRIEVAL-AUGMENTED GENERATION FOR ADVANCED QUESTION ANSWERING

Chapter 4 introduces KERAG, a novel KB-based RAG approach focusing on improving on existing KBQA approaches. Recently, Retrieval-Augmented Generation (RAG) becomes a focus for both academia and industry as a method to mitigate hallucination in Large Language Models (LLMs) by incorporating external data, with Knowledge Bases (KBs) offering crucial information for question answering. Traditional Knowledge Base Question Answering (KBQA) methods rely on semantic parsing, which typically retrieves knowledge strictly necessary for answer generation, thus often suffer from low coverage due to rigid schema requirements and semantic ambiguity. We present KERAG, a novel KB-based RAG pipeline that enhances QA coverage by retrieving a broader sub-graph likely to contain relevant information. Our retrieval-filtering-summarization approach, combined with fine-tuned LLMs for Chain-of-Thought reasoning on knowledge sub-graphs, reduces noises and improves QA for both simple and complex questions. Experiments demonstrate that KERAG surpasses state-of-the-art solutions by about 7% in quality and exceeds GPT-4o (Tool) by 10-21%.

4.1 Introduction

Retrieval-Augmented Generation (RAG) has been extensively explored recently in both academia and industry, as a means to mitigate the hallucination issues of Large Language Models (LLMs) [5, 6, 7, 8]. RAG enhances LLM outputs by incorporating relevant data from external sources, thereby grounding the generated content with factual knowledge. Knowledge graphs serve as a crucial data source, providing efficient access to rich and precise information for question answering [102, 103].

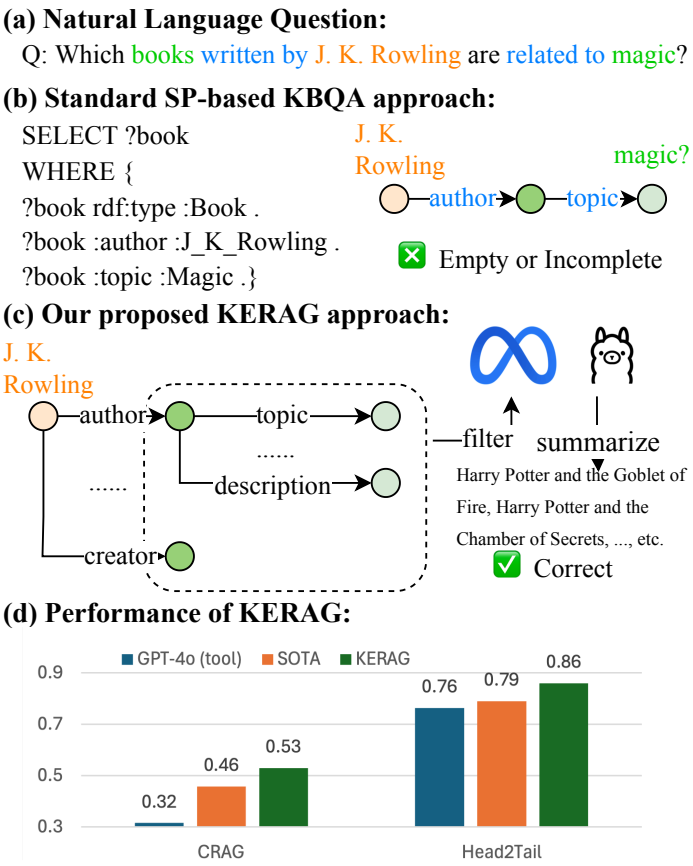


Figure 4.1: (a) An exemplar query. (b) On the left, we show the SPARQL query generated by the triple-level SP-based approach. On the right, we display the linked KB content. This design leads to empty or incomplete answers since: 1) The book entities do not necessarily have an attribute named "topic"; 2) "magic" may not be explicitly presented in the topic attributes, but may be embedded in other attributes such as description. (c) The rationale behind our KERAG approach: entity-level SP with LLM-based post-processing of the retrieved KB content. (d) Trustfulness of KERAG against state-of-the-arts (trustfulness=accuracy-hallucination).

Knowledge Graph Question Answering (KGQA) is a well-established field that has been extensively studied long before the advent of LLMs. Most solutions utilize the technique of *semantic parsing*, where a natural language question is converted into a structured query, in the form of either a SPARQL query [104, 105, 106, 107, 108, 109, 110, 111], or a retrieval path that first identifies the queried entity and then follows a particular predicate to a neighboring entity or attribute value as the answer [112, 113, 114, 3]. When successful, these solutions produce precise and concise answers. However, they require rigorous schemas (ontologies) and have low tolerance to parsing errors (see Figure 4.1 for an example). As a result, whereas KGQA generally provides highly accurate answers, it often suffers from low coverage, missing answers for many questions. With LLMs emerging, re-

cent works [115, 116] leverage the reasoning ability of LLMs to generate *multiple* retrieval paths; however, the small number of paths (e.g., up to 3 in [116]) does not fully address the retrieval recall issue.

LLMs, with their strong summarization and reasoning capabilities, offer a new perspective to pursue KGQA and allow us to push *reasoning* from retrieval-path generation to the summarization step. At retrieval time, instead of following up to a few retrieval *paths*, we gather from the neighborhood of the topic entity *all* information potentially relevant to the question. At answer generation time, we leverage LLM's reasoning capabilities to identify relevant information and generate accurate answers. By focusing on the whole neighborhood rather than a few KG paths, the complexity of generating logical forms for KG querying is significantly reduced and the retrieval recall can significantly increase.

Although the idea appears straightforward, there are three prominent challenges. First, entities, particularly head entities, can be associated with a substantial amount of knowledge (2 million triples per head entity [117]). All knowledge may not fit within the context window, and even if it does, the noises can lead to confusion during summarization. Second, some questions require information from multi-hop neighbors, complicating the determination of retrieval boundaries, and further increasing the volume of retrieved information. Lastly, complex questions that necessitate aggregation and reasoning remain challenging.

We proposed KERAG, a novel Knowledge-Enhanced RAG pipeline that employs a *retrieval-filter-summarization* paradigm. The *retrieval* part gathers information about the topic entity mentioned in the question; the *filtering* part eliminates information irrelevant to the question; and the *summarization* part reasons over the remaining information to generate the answer. Rather than retrieving the entities and triples strictly necessary for generating an answer, our method retrieves a broader subgraph that is likely to contain relevant information for answer generation.

There are two key innovations that ensure high-quality answers. First, in our planning phase we interleave multi-hop neighborhood retrieval with filtering according to the KG schema, allowing us to filter at the predicate level rather than overwhelming the summarizer with irrelevant triples, and meanwhile ensuring that all information necessary for question

answering is retrieved from the neighborhood. Second, we fine-tune the LLM for Chain-of-Thought (CoT) reasoning over the neighborhood sub-graph, enabling it to generate accurate answers for both simple questions and complex questions. An additional advantage of our method is its applicability not only to graph databases that support SPARQL, but also to API-based knowledge-providing systems, which are commonly used in production [6].

To summarize, we make three contributions.

- We introduce a novel KG-based RAG pipeline, KERAG, which retrieves information at the entity level, rather than at the triple level as in traditional semantic parsing solutions. This approach significantly enhances QA coverage.
- We design a retrieval-filter-summarization paradigm, complemented by fine-tuning LLMs for Chain-of-Thought-based summarization. This solution reduces the noises in summarization and improves answer generation for both simple and complex questions.
- We conducted extensive experiments on two KG benchmarks (API-based and SPARQL-based), showing that KERAG outperforms state-of-the-art solutions by 7% in quality, and outperforms GPT-4o (Tool) by 10-21%.

4.2 Related Work

One major line of research of KGQA is semantic-parsing-based (SP-based) KGQA approaches, which primarily transform natural language queries into logical forms and initiate structured queries to access the knowledge graph content. The SP-based KGQA approaches can be further divided into multi-step and seq2seq approaches. The multi-step approaches generally formulate the semantic parsing problem as a multi-step search problem: identifying the core entity, generating and expanding the query graph based on the relationships between the attributes of the entities and the predicates of the queries [114, 118, 119]. The seq2seq approaches primarily consider the semantic parsing problem in a seq2seq manner with the help of fine-tuned language models: directly generating the complete semantic expression based on the query and retrieving the relevant knowledge graph con-

tent [105, 107, 108, 109, 3, 111, 120, 110]. Specifically, Pangu [120] presents an innovative framework for grounded language understanding that integrates a symbolic agent with a neural language model. This system enables the stepwise formulation of valid plans while employing the language model to evaluate the viability of these logical plans in an online manner. WikiSP is a novel few-shot seq2seq semantic parser developed by [110], which fine-tunes a Llama model on modified SPARQL queries.

Another line of research in KGQA is the information-retrieval-based KGQA approaches, where a subgraph is generated based on the entities and relations in the query, graph matching and retrieval techniques are applied to retrieve relevant graphs from the KG to answer the query [121, 122, 123, 124, 125, 126]. For instance, [126] introduces a novel approach with a trainable subgraph retriever (SR) designed to operate separately from the reasoning process, identifying relevant subgraphs and prioritizing smaller and more relevant subgraphs. The subgraphs identified by our approach are more relaxed compared to [126], which provide a broader context for the summarizer to generate accurate answers.

Recently, the KGQA community focuses on LLM-based approaches, which utilize large language models (LLMs) to perform KGQA [115, 116, 127]. As for the LLM-based KGQA approaches, several state-of-the-art LLMs such as Llama-3.1s [128] support tool calling, which provides the opportunity to directly prompt the LLMs to generate API calls or SPARQL queries. However, retrieving directly with the tool call plan given by the LLM would propagate planning errors into the retrieval process, resulting in insufficient KG content being retrieved. We provide a comparison of performance between our KERAG pipeline and the Llama-3.1-70B-Instruct (tool) model (which can freely plan and initiate multiple rounds of API calls) in Tables 4.3 and 4.4.

Beyond naive LLM tool calling, ToG [115] introduces beam search on the knowledge graph to iteratively explore reasoning paths for LLMs. ToG-2 [116] links to external Wikipedia pages to get additional context, the retrieval and reasoning are conducted between KG and external pages. StructGPT [127] designs an interface for data retrieval and a unique process that involves invoking, linearizing, and generating outputs. Our method is fundamentally different: these approaches retrieve information from KGs based on re-

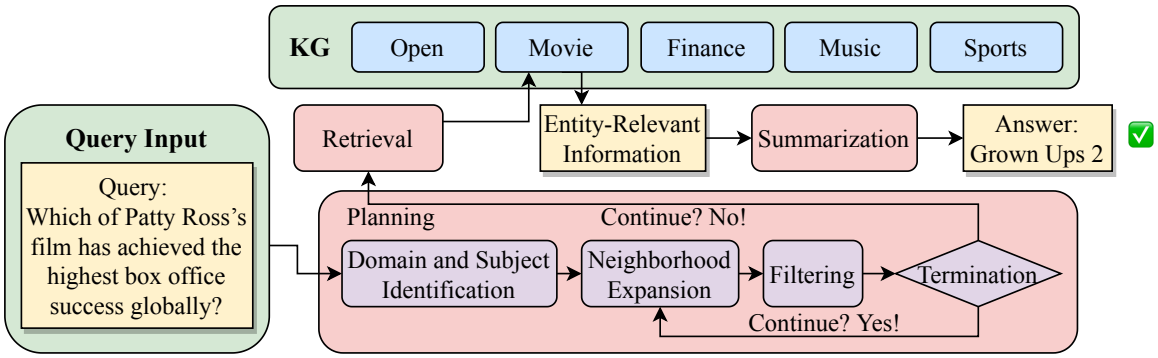


Figure 4.2: The overall pipeline of our KERAG approach.

trieval paths, whereas KERAG retrieves all potentially relevant information in a neighborhood, thus improving on the retrieval recall (~11% over ToG). As such, KERAG achieves a higher accuracy on all benchmarks, improving over ToG and ToG-2 by up to 20% and 4% respectively (Table 4.5), and outperforming StructGPT by 7% in truthfulness (Table 4.4). Additionally, KERAG has the flexibility to apply both on KGs and APIs, allowing experiments on datasets like Head2Tail [129] and CRAG [6].

4.3 Problem Statement

Knowledge Base: A *Knowledge Base (KB)* K can be defined as $K = (E, R, D)$, where E is a set of *entities*, R is a set of *relations* and optionally D is a set of *domains* covered by K. Each entity $e_i \in E$ can have properties. Each relation $r_j \in R$ can be represented as a binary relation $r_j : E \times E \to \mathbb{B}$. A KB is normally accompanied with an *ontology*, describing the entity types and relationships between different types of entities.

KB-based RAG: Given a natural language question Q with a KB K, *KB-based RAG* aims to answer the question Q leveraging knowledge in K.

4.4 Methodology

4.4.1 Solution overview

We start with an overview of our solution. Given a question Q and a knowledge base K, KERAG retrieves knowledge from K and uses it to generate the answer for Q. As illustrated in Figure 4.2, the pipeline contains three major steps.

1. **Planning:** The *Planning* step formulates a retrieval plan by selecting a subject entity E as a starting point and determining the scope of its neighborhood for retrieval. The scope includes the number of hops h for neighborhood exploration and relations $\bar{R}$ to filter.

2. **Retrieval:** The *Retrieval* step translates the retrieval plan to KB queries or API calls to retrieve relevant knowledge.

3. **Summarization:** Finally, the *Summarization* step reasons over the retrieved knowledge and produces the final answer.

The Retrieval step is fairly easy and varies based on the underlying KB system. In the rest of the section, we describe the planning and summarization steps in detail, with a focus on how they address the three main challenges: managing *knowledge overloading*, determining *multi-hop retrieval boundaries*, and handling *complex questions*.

4.4.2 Planning

The planning step takes a question Q as input, and decides a retrieval plan as a quadruplet $(D, E, \bar{R}, h)$, where D denotes the domain, E denotes the topic entity as the starting point, $\bar{R}$ denotes the relations to filter, and h denotes the number of hops to explore in the neighborhood.

We distinguish the retrieval plan from a retrieval query (e.g., SPARQL) or a retrieval path generated by traditional semantic parsing. Normal retrieval queries return a particular node (or a set of nodes) strictly necessary to answer the question. Our retrieval plan, instead,

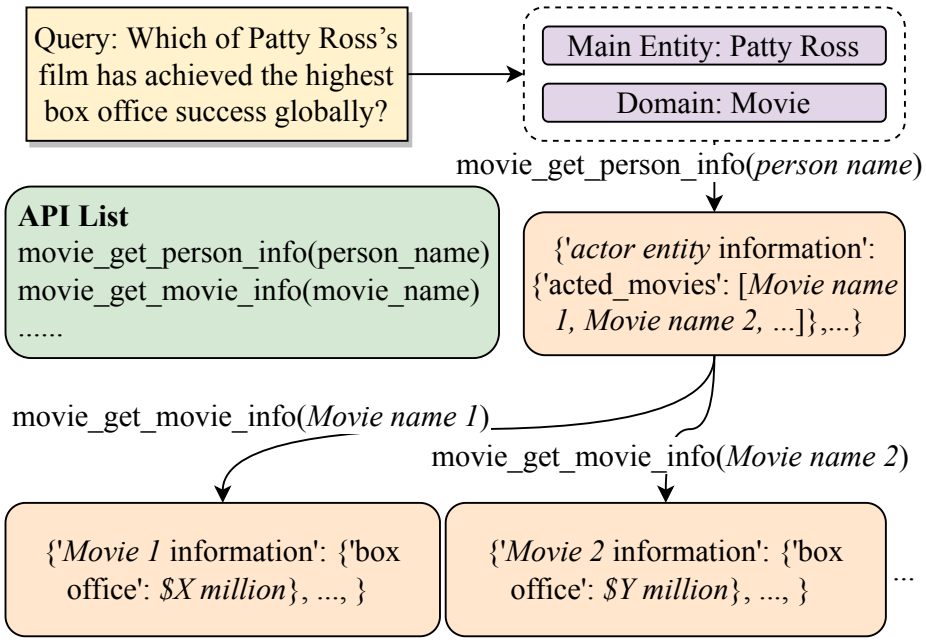


Figure 4.3: An exemplar query illustrating the planning step.

returns a neighborhood scope that might be needed for answering a question; only relations that are absolutely irrelevant to a question are excluded (e.g., *Oscar awards* for a question regarding movie *box office*).

Our planning proceeds in four steps.

- *Step 1:* We start with prompting an LLM to decide the main topic entity E and the domain D from the input question Q. Consider an example in Figure 4.3, *"Which of Patty Ross' s film has achieved the highest box office success globally?"*. The topic entity is *Patty Ross* and the queried domain is *Movie*.

- *Step 2:* Knowing the type of the topic entity, we leverage the schema to explore one hop in the neighborhood, identifying the relations and the types of the neighbor entities. From our example entity *Patty Ross*, the neighborhood includes relations such as acted movies, birthday, etc., with neighbors of types *entity (movie)*, *attribute (date)*, etc.

- *Step 3:* We feed the neighborhood information into an LLM to decide which relations are irrelevant, and whether enough information has been collected to answer question Q. For example, facts regarding *birth_date* and *birth_place* are irrelevant to answer

the question; just knowing the movie neighbors without further information about these movies is not enough to answer the question.

- *Step 4:* We terminate if LLM decides that there is enough information to answer the question, updating h (#hops) accordingly; otherwise, we go to Step 2 to explore the next hop of the neighborhood, unless we have exhausted relevant relations or reached a predetermined maximum number of hops. In our example, we will continue to retrieve information about movies necessary to answer the question.

Next, we describe each step in more detail.

Domain and Subject Identification: We leverage LLMs for this step. We apply few-shot learning and include necessary schema information to guide LLM, including the explanation and examples of domains. We use the following template.

Planning Prompts Skeleton

You are given a Query [Q], please extract the main entity [E] from the Query.
Determine the domain the query is about. The domain should be one of the following: [Set of KB domains].
[Examples] (Optional)
[Further Instructions] (Optional)

Neighborhood Expansion: This step simply explores the schema of the knowledge base, to understand relations and entity types in the next hop of the neighborhood to prepare the next step.

Filtering: This step decides 1) which relations in the obtained neighborhood content are absolutely irrelevant to answer the question Q thus can be filtered, and 2) whether we have obtained enough information for question answering, addressing both the *knowledge overloading* challenge and *multi-hop boundary* challenge. We prepare both LLM-based filtering and similarity-based filtering [130] to filter according to the semantics of the question and the ontology. We apply the following LLM-based filtering template:

Filtering Prompts Skeleton

Given the user query [Q] with detected main entity [E] and the current [retrieval plan], you will also have access to the following functions [a set of functions/predicates with ontology] to perform additional hops based on the current plan.
Decide which functions/predicates are relevant to answer the query based on the ontology and if the composed retrieval plan sufficient for the question answering.
[Examples] (Optional)
[Further Instructions] (Optional)

Termination: This step takes the LLM output to decide whether to continue planning or terminate. Once the LLM decides all necessary information is collected for question answering, we can proceed to the retrieval step.

Our planning step has the following advantages. First, we enlarge the scope of retrieval compared to traditional semantic parsing, thus allowing us to lift the retrieval recall. Second, we allow multi-hop retrieval, such that we are able to answer complex questions regarding multi-hop reasoning. Third, we apply filtering to remove irrelevant content to avoid distracting the final summarization step. Finally, hopping and filtering decisions are made according to the ontology, significantly reduced the cost for full data retrieval (in case ontology is absent, we can easily adjust the algorithm to retrieve and reason about the data in each hop of neighborhood). On the CRAG dataset, multi-hop retrieval improved truthfulness by 5.5% and filtering improved by 3.9% (Table 4.6).

4.4.3 CoT Summarization

The final *Summarization* step takes the question Q and the retrieval results, applies CoT reasoning to generate the answer. The key challenge is to answer complex questions leveraging the retrieved knowledge, oftentimes still in large volume.

Complex questions, such as aggregation questions (e.g., *What is the total points scored by the New Orleans Pelicans in 2022-01?*) and reasoning questions (e.g., *Is memory management in Python handled through garbage collection?*), often require cross-referencing multiple pieces of KB information. We employ CoT prompting for the reasoning, the rough template is as follows.

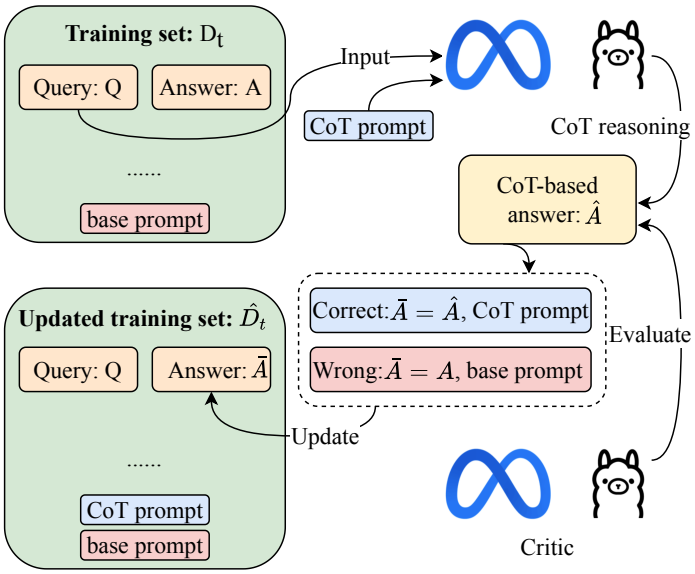


Figure 4.4: Generation of the fine-tuning data.

Summarization Prompts Skeleton

Please provide a brief answer to the question [Q] based on your knowledge and the following content. Answer "I don't know" if you are not confident of your answer. Please think step by step.
[Retrieved Content (Triple form)]

To further enhance performance, we fine-tune the CoT-based summarizer. Figure 4.4 describes how we generate the fine-tuning data with reasoning.

1. For each query Q in the training set, we first prompt an LLM model to generate CoT-style reasoning together with the answer $\hat{A}$.

2. We then prompt LLM to decide the correctness of $\hat{A}$ by comparing with the ground truth answer A.

3. Based on the correctness judgment, we enhance the training data as follows. If the generated answer $\hat{A}$ is correct, we will take $\hat{A}$ together with the generated reasoning as the ground truth; otherwise, we retain the original ground truth and replace the CoT prompt with stand non-CoT prompt.

4. The queries, the updated prompts, and the updated answers formulate our updated training set for Supervised Fine Tuning (SFT).

Our proposed CoT-based summarizer together with the fine-tuning data generation scheme is the biggest contributor to the end-to-end answer quality improvement. On the CRAG dataset, using plain CoT reasoning refrained from generating hallucinated answers for 18% of questions (mostly saying "I don't know" instead), and our fine-tuning further increased accuracy by 10% (Table 4.6).

4.5 Experiments

4.5.1 Datasets

To evaluate the performance of our method, we consider the widely used CRAG [6] benchmark, where the KBs are accessed through API functions. The questions in CRAG can be divided into KB and web questions based on the data source. We preserve the KB questions in CRAG for conducting experiments.

We further adapted our method to the Open domain question set of the Head-to-tail benchmark (Head2Tail) [129], where the DBPedia KB is utilized, and the content is accessed through SPARQL queries to showcase the generalizability of our approach under different KBQA settings.

Both the CRAG and Head2Tail datasets categorize the questions into head, torso, and tail categories based on the popularity of the main entity in the questions. We preserve this categorization.

We adopt the public validation set of CRAG as the training set and use the private testing set as the testing set. The statistics of training and testing sets in the CRAG KB subset is presented in Table 4.1.

Table 4.1: Training and testing sets statistics of CRAG.

	Training	**Testing**
head	219	187
torso	197	203
tail	188	188
overall	604	578

Table 4.2: statistics of each question type in different question splits

	head	**torso**	**tail**
open	48	48	48
finance	125	134	112
movie	121	120	118
music	35	41	14
sports	77	57	84
simple	205	218	187
simple_w_condition	59	43	57
set	19	12	13
comparison	41	44	34
aggregation	33	35	38
multi-hop	14	13	11
post-processing	6	6	6
false premise	29	29	30
real-time	78	85	72
fast-changing	67	59	76
slow-changing	46	51	45
static	215	205	183

The CRAG benchmark includes KBs from open, movie, finance, sports, and music domains. The question types include simple, simple with condition, set, comparison, aggregation, multi-hop, post-processing, and false premise questions. There are also four dynamism considered, including real-time, fast-changing, slow-changing, and static questions. A detailed statistics about each question type is presented in Table 4.2.

The Open domain question set of the Head2Tail benchmark consists of in total 9219 questions, each of the head, torso, tail data split consist of 3073 questions. We randomly sampled 375 questions from each data split to formulate the testing set, which guarantees a margin of error of less than 5%. The rest of the Head2Tail data is treated as training set.

We also included other SPARQL-based datasets used by the state-of-the-art approaches: QALD-10-en [131], WebQSP [104], and AdvHotpotQA [132].

4.5.2 Baselines

We compare our approach against the following baselines on both the CRAG and Head2Tail datasets.

- *Base models:* GPT-4o [133], Llama-3.1-70B-Instruct [128], GPT-4o (tool), and Llama-3.1-70B-Instruct (tool). The GPT-4o and Llama-3.1-70B-Instruct models, along with their tool-use variants, serve as representatives for closed-source and open-source LLM solutions.

- *KDD Cup winners:* For CRAG, we include the top-2 KDD Cup CRAG winning solutions, db3 [134] and apex [135], which are state-of-the-art solutions on the CRAG benchmark.

- *Other state-of-the-art models:* We consider the following state-of-the-art methods on the SPARQL-based Head2Tail dataset: WikiSP [110], StructGPT [127], and ToG [115]. Note that since CRAG does not support SPARQL querying, so we ran them only on Head2Tail. We further compared with the state-of-the-art results mentioned in ToG-2 [116] on QALD-10-en, WebQSP, AdvHotpotQA, and CWQ datasets. We did not evaluate ToG-2 on Head2Tail because ToG-2 is designed based on Wikidata and requires external pages to provide context, which is not compatible with the DBPeida KG used by Head2Tail. Besides, the extra information introduced by external pages would influence fairness of evaluation.

Finally, for fair comparison with KDD Cup winning solutions, which may use 8B models, we also included KERAG (8B), where all modules are developed based on Llama-3.1-8B-Instruct.

4.5.3 Evaluation

To evaluate the performance of the approaches, we adopted a similar auto-evaluation template used by CRAG [6]. The Llama-3.1-70B-Instruct model is employed as the critic. Following the settings in CRAG, we classify the answers into three categories: accurate, missing, and hallucination, corresponding to the cases where the methods respond correctly, with "I don't know," and incorrectly, respectively. We report the accuracy (A), miss rate (M), hallucination rate (H), and additionally the *truthfulness* score (T), which is defined as $T = 1 * A + 0 * M - 1 * H$, penalizing the cases where the methods hallucinate.

For QALD-10-en, WebQSP, and AdvHotpotQA datasets, we align with their respective evaluation methods and metrics.

4.5.4 Configurations

All experiments were performed on four NVIDIA A100 GPUs (80GB).

For the CRAG benchmark, the Llama-3.1-70B-Instruct model is used as the base model for planning. We adopted FSDP [136] and LoRA [137] to efficiently fine-tune the summarizer, which is based on the Llama-3.1-8B-Instruct model. The temperatures are set to 0 for all modules. The summarizer is fine-tuned for 20 epochs (in total 6 GPU hours), with a learning rate of 5e-5.

For the configurations of KERAG on Head2Tail dataset, Llama-3-8B-Instruct is the base model used for planning, and summarization. Chain-of-Thought reasoning and summarizer finetuning are not applied to the Head2Tail dataset. Temperatures of the models are set to 0 for stable generation of the answer. We include a 3-gram similarity based method to first identify the potential (entity, predicate) pairs (maximum three candidates per query) based on the entity and relation lists in Head2Tail and then apply the Main Entity Extraction prompts to identify the main entity of the query. Since there is only one DBPeida KB in Head2Tail and the questions in Head2Tail are not time-sensitive, we don't need to perform domain and time information extraction. The similarity-based filtering is applied. Specifically, we use DPR [130] to compute and rank the similarity between the candidate predicates in the retrieved triples and the query with the main entity removed. Top-30 predicates are preserved in the retrieval step for the summarizer to generate responses.

The configurations of KERAG on the QALD-10-en, WebQSP, and AdvHotpotQA CWQ datasets are as follow: We use Llama-3-8B-Instruct as the base model for the planning step. The llama-7b-wikiwebquestions-qald7 model[1] [110] is used to aid the process of main entity extraction. The plain GPT-4o model is used as the summarizer of the model. Chain-of-Thought reasoning and summarizer finetuning are not applied. Similarity-based DPR filtering is used to filter irrelevant predicates. The planning and summarization

[1]https://huggingface.co/stanford-oval/llama-7b-wikiwebquestions-qald7

prompts are similar to those used on Head2Tail.

4.5.5 Experimental Results

We present the experimental results for the CRAG dataset in Table 4.3. We first notice that KERAG outperforms all baselines in CRAG, with an overall improvement of 7.1% in terms of the truthfulness score. Specifically, we observe that the LLM baselines (GPT-4o and Llama) perform poorly (accuracy < 40%) on the CRAG benchmark, which highlights the difficulty of the questions. In addition, although the tool-use variants of LLM have slightly lower hallucination rates, they fail to achieve high accuracy and have >55% missing rates. We attribute the high missing rate of tool-use variants to the fact that direct function chaining is likely to propagate planning errors to the retrieval stage, leading to low recall of relevant KB content. The KDD Cup winning solutions significantly outperformed baseline LLM solutions, obtaining higher accuracy (40-65%), at the price of much higher hallucination rate (~20%). Our KERAG solution achieves much higher accuracy (>70%), with very minor increase of hallucination rate; showcasing the superiority of multi-hop retrieval and filtering design by providing high recall of relevant content, thereby reducing the miss rate. Moreover, the overall performance difference between KERAG (8B) and KERAG is below 3%, demonstrating the robustness of our pipeline design.

We observed similar trends on the Head2Tail benchmark, shown in Table 4.4. First, KERAG outperforms all the baselines in terms of overall accuracy and truthfulness scores, in particular significantly outperforming GPT-4o and Llama-3.1-70B-Instruct across all data splits. Furthermore, although we allow LLMs to write SPARQL queries to access the knowledge base (tool variants), KERAG still shows overall trustfulness improvements of 10-15% compared to GPT-4o (tool) and Llama-3.1-70B-Instruct (tool). Additionally, KERAG outperforms WikiSP and StructGPT by ~8% on truthfulness scores, with lower hallucinations. We report that the retrieval recall of our approach is 0.952, while the recall of ToG is 0.844, validating the advantages of our neighborhood-based exploration against the path-based exploration of ToG. Finally, we observed 91% accuracy on this dataset, which mainly contains simple questions, showing that KERAG has mostly conquered simple-

Table 4.3: KERAG improves over state-of-the-art by 7.1% on CRAG benchmark.

Model	Accuracy	Hallucination	Miss	Trustfulness
GPT-4o	0.341	0.090	0.569	0.251
Llama	0.306	0.080	0.614	0.227
GPT-4o (tool)	0.362	**0.047**	0.592	0.315
Llama (tool)	0.220	0.057	0.723	0.163
apex [135]	0.652	0.194	0.154	0.458
db3 [134]	0.510	0.173	0.317	0.337
KERAG (8B)	0.713	0.208	0.080	0.505
KERAG	**0.732**	0.202	**0.066**	**0.529**

Table 4.4: KERAG improves over state-of-the-art by 7% on Head2Tail benchmark.

Model	Accuracy	Hallucination	Miss	Trustfulness
GPT-4o	0.502	0.160	0.338	0.342
Llama	0.452	0.163	0.385	0.290
GPT-4o (tool)	0.799	0.035	0.166	0.764
Llama (tool)	0.752	**0.031**	0.217	0.721
WikiSP [110]	0.858	0.066	0.076	0.782
StructGPT [127]	0.895	0.105	**0.000**	0.790
ToG [115]	0.833	0.063	0.104	0.770
KERAG	**0.908**	0.049	0.043	**0.860**

Table 4.5: Experimental results on QALD-10-en, WebQSP, and AdvHotpotQA datasets. The prior SOTA include the best known methods on each dataset: α: [2]; β: [3]; γ: [4]. The baseline performance is directly taken from ToG and ToG-2. ToG-2 additionally links to external Wikipedia pages to get context information.

Model	QALD-10-en	WebQSP	AdvHotpotQA
Prior SOTA	0.454^{α}	0.821^{β}	0.354^{γ}
ToG	0.502	0.762	0.263
ToG-2	0.541	0.811	0.429
KERAG	**0.558**	**0.843**	**0.461**

question answering.

We further present the results on QALD-10-en, WebQSP, and AdvHotpotQA datasets in Table 4.5: our approach outperforms SOTA methods on QALD-10-en, WebQSP, and AdvHotpotQA datasets.

Table 4.6: Ablation Study on CRAG

Model	Accuracy	Hallucination	Miss	Trustfulness
KERAG	0.732	0.202	0.066	0.529
- multihop	0.587	0.112	0.301	0.474
- filter	0.721	0.232	0.047	0.490
- finetuning	0.626	0.173	0.201	0.453
- CoT	0.649	0.348	0.003	0.301

4.5.6 Ablation Study

We conducted an ablation study to explore the effectiveness of each component in KERAG and present the results in Table 4.6: 1) -multihop: considers only one-hop neighbors in the retriever; 2) -filter: without performing the filtering step; 3) -finetuning: without fine-tuning the CoT-based summarizer; 4) -CoT: replaces the CoT-based summarizer with a plain tuned summarizer.

We observed that the overall truthfulness of these four variants decreases relative to KERAG by 7.4%, 10.4%, 14.4%, 43.1%. As expected, multi-hop retrieval increased missing rate by 25%, with the price of increased hallucination rate (by 9%). Filtering reduced hallucination rate by removing noises in the retrieval results. The Chain-of-Thought prompting of the summarizer and the fine-tuning process improved the quality most, increasing accuracy by 9% and reducing hallucinations by 15%; fine-tuning played an important role in reducing missing answers, especially for complex questions. Combining the four designs maximizes the performance of KERAG.

We further explored the effect of multi-hop retrieval on KBs by comparing the performance difference of our multi-hop retrieval strategy for one-hop, two-hop, and even n-hop on CRAG[2]. We present the results in Figure 4.5. We observe that the accuracy increased significantly (15%) as we move from one-hop to exhaustive retrieval; correspondingly, missing rate reduced significantly from 30% for one-hop to 7% for exhaustive. On the other hand, we also observe an increased hallucination rate from one-hop to two-hop (9%), brought by retrieval noises, thereby validating the necessity of a proper content filter and a summarizer with high reasoning ability in the pipeline.

[2]In our case, the API schema of CRAG limits the maximum number of hops for API chaining to three.

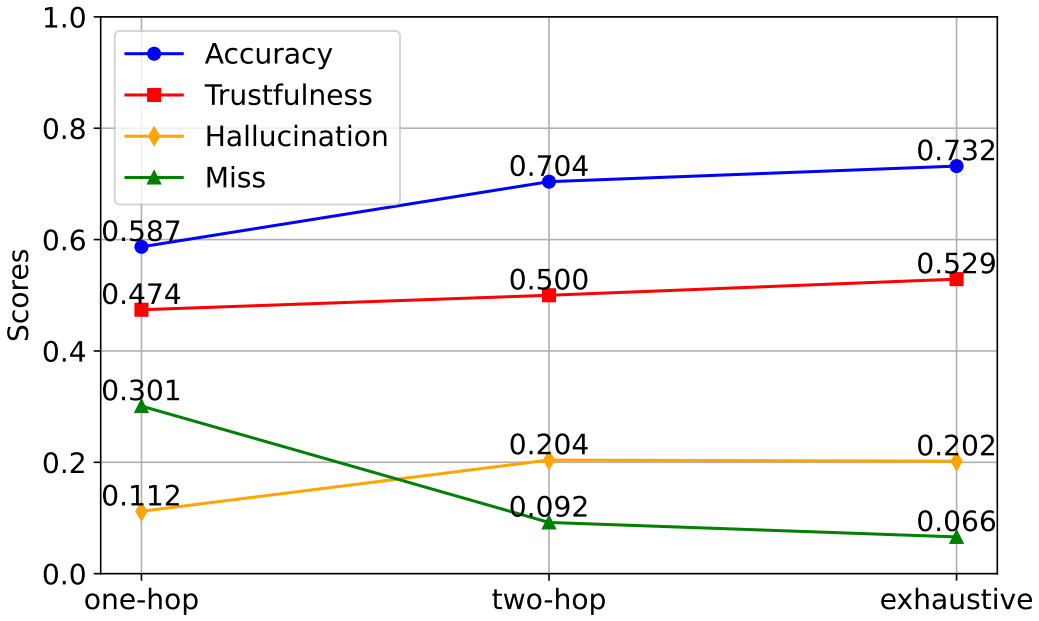


Figure 4.5: Comparison between one-hop, two-hop, and exhaustive retrieval on CRAG.

4.5.7 Robustness

We explore the robustness of KERAG on head/torso/tail categorization. As shown in Tables 4.7 and 4.8, GPT-4o and Llama-3.1-70B-Instruct models present a significant decreasing trend in terms of accuracy and truthfulness going from head to torso to tail categories, even with KB-based RAG. In contrast, KERAG presents a relatively stable performance, narrowing down the gaps and demonstrating robustness. Interestingly, on Head2Tail KERAG has lower performance on head entities, because of challenges like knowledge overloading on head entities and ambiguous entity names.

4.5.8 Clarifications of the Results

We observe that the improvement of our approach on API-based KGs is significantly higher than that on SPARQL-based KGs. In this section, we would like to analyze this phenomenon. Before the construction of CRAG benchmark, most of the KGQA datasets focus on SPARQL-based KGs, as a result, the community spends a lot of efforts to resolve KGQA problems on the SPARQL-based KGs. However, due to the recent advancement of

Table 4.7: Head/torso/tail experimental results on CRAG

	Model	Accuracy	Hallucination	Miss	Trustfulness
head	GPT-4o	0.401	**0.086**	0.513	0.316
	Llama	0.390	0.096	0.513	0.294
	KERAG (8B)	0.738	0.225	0.037	0.513
	KERAG	**0.754**	0.219	**0.027**	**0.535**
torso	GPT-4o	0.355	**0.089**	0.557	0.266
	Llama	0.305	**0.089**	0.606	0.217
	KERAG (8B)	0.714	0.192	0.094	0.522
	KERAG	**0.734**	0.207	**0.059**	**0.527**
tail	GPT-4o	0.266	0.096	0.638	0.170
	Llama	0.223	**0.053**	0.723	0.170
	KERAG (8B)	0.686	0.207	**0.106**	0.479
	KERAG	**0.707**	0.181	0.112	**0.527**

Table 4.8: Head/torso/tail experimental results on Head2Tail

	Model	Accuracy	Hallucination	Miss	Trustfulness
head	GPT-4o	0.613	0.181	0.205	0.432
	Llama	0.629	0.197	0.173	0.432
	KERAG	**0.883**	**0.067**	**0.051**	**0.816**
torso	GPT-4o	0.517	0.133	0.349	0.384
	Llama	0.440	0.139	0.421	0.301
	KERAG	**0.917**	**0.045**	**0.037**	**0.872**
tail	GPT-4o	0.376	0.165	0.459	0.211
	Llama	0.288	0.152	0.560	0.136
	KERAG	**0.925**	**0.035**	**0.040**	**0.891**

tool-calling and planning abilities of LLMs, the research community identifies the needs of analyzing the KGQA problem on API-based KGs, and develops the CRAG benchmark. Since the API-based KG dataset was only recently established and gained attention, the current state-of-the-art baselines are not as strong as those on the SPARQL-based KGs. As a result, our proposed method presents a better improvement on the API-based KGs in comparison with the SPARLQ-based KGs.

4.6 Conclusion

In conclusion, our proposed KERAG pipeline addresses key challenges in KB-based RAG by leveraging entity-level information, significantly reducing reliance on triple-level semantic parsing. By integrating a novel data generation scheme for fine-tuning chain-of-thought summarizers, KERAG enhances the ability to handle complex queries effectively. Its flexibility across various knowledge bases, including those accessed via SPARQL or APIs, demonstrates broad applicability and generalizability. Experimental results confirm that KERAG outperforms existing state-of-the-art methods, making it a promising advancement in the field of KBQA.

CHAPTER 5

CONCLUSION AND FUTURE WORK

In this thesis, we introduced three works on the three phases of data curation and RAG.

Firstly, we delve into the challenge of the lack of a proper annotator training and selection scheme in the data annotation process. We designed a novel cross-domain-aware worker selection with training framework that properly allocates the training tasks and selects and trains the top-performing data annotators iteratively. Specifically, we adopted multi-variate normal distribution to model the correlations between prior and target domains. Moreover, we introduced item response theory to model the dynamic change of annotators throughout the worker training process. The proposed worker selection scheme assists users in identifying the top-performing annotators from the candidate pool, ensuring the effectiveness and quality of the data annotation pipeline.

Then we explore how to enhance the data preparation process by developing advanced table column semantic type annotation methods. Specifically, we proposed a novel inter-table-context-aware column type annotation framework RECA. RECA discovers the relationship between different tables and identifies the potentially related data columns. By jointly considering multiple related columns when making the annotation, RECA is able to capture broader table context and thus provides more accurate annotation of the column types.

Finally, we investigate effective methods for knowledge-based Retrieval-Augmented Generation (RAG). We introduce KERAG, an innovative framework that enhances retrieval-augmented generation through knowledge integration. The KERAG pipeline specifically addresses critical challenges in knowledge base (KB)-based RAG by utilizing entity-level information, which significantly diminishes the dependence on triple-level semantic parsing. Additionally, KERAG incorporates a novel data generation scheme designed for fine-tuning chain-of-thought summarizers, greatly improving its capability to

manage complex queries. Its adaptability across various knowledge bases, whether accessed through SPARQL or APIs, highlights its broad applicability and versatility. Experimental findings indicate that KERAG surpasses existing state-of-the-art approaches, establishing it as a significant advancement in the realm of knowledge base question answering (KBQA).

In the future, we plan to explore other emerging directions in data curation and RAG. For example, if multiple data sources with different modality and different data curation process are involved, how to select the most trustworthy and effective data source based on the question context? How to resolve the conflicts between multiple data sources? Our initial exploration is to design an LLM-based router, which has access to the current question context and previous QA history. Throughout the historical interaction records, we can generate representations such as embeddings or textual descriptions for each data source and evaluate the relevance, diversity, reliability by comparing the question context against the representations of different data sources. We also need to develop a comprehensive metric to capture the relevance, diversity, and reliability of different data sources so that the most suitable data source can be selected from multiple candidate data sources. Timestamps, locations, consistency of different data sources may also be considered in the metric when we would like to resolve the conflicts between different sources. As such, we believe we can adaptively route the user question to the most suitable data source and obtain high quality answers from the RAG pipelines. Research works related to recommendation system and truth discovery may be helpful for developing such a multi-source RAG approach.

BIBLIOGRAPHY

[1] M. Honnibal and I. Montani, "spacy 2: Natural language understanding with bloom embeddings, convolutional neural networks and incremental parsing," *To appear*, vol. 7, no. 1, pp. 411–420, 2017.

[2] M. Borroto, F. Ricca, B. Cuteri, and V. Barbara, "Sparql-qa enters the qald challenge," in *Proceedings of the 7th Natural Language Interfaces for the Web of Data (NLIWoD) co-located with the 19th European Semantic Web Conference, Hersonissos, Greece*, vol. 3196, 2022, pp. 25–31. [Online]. Available: https://ceur-ws.org/Vol-3196/paper3.pdf

[3] D. Yu, S. Zhang, P. Ng, H. Zhu, A. H. Li, J. Wang, Y. Hu, W. Y. Wang, Z. Wang, and B. Xiang, "Decaf: Joint decoding of answers and logical forms for question answering over knowledge bases," in *The Eleventh International Conference on Learning Representations*, 2022. [Online]. Available: https://openreview.net/pdf?id=XHc5zRPxqV9

[4] X. Li, R. Zhao, Y. K. Chia, B. Ding, S. Joty, S. Poria, and L. Bing, "Chain-of-knowledge: Grounding large language models via dynamic knowledge adapting over heterogeneous sources," in *The Twelfth International Conference on Learning Representations*, 2023.

[5] Y. Sun, H. Xin, K. Sun, Y. E. Xu, X. Yang, X. L. Dong, N. Tang, and L. Chen, "Are large language models a good replacement of taxonomies?" *Proc. VLDB Endow.*, vol. 17, no. 11, p. 2919–2932, Aug. 2024. [Online]. Available: https://doi.org/10.14778/3681954.3681973

[6] X. Yang, K. Sun, H. Xin, Y. Sun, N. Bhalla, X. Chen, S. Choudhary, R. D. Gui, Z. W. Jiang, Z. Jiang *et al.*, "Crag-comprehensive rag benchmark," *arXiv preprint arXiv:2406.04744*, 2024.

[7] W. Fan, Y. Ding, L. Ning, S. Wang, H. Li, D. Yin, T.-S. Chua, and Q. Li, "A survey on rag meeting llms: Towards retrieval-augmented large language models," in *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024, pp. 6491–6501.

[8] B. Ni, Z. Liu, L. Wang, Y. Lei, Y. Zhao, X. Cheng, Q. Zeng, L. Dong, Y. Xia, K. Kenthapadi *et al.*, "Towards trustworthy retrieval augmented generation for large language models: A survey," *arXiv preprint arXiv:2502.06872*, 2025.

[9] Y. Sun, J. Wang, P. Cheng, L. Zheng, L. Chen, and J. Yin, " Cross-Domain-Aware Worker Selection with Training for Crowdsourced Annotation ," in *2024 IEEE 40th International Conference on Data Engineering (ICDE)*. Los Alamitos, CA, USA: IEEE Computer Society, May 2024, pp. 249–262. [Online]. Available: https://doi.ieeecomputersociety.org/10.1109/ICDE60146.2024.00026

[10] Y. Sun, H. Xin, and L. Chen, "Reca: Related tables enhanced column semantic type annotation framework," *Proceedings of the VLDB Endowment*, vol. 16, no. 6, pp. 1319–1331, 2023.

[11] K. Sun, Y. Xu, H. Zha, Y. Liu, and X. L. Dong, "Head-to-tail: How knowledgeable are large language models (llms)? aka will llms replace knowledge graphs?" in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 311–325.

[12] A. Jain, H. Patel, L. Nagalapatti, N. Gupta, S. Mehta, S. Guttula, S. Mujumdar, S. Afzal, R. Sharma Mittal, and V. Munigala, "Overview and importance of data quality for machine learning tasks," in *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. ACM, 2020, pp. 3561–3562.

[13] S. Shen, M. Ji, Z. Wu, and X. Yang, "An optimization approach for worker selection in crowdsourcing systems," *Computers & Industrial Engineering*, vol. 173, 2022.

[14] C. C. Cao, J. She, Y. Tong, and L. Chen, "Whom to ask? jury selection for decision making tasks on micro-blog services," *Proceedings of the VLDB Endowment*, vol. 5, no. 11, pp. 1495–1506, 2012.

[15] Y. Zhao, G. Liu, K. Zheng, A. Liu, Z. Li, and X. Zhou, "A context-aware approach for trustworthy worker selection in social crowd," *World Wide Web*, vol. 20, no. 6, pp. 1211–1235, 2017.

[16] H. Li and Q. Liu, "Cheaper and better: Selecting good workers for crowdsourcing," in *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing*, vol. 3. AAAI, 2015, pp. 20–21.

[17] G. Wu, Z. Chen, J. Liu, D. Han, and B. Qiao, "Task assignment for social-oriented crowdsourcing," *Frontiers of Computer Science*, vol. 15, no. 2, 2021.

[18] A. Yadav, S. Mishra, and A. S. Sairam, "A multi-objective worker selection scheme in crowdsourced platforms using nsga-ii," *Expert Systems with Applications*, vol. 201, 2022.

[19] "Jd-platform," 2023. [Online]. Available: https://biao.jd.com/

[20] "Alibaba-platform," 2023. [Online]. Available: https://zhongbao.aliyun.com/

[21] "Baidu-platform," 2023. [Online]. Available: https://zhongbao.baidu.com/

[22] T. Awwad, N. Bennani, K. Ziegler, V. Sonigo, L. Brunie, and H. Kosch, "Efficient worker selection through history-based learning in crowdsourcing," in *2017 IEEE 41st Annual Computer Software and Applications Conference*, vol. 1. IEEE, 2017, pp. 923–928.

[23] C. Shan, N. Mamoulis, G. Li, R. Cheng, Z. Huang, and Y. Zheng, "A crowdsourcing framework for collecting tabular data," *IEEE Transactions on Knowledge and Data Engineering*, vol. 32, no. 11, pp. 2060–2074, 2019.

[24] Z. Zhao, F. Wei, M. Zhou, W. Chen, and W. Ng, "Crowd-selection query processing in crowdsourcing databases: A task-driven approach," in *Proceedings 26th Interna-*

tional Conference on Extending Database Technology. OpenProceedings.org, 2015, pp. 397–408.

[25] Q. Liu, A. T. Ihler, and M. Steyvers, "Scoring workers in crowdsourcing: How many control questions are enough?" in *Proceedings of the 26th International Conference on Neural Information Processing Systems*, vol. 2. Curran Associates Inc., 2013, pp. 1914–1922.

[26] U. Gadiraju, B. Fetahu, and R. Kawase, "Training workers for improving performance in crowdsourcing microtasks," in *European Conference on Technology Enhanced Learning*. Springer, 2015, pp. 100–114.

[27] D. Haas, J. Ansel, L. Gu, and A. Marcus, "Argonaut: Macrotask crowdsourcing for complex data processing," *Proceedings of the VLDB Endowment*, vol. 8, no. 12, pp. 1642–1653, 2015.

[28] G. Rasch, *Probabilistic models for some intelligence and attainment tests*. ERIC, 1993.

[29] H. Li, B. Zhao, and A. Fuxman, "The wisdom of minority: Discovering and targeting the right group of workers for crowdsourcing," in *Proceedings of the 23rd International Conference on World Wide Web*. ACM, 2014, pp. 165–176.

[30] E. Even-Dar, S. Mannor, Y. Mansour, and S. Mahadevan, "Action elimination and stopping conditions for the multi-armed bandit and reinforcement learning problems," *Journal of Machine Learning Research*, vol. 7, no. 6, pp. 1079–1105, 2006.

[31] W. Cao, J. Li, Y. Tao, and Z. Li, "On top-k selection in multi-armed bandits and hidden bipartite graphs," in *Proceedings of the 28th International Conference on Neural Information Processing Systems*, vol. 1. MIT Press, 2015, pp. 1036–1044.

[32] A. T. Corbett and J. R. Anderson, "Knowledge tracing: Modeling the acquisition of procedural knowledge," *User Modeling and User-Adapted Interaction*, vol. 4, no. 4, pp. 253–278, 1994.

[33] G. Abdelrahman, Q. Wang, and B. P. Nunes, "Knowledge tracing: A survey," *ACM Computing Surveys*, 2022, just Accepted.

[34] S. Minn, Y. Yu, M. C. Desmarais, F. Zhu, and J.-J. Vie, "Deep knowledge tracing and dynamic student classification for knowledge tracing," in *2018 IEEE International Conference on Data Mining*. IEEE, 2018, pp. 1182–1187.

[35] M. Khajah, R. Wing, R. V. Lindsey, and M. Mozer, "Integrating latent-factor and knowledge-tracing models to predict individual differences in learning," in *Proceedings of the 7th International Conference on Educational Data Mining*. IEDMS, 2014, pp. 99–106.

[36] J. I. Lee and E. Brunskill, "The impact on individualizing student models on necessary practice opportunities," in *Proceedings of the 5th International Conference on Educational Data Mining*. IEDMS, 2012, pp. 118–125.

[37] M. V. Yudelson, K. R. Koedinger, and G. J. Gordon, "Individualized bayesian knowledge tracing models," in *International Conference on Artificial Intelligence in Education*. Springer, 2013, pp. 171–180.

[38] R. S. d Baker, A. T. Corbett, and V. Aleven, "More accurate student modeling through contextual estimation of slip and guess probabilities in bayesian knowledge tracing," in *International Conference on Intelligent Tutoring Systems*. Springer, 2008, pp. 406–415.

[39] Z. A. Pardos and N. T. Heffernan, "Kt-idem: Introducing item difficulty to the knowledge tracing model," in *International Conference on User Modeling, Adaptation, and Personalization*. Springer, 2011, pp. 243–254.

[40] K. H. Wilson, Y. Karklin, B. Han, and C. Ekanadham, "Back to the basics: Bayesian extensions of irt outperform neural networks for proficiency estimation," in *Proceedings of the 9th International Conference on Educational Data Mining*. ERIC, 2016, pp. 539–544.

[41] P. I. Pavlik, H. Cen, and K. R. Koedinger, "Performance factors analysis–a new alternative to knowledge tracing," in *Proceedings of the 2009 conference on Artificial Intelligence in Education: Building Learning Systems that Care: From Knowledge Representation to Affective Modelling*. IOS Press, 2009, pp. 531–538.

[42] C. Piech, J. Bassen, J. Huang, S. Ganguli, M. Sahami, L. J. Guibas, and J. Sohl-Dickstein, "Deep knowledge tracing," in *Proceedings of the 28th International Conference on Neural Information Processing Systems*, vol. 1. MIT Press, 2015, pp. 505–513.

[43] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[44] J. Zhang, X. Shi, I. King, and D.-Y. Yeung, "Dynamic key-value memory networks for knowledge tracing," in *Proceedings of the 26th International Conference on World Wide Web*. International World Wide Web Conferences Steering Committee, 2017, pp. 765–774.

[45] A. Miller, A. Fisch, J. Dodge, A.-H. Karimi, A. Bordes, and J. Weston, "Key-value memory networks for directly reading documents," in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, 2016, pp. 1400–1409. [Online]. Available: https://aclanthology.org/D16-1147.pdf

[46] G. Abdelrahman and Q. Wang, "Knowledge tracing with sequential key-value memory networks," in *Proceedings of the 42nd International ACM SIGIR Conference on Research and Development in Information Retrieval*. ACM, 2019, pp. 175–184.

[47] Y. L. Tong, *The multivariate normal distribution*. Springer Science & Business Media, 2012.

[48] "Technical report of the crowdsourced paper," 2024. [Online]. Available: https://github.com/ysunbp/crowdsourcing/blob/main/C4U_ICDE_technical_report.pdf

[49] Q. Inc., "Qualtrics," 2023, retrieved on Jan 1, 2023. [Online]. Available: https://www.qualtrics.com

[50] Z. Chen, R. Fu, Z. Zhao, Z. Liu, L. Xia, L. Chen, P. Cheng, C. C. Cao, Y. Tong, and C. J. Zhang, "gmission: A general spatial crowdsourcing platform," *Proceedings of the VLDB Endowment*, vol. 7, no. 13, pp. 1629–1632, 2014.

[51] K. Green, "Multiple choice and true-false: reliability and validity compared," *The Journal of Experimental Education*, vol. 48, no. 1, pp. 42–44, 1979.

[52] A. Dudley, "Multiple dichotomous-scored items in second language testing: Investigating the multiple true-false item type under norm-referenced conditions," *Language Testing*, vol. 23, no. 2, pp. 198–228, 2006.

[53] Vivek, "Kaggle elephant data," 2022, retrieved on Jan 1, 2023. [Online]. Available: https://www.kaggle.com/datasets/vivmankar/asian-vs-african-elephant-image-classification

[54] Koto, "Heywhale marine life data," 2020, retrieved on Jan 1, 2023. [Online]. Available: https://www.heywhale.com/mw/dataset/5e55f7960e2b66002c245df5

[55] B. J. Boom, P. X. Huang, C. Beyan, C. Spampinato, S. Palazzo, J. He, E. Beauxis-Aussalet, S.-I. Lin, H.-M. Chou, G. Nadarajan *et al.*, "Long-term underwater camera surveillance for monitoring and analysis of fish populations," in *Proceedings of Workshop on Visual Observation and Analysis of Animal and Insect Behavior*. Curran Associates Inc., 2012.

[56] B. J. Boom, P. X. Huang, J. He, and R. B. Fisher, "Supporting ground-truth annotation of image datasets using clustering," in *Proceedings of the 21st International Conference on Pattern Recognition*. IEEE, 2012, pp. 1542–1545.

[57] S. Maji, E. Rahtu, J. Kannala, M. Blaschko, and A. Vedaldi, "Fine-grained visual classification of aircraft," *Arxiv Preprint*, 2013.

[58] M.-E. Nilsback and A. Zisserman, "Automated flower classification over a large number of classes," in *2008 Sixth Indian Conference on Computer Vision, Graphics & Image Processing*. IEEE, 2008, pp. 722–729.

[59] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "Imagenet: A large-scale hierarchical image database," in *2009 IEEE conference on computer vision and pattern recognition*. Ieee, 2009, pp. 248–255.

[60] P. J. Werbos, "Backpropagation through time: what it does and how to do it," *Proceedings of the IEEE*, vol. 78, no. 10, pp. 1550–1560, 1990.

[61] L. Zheng and L. Chen, "Dlta: A framework for dynamic crowdsourcing classification tasks," *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 5, pp. 867–879, 2018.

[62] M. Dontcheva, R. R. Morris, J. R. Brandt, and E. M. Gerber, "Combining crowdsourcing and learning to improve engagement and performance," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2014, pp. 3379–3388.

[63] R. de Leon Pereira, A. Tan, A. Bunt, and O. Tremblay-Savard, "Increasing player engagement, retention and performance through the inclusion of educational content in a citizen science game," in *Proceedings of the 16th International Conference on the Foundations of Digital Games*, 2021, pp. 1–12.

[64] J. R. Hackman and G. R. Oldham, "Motivation through the design of work: Test of a theory," *Organizational behavior and human performance*, vol. 16, no. 2, pp. 250–279, 1976.

[65] T. L.-P. Tang and L. Sarsfield-Baldwin, "The effects of self-esteem, task label, and performance feedback on task liking and intrinsic motivation," *The Journal of Social Psychology*, vol. 131, no. 4, pp. 567–572, 1991.

[66] M. Hulsebos, K. Hu, M. Bakker, E. Zgraggen, A. Satyanarayan, T. Kraska, Ç. Demiralp, and C. Hidalgo, "Sherlock: A deep learning approach to semantic data type detection," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*. New York, NY, USA: ACM, 2019, pp. 1500–1508.

[67] E. Rahm and P. A. Bernstein, "A survey of approaches to automatic schema matching," *the VLDB Journal*, vol. 10, no. 4, pp. 334–350, 2001.

[68] V. Raman and J. M. Hellerstein, "Potter's wheel: An interactive data cleaning system," in *VLDB*, vol. 1. San Francisco, CA, USA: Morgan Kaufmann, 2001, pp. 381–390.

[69] S. Kandel, A. Paepcke, J. Hellerstein, and J. Heer, "Wrangler: Interactive visual specification of data transformation scripts," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. New York, NY, USA: ACM, 2011, pp. 3363–3372.

[70] P. Venetis, A. Y. Halevy, J. Madhavan, M. Pasca, W. Shen, F. Wu, and G. Miao, "Recovering semantics of tables on the web," *Proceedings of the VLDB Endowment*, vol. 4, no. 9, pp. 528–538, 2011.

[71] D. Zhang, M. Hulsebos, Y. Suhara, Ç. Demiralp, J. Li, and W.-C. Tan, "Sato: contextual semantic type detection in tables," *Proceedings of the VLDB Endowment*, vol. 13, no. 12, pp. 1835–1848, 2020.

[72] P. Yin, G. Neubig, W.-t. Yih, and S. Riedel, "Tabert: Pretraining for joint understanding of textual and tabular data," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. ACL, 2020, pp. 8413–8426.

[73] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "Bert: Pre-training of deep bidirectional transformers for language understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. ACL, 2019, pp. 4171–4186.

[74] H. Iida, D. Thai, V. Manjunatha, and M. Iyyer, "Tabbie: Pretrained representations of tabular data," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. ACL, 2021, pp. 3446–3456.

[75] Y. Suhara, J. Li, Y. Li, D. Zhang, Ç. Demiralp, C. Chen, and W.-C. Tan, "Annotating columns with pre-trained language models," in *Proceedings of the 2022 International Conference on Management of Data*. New York, NY, USA: ACM, 2022, pp. 1493–1503.

[76] R. J. Miller, "Open data integration," *Proceedings of the VLDB Endowment*, vol. 11, no. 12, pp. 2130–2139, 2018.

[77] D. Wang, P. Shiralkar, C. Lockard, B. Huang, X. L. Dong, and M. Jiang, "Tcn: Table convolutional network for web table interpretation," in *Proceedings of the Web Conference 2021*. New York, NY, USA: ACM, 2021, pp. 4020–4032.

[78] V. Cutrona, F. Bianchi, E. Jiménez-Ruiz, and M. Palmonari, "Tough tables: Carefully evaluating entity linking for tabular data," in *International Semantic Web Conference*. Cham, Switzerland: Springer, 2020, pp. 328–343.

[79] K. Hu, S. Gaikwad, M. Hulsebos, M. A. Bakker, E. Zgraggen, C. Hidalgo, T. Kraska, G. Li, A. Satyanarayan, and Ç. Demiralp, "Viznet: Towards a large-scale visualization learning and benchmarking repository," in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. New York, NY, USA: ACM, 2019.

[80] S. K. Ramnandan, A. Mittal, C. A. Knoblock, and P. Szekely, "Assigning semantic labels to data sources," in *European Semantic Web Conference*. Cham, Switzerland: Springer, 2015, pp. 403–417.

[81] E. L. Lehmann, J. P. Romano, and G. Casella, *Testing statistical hypotheses*. New York, NY, USA: Springer, 2005, vol. 3.

[82] M. Pham, S. Alse, C. A. Knoblock, and P. Szekely, "Semantic labeling: a domain-independent approach," in *International Semantic Web Conference*. Cham, Switzerland: Springer, 2016, pp. 446–462.

[83] X. Deng, H. Sun, A. Lees, Y. Wu, and C. Yu, "Turl: table understanding through

representation learning," *Proceedings of the VLDB Endowment*, vol. 14, no. 3, pp. 307–319, 2020.

[84] J. Herzig, P. K. Nowak, T. Mueller, F. Piccinno, and J. Eisenschlos, "Tapas: Weakly supervised table parsing via pre-training," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. ACL, 2020, pp. 4320–4333.

[85] Y. Zhou, S. Singh, and C. Christodoulopoulos, "Tabular data concept type detection using star-transformers," in *Proceedings of the 30th ACM International Conference on Information & Knowledge Management*. New York, NY, USA: ACM, 2021, pp. 3677–3681.

[86] Q. Guo, X. Qiu, P. Liu, Y. Shao, X. Xue, and Z. Zhang, "Star-transformer," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*. ACL, 2019, pp. 1315–1325.

[87] A. Backurs and P. Indyk, "Edit distance cannot be computed in strongly subquadratic time (unless seth is false)," in *Proceedings of the forty-seventh annual ACM symposium on Theory of computing*. New York, NY, USA: ACM, 2015, pp. 51–58.

[88] F. P. Miller, A. F. Vandome, and J. McBrewster, "Levenshtein distance: Information theory, computer science, string (computer science), string metric, damerau? levenshtein distance, spell checker, hamming distance," 2009.

[89] P. Jaccard, "The distribution of the flora in the alpine zone. 1," *New phytologist*, vol. 11, no. 2, pp. 37–50, 1912.

[90] E. Jiménez-Ruiz, O. Hassanzadeh, V. Efthymiou, J. Chen, K. Srinivas, and V. Cutrona, "Results of semtab 2020," in *CEUR Workshop Proceedings*, vol. 2775, 2020, pp. 1–8.

[91] B. Fetahu, A. Anand, and M. Koutraki, "Tablenet: An approach for determining fine-grained relations for wikipedia tables," in *The World Wide Web Conference*. New York, NY, USA: ACM, 2019, pp. 2736–2742.

[92] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz *et al.*, "Huggingface's transformers: State-of-the-art natural language processing," *arXiv preprint*, p. 1, 2019.

[93] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in neural information processing systems*, 2017, pp. 5998–6008.

[94] V. Cutrona, J. Chen, V. Efthymiou, O. Hassanzadeh, E. Jiménez-Ruiz, J. Sequeda, K. Srinivas, N. Abdelmageed, M. Hulsebos, D. Oliveira *et al.*, "Results of semtab 2021," *Proceedings of the Semantic Web Challenge on Tabular Data to Knowledge Graph Matching*, vol. 3103, pp. 1–12, 2022.

[95] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg *et al.*, "Scikit-learn: Machine learning in python," *the Journal of machine Learning research*, vol. 12, pp. 2825–2830, 2011.

[96] "Sherlock," 2019. [Online]. Available: https://github.com/mitmedialab/sherlock-project

[97] "Tabert," 2020. [Online]. Available: https://github.com/facebookresearch/TaBERT

[98] A. Koleva, M. Ringsquandl, M. Joblin, and V. Tresp, "Generating table vector representations," *arXiv preprint*, p. 5, 2021.

[99] "Tabbie," 2021. [Online]. Available: https://github.com/SFIG611/tabbie

[100] "Doduo," 2022. [Online]. Available: https://github.com/megagonlabs/doduo

[101] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint*, p. 5, 2014.

[102] X. L. Dong, "The journey to a knowledgeable assistant with retrieval-augmented generation (rag)," in *Companion of the 2024 International Conference on Management of Data*, ser. SIGMOD/PODS '24. New York, NY, USA:

Association for Computing Machinery, 2024, p. 3. [Online]. Available: https://doi.org/10.1145/3626246.3655999

[103] A. Hogan, X. L. Dong, D. Vrandečić, and G. Weikum, "Large language models, knowledge graphs and search engines: A crossroads for answering users' questions," *arXiv preprint arXiv:2501.06699*, 2025.

[104] W.-t. Yih, M. Richardson, C. Meek, M.-W. Chang, and J. Suh, "The value of semantic parse labeling for knowledge base question answering," in *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, 2016, pp. 201–206. [Online]. Available: https://aclanthology.org/P16-2033.pdf

[105] R. Das, M. Zaheer, D. Thai, A. Godbole, E. Perez, J.-Y. Lee, L. Tan, L. Polymenakos, and A. Mccallum, "Case-based reasoning for natural language queries over knowledge bases," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 9594–9611.

[106] Y. Gu and Y. Su, "Arcaneqa: Dynamic program induction and contextualized encoding for knowledge base question answering," in *Proceedings of the 29th International Conference on Computational Linguistics*, 2022, pp. 1718–1731. [Online]. Available: https://aclanthology.org/2022.coling-1.148/

[107] Y. Shu, Z. Yu, Y. Li, B. Karlsson, T. Ma, Y. Qu, and C.-Y. Lin, "Tiara: Multi-grained retrieval for robust question answering over large knowledge base," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 8108–8121.

[108] X. Hu, X. Wu, Y. Shu, and Y. Qu, "Logical form generation via multi-task learning for complex question answering over knowledge bases," in *Proceedings of the 29th International Conference on Computational Linguistics*, 2022, pp. 1687–1696. [Online]. Available: https://aclanthology.org/2022.coling-1.145/

[109] T. Xie, C. H. Wu, P. Shi, R. Zhong, T. Scholak, M. Yasunaga, C.-S. Wu, M. Zhong,

P. Yin, S. I. Wang *et al.*, "Unifiedskg: Unifying and multi-tasking structured knowledge grounding with text-to-text language models," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 602–631.

[110] S. Xu, S. Liu, T. Culhane, E. Pertseva, M.-H. Wu, S. Semnani, and M. Lam, "Fine-tuned llms know more, hallucinate less with few-shot sequence-to-sequence semantic parsing over wikidata," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 5778–5791.

[111] L. Zhang, J. Zhang, Y. Wang, S. Cao, X. Huang, C. Li, H. Chen, and J. Li, "Fc-kbqa: A fine-to-coarse composition framework for knowledge base question answering," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 1002–1017.

[112] S. W.-t. Yih, M.-W. Chang, X. He, and J. Gao, "Semantic parsing via staged query graph generation: Question answering with knowledge base," in *Proceedings of the Joint Conference of the 53rd Annual Meeting of the ACL and the 7th International Joint Conference on Natural Language Processing of the AFNLP*, 2015. [Online]. Available: https://aclanthology.org/P15-1128.pdf

[113] K. Luo, F. Lin, X. Luo, and K. Zhu, "Knowledge base question answering via encoding of complex query graphs," in *Proceedings of the 2018 conference on empirical methods in natural language processing*, 2018, pp. 2185–2194.

[114] Y. Lan, S. Wang, and J. Jiang, "Knowledge base question answering with topic units.(2019)," in *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, 2019, pp. 5046–5052. [Online]. Available: https://www.ijcai.org/proceedings/2019/0701.pdf

[115] J. Sun, C. Xu, L. Tang, S. Wang, C. Lin, Y. Gong, L. Ni, H.-Y. Shum, and J. Guo, "Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=nnVO1PvbTv

[116] S. Ma, C. Xu, X. Jiang, M. Li, H. Qu, C. Yang, J. Mao, and J. Guo, "Think-on-graph 2.0: Deep and faithful large language model reasoning with knowledge-guided retrieval augmented generation," in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: https://openreview.net/forum?id=oFBu7qaZpS

[117] X. L. Dong, E. Gabrilovich, G. Heitz, W. Horn, K. Murphy, S. Sun, and W. Zhang, "From data fusion to knowledge fusion," *Proceedings of the VLDB Endowment*, vol. 7, no. 10, pp. 881–892, 2014.

[118] Y. Lan and J. Jiang, "Query graph generation for answering multi-hop complex questions from knowledge bases," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 969–974.

[119] B. Oguz, X. Chen, V. Karpukhin, S. Peshterliev, D. Okhonko, M. Schlichtkrull, S. Gupta, Y. Mehdad, and S. Yih, "Unik-qa: Unified representations of structured and unstructured knowledge for open-domain question answering," in *Findings of the Association for Computational Linguistics: NAACL 2022*, 2022, pp. 1535–1546.

[120] Y. Gu, X. Deng, and Y. Su, "Don ' t generate, discriminate: A proposal for grounding language models to real-world environments," in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 4928–4949.

[121] A. Saxena, A. Tripathi, and P. Talukdar, "Improving multi-hop question answering over knowledge graphs using knowledge base embeddings," in *Proceedings of the 58th annual meeting of the association for computational linguistics*, 2020, pp. 4498–4507.

[122] G. He, Y. Lan, J. Jiang, W. X. Zhao, and J.-R. Wen, "Improving multi-hop knowledge base question answering by learning intermediate supervision signals," in *Proceedings of the 14th ACM international conference on web search and data mining*, 2021, pp. 553–561.

[123] P. Sen, A. Oliya, and A. Saffari, "Expanding end-to-end question answering on differentiable knowledge graphs with intersection," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 8805–8812.

[124] J. Shi, S. Cao, L. Hou, J. Li, and H. Zhang, "Transfernet: An effective and transparent framework for multi-hop question answering over relation graph," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 4149–4158.

[125] C. Mavromatis and G. Karypis, "Rearev: Adaptive reasoning for question answering over knowledge graphs," in *Findings of the Association for Computational Linguistics: EMNLP 2022*, 2022, pp. 2447–2458.

[126] J. Zhang, X. Zhang, J. Yu, J. Tang, J. Tang, C. Li, and H. Chen, "Subgraph retrieval enhanced model for multi-hop knowledge base question answering," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2022, pp. 5773–5784.

[127] J. Jiang, K. Zhou, Z. Dong, K. Ye, W. X. Zhao, and J.-R. Wen, "Structgpt: A general framework for large language model to reason over structured data," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, 2023, pp. 9237–9251.

[128] A. Dubey, A. Jauhri, A. Pandey, A. Kadian, A. Al-Dahle, A. Letman, A. Mathur, A. Schelten, A. Yang, A. Fan *et al.*, "The llama 3 herd of models," *arXiv preprint arXiv:2407.21783*, 2024.

[129] K. Sun, Y. Xu, H. Zha, Y. Liu, and X. L. Dong, "Head-to-tail: How knowledgeable are large language models (llms)? aka will llms replace knowledge graphs?" in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, 2024, pp. 311–325.

[130] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the*

2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020, pp. 6769–6781.

[131] R. Usbeck, X. Yan, A. Perevalov, L. Jiang, J. Schulz, A. Kraft, C. Möller, J. Huang, J. Reineke, A.-C. N. Ngomo, M. Saleem, and A. Both, "Qald-10 –the 10th challenge on question answering over linked data," *Semantic Web*, 2023. [Online]. Available: https://api.semanticscholar.org/CorpusID:265577096

[132] X. Ye and G. Durrett, "The unreliability of explanations in few-shot prompting for textual reasoning," *Advances in neural information processing systems*, vol. 35, pp. 30 378–30 392, 2022. [Online]. Available: https://openreview.net/pdf?id=Bct2f8fRd8S

[133] A. Hurst, A. Lerer, A. P. Goucher, A. Perelman, A. Ramesh, A. Clark, A. Ostrow, A. Welihinda, A. Hayes, A. Radford *et al.*, "Gpt-4o system card," *arXiv preprint arXiv:2410.21276*, 2024.

[134] Y. Xia, J. Chen, and J. Gao, "Winning solution for meta kdd cup'24," *arXiv preprint arXiv:2410.00005*, 2024. [Online]. Available: https://openreview.net/forum?id=oWNPeoP1uC

[135] J. Ouyang, Y. Luo, M. Cheng, D. Wang, S. Yu, Q. Liu, and E. Chen, "Revisiting the solution of meta kdd cup 2024: Crag," *arXiv preprint arXiv:2409.15337*, 2024. [Online]. Available: https://openreview.net/forum?id=PUzLjWIgqC

[136] Y. Zhao, A. Gu, R. Varma, L. Luo, C.-C. Huang, M. Xu, L. Wright, H. Shojanazeri, M. Ott, S. Shleifer *et al.*, "Pytorch fsdp: Experiences on scaling fully sharded data parallel," *Proceedings of the VLDB Endowment*, vol. 16, no. 12, pp. 3848–3860, 2023.

[137] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," *arXiv preprint arXiv:2106.09685*, 2021.